

THÈSE DE DOCTORAT

Titre français

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Résumé

Les macarons c'est bon par nature, c'est le propre même du macaron que d'être bon. Au vue de l'affluence constante aux divers magasins Ladurée on peut supposer que les macarons qu'on y trouve sont meilleurs qu'ailleurs. Cette thèse vise à donner des idées et pistes sur le pourquoi du comment que les macarons de Ladurée sont si bons.

Mots-clés : Pâtisserie, Macaron, Ladurée.

English Title

Abstract

Macarons are so good.

Keywords: Pastry, Macaron, Ladurée.

Acknowledgement

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CHAPTER 1

Introduction

Automation is a central component of modern proof assistants such as *Rocq*. Mechanisms like type-class resolution allow the system to infer implicit arguments and build proof terms with limited user input. Operationally, this process can be seen as a form of logic programming, where goals are solved by searching for instances in a database. This view suggests using a meta-language with strong support for search and unification.

In this work, we study the integration of *Rocq* with the logic-programming language *Elpi*. The language follows a *Prolog*-like execution model based on higher-order unification, backtracking, and control operators such as cut. These features make it suitable for implementing a type-class solver for *Rocq* in *Elpi*.

A first point we emphasize is the simplicity of this approach. A basic compiler for type classes can be written in only a few lines, while the solver directly relies on the existing interpreter. We then extend this basic translation to support more complex instances as found in practice, and evaluate the resulting solver against the native one of *Rocq*.

A difficulty arises from the mismatch between the object language and the meta-language. Although both support higher-order unification and $\beta\eta$ -conversion, the representation of terms in *Elpi* differs from that of *Rocq*. As a result, some terms that unify in *Rocq* may fail to unify in *Elpi* after their translation. A direct reuse of meta-level unification is therefore not sufficient.

To overcome this issue, we introduce a compilation scheme that preserves most of the structure of compiled terms while isolating problematic parts. These subterms are replaced only when needed and recorded as *links*, which represent delayed constraints. They are checked during execution, allowing us to recover the intended behavior without reimplementing unification.

This design is also motivated by efficiency. Since most of the term structure is kept, the execution can benefit from indexing and mode unification strategy provided by *Elpi*. This is important for type-class resolution, where the search space can be large.

An additional contribution of this work is a determinacy analysis for *Elpi* programs. Determinacy aims to ensure that certain predicates do not introduce choice points and produce at most one solution. This property is useful to understand and control the behavior of logic programs.

We present an analysis that guarantees determinacy by inspecting clauses and tracking argument modes. The algorithm is designed to cover most features of *Elpi*, including higher-order predicates, dynamic programming and the cut operator. Handling cut is essential, as it can enforce deterministic behavior but interacts in subtle ways with the logical reading of programs.

To support this development, we also study a first-order fragment of the language, where terms do not contain higher-order abstractions. In this setting, unification is simpler and can be treated formally. We mechanize (in *Rocq*) the determinacy checker for this fragment and prove its main properties.

This mechanization provides a formal basis for the analysis and serves as a reference for the full algorithm. Even if it does not cover all higher-order features, it captures the essential behavior and increases confidence in the approach.

1.1 Contributions and structure of the thesis

The remainder of this thesis is organized as follows. We first present the context of the work (chapter 2). In particular, we describe the type-class resolution mechanism implemented in *Rocq*, together with its main options. Then, we introduce the *Elpi* language and give a formal description of its syntax and semantics. Finally, we present the *Rocq-Elpi* plugin, which enables the interaction between *Elpi* and *Rocq*.

This thesis contains four main contributions, organized as follows:

Chapter 3 We study the implementation of a type-class solver written in *Elpi* for *Rocq*.

We present the core part of the compiler that translates *Rocq* classes and instances into an *Elpi* database. This database is then used by the interpreter to search for inhabitants of type classes. We also compare our solver with the one of *Rocq*, focusing on performance, modes, and unification issues.

Chapter 4 We analyze unification problems arising from the higher-order abstract syntax used to represent *Rocq* terms in *Elpi*. In *Rocq*, terms that are equal up to $\beta\eta$ -conversion can unify, while in *Elpi* such unification may fail, even if its algorithm supports these conversions. Based on our work [16], we explain how to address this issue by delaying problematic unifications using a system of links.

Chapter 5 We present a static determinacy checker implemented in *Elpi*, available since version 3. This tool verifies that predicates declared as deterministic by the user effectively behave deterministically, meaning that they never create additional choice points. This result is based on our work [17].

Chapter 6 We mechanize the determinacy checker in the *Rocq* proof assistant. The formalization considers a restricted fragment of *Elpi*, namely its first-order part, where variables only represent data. Within this setting, we prove the main determinacy property: in a program that satisfies the checker, any call to a deterministic predicate produces at most one solution.

CHAPTER 2

Context: languages and tools

At the core of this manuscript, there are two important pieces of software coming from different areas of computer science. On the one hand, we consider the *Rocq* proof assistant (formerly known as *Coq*); on the other hand, we consider the logic programming language *Elpi*.

Rocq [5] is one of the main proof assistants. It is based on intuitionistic logic and allows formal reasoning about mathematical statements and program specifications. In this setting, logical propositions are interpreted as types, and proofs are interpreted as programs (or terms) of these types. The underlying type theory of *Rocq* ensures that the coherence between a program and its type is rigorously checked, thus providing strong guarantees of correctness.

Proof development in *Rocq* is carried out using a system of tactics, which allows the user to construct proof terms interactively.

On the other hand, *Elpi* [14, 53] is a language from the field of logic programming. In particular, it is a dialect of λ -*Prolog* which features backtracking, higher-order programming and unification, backtracking control through the `Cut` operator, unification control through predicates modes. This language provides a plugin, called *Rocq-Elpi*, which allows one to extend *Rocq* with new commands and tactics [32, 57, 24].

Commands extend the *Rocq* language by introducing new definitions or theorems that are automatically generated by *Elpi* programs. Examples of libraries that intensively use this feature include *derive* [54], which derives equality theorems for inductive types, and *Hierarchy Builder*, which helps define complex algebraic structures while automatically generating associated theory. *Hierarchy Builder* supports the Mathematical Components ecosystem [36] and mechanizations such as the Odd Order and Four Color theorems [23, 22].

Tactics extend the proof system of *Rocq*. They rely on the *Elpi* interpreter, which uses a Prolog-like SLD resolution strategy [21, Chap. 9] to search for proofs in a predefined *Elpi* database. For example, the *Trocq* [6, 9] library provides an *Elpi*-based tactic for automatic reasoning via proof transfer.

In the following sections, we introduce the key concepts that form the foundation of chapters 3 to 6. In particular, in section 2.1, we briefly discuss about elaboration and in particular about type-classes, a central component of a proof assistant such as *Rocq*. In section 2.2, we present the core aspects of the *Elpi* language, including its syntax and semantics, which are necessary to understand its execution model. Finally, in section 2.3, we describe the main ideas behind the *Rocq-elpi* plugin.

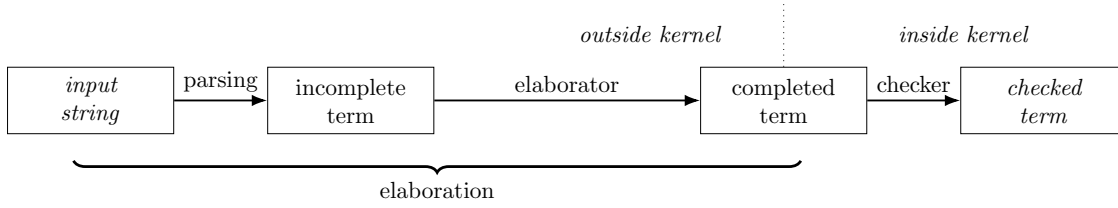


Figure 2.1: Elaboration scheme

2.1 Elaboration in the *Rocq* proof assistant

A standard introduction to *Rocq* usually starts with the foundations of the language, such as type theory, the λ -calculus, and the basic constructs used to define functions and state theorems. These topics are covered in detail books, such as [5], and papers [10]. Additional resources can be found at: <https://rocq-prover.org/papers>.

In this section, we focus on a more technical aspect that enables communication between the user and the *Rocq* system: the *elaborator*. This component is part of all proof assistants. Its purpose is to allow the user to write incomplete or ambiguous expressions, leaving the system to infer the missing information.

In the terminology of *Rocq*, missing information is called *holes*, usually denoted by underscores.

As a simple example, consider the term `length [1,0]`. This expression uses notations for natural numbers and lists, and it omits several implicit type annotations. The fully elaborated term is:

```
@length nat (@cons nat (S 0) (@cons nat 0 (@nil nat)))
```

Thus, a short expression of 12 characters is expanded into a fully explicit term of 62 characters. This automation significantly reduces the effort required from the developer.

Figure 2.1, taken from [56], summarizes the main steps of the elaboration process. Starting from a raw input string, the system parses it into a *Rocq* term that may still be incomplete, and then elaborates it into a fully specified term. This final term is passed to the kernel, which checks its type correctness. If the check succeeds, the term is a valid inhabitant of its type.

The elaboration process includes several key tasks: notation resolution, type inference, coercion insertion, and inference of overloaded symbols.

Notation resolution removes syntactic ambiguities by replacing user-defined notations with their corresponding logical symbols in the abstract syntax tree (AST). The resulting term is unambiguous but may still contain holes, which are then filled during later stages.

Type inference is one of the main mechanisms used to fill these holes. The type-checker can infer missing types when symbols are not overloaded, that is, when they have a unique interpretation.

	Symbol	Type
$\frac{s : \tau \in \Gamma}{\Gamma \vdash s : \tau} \text{const}$	<code>cons</code>	$\forall T, T \rightarrow \text{list } T \rightarrow \text{list } T$
$\frac{\Gamma \vdash f : \tau_1 \rightarrow \tau_2 \quad \Gamma \vdash a : \tau_1}{\Gamma \vdash f \ a : \tau_2} \text{app}$	<code>nil</code>	$\forall T, \text{list } T$
	<code>0</code>	<code>nat</code>

In the simple setting displayed above, using the typing rules for symbols and function application, and assuming a context Γ that contains the types of the symbols `cons`, `nil`, and `0`, the type checker can infer that both holes in the term `@cons \_ 0 (@nil \_)` have type `nat`.

Another important feature of the elaborator is the coercion mechanism [3]. Coercions allow one to implicitly convert a term of type t_1 into a term of type t_2 by inserting a function of type $t_1 \rightarrow t_2$.

For instance, suppose that `nat_to_bool` maps `0` to `false` and any other natural number to `true`, and is declared as a coercion from `nat` to `bool`. Then the term `negb 0`, where `negb` expects an argument of type `bool`, is automatically elaborated into `negb (nat_to_bool 0)`.

Beyond type inference and coercions, a key feature of elaboration is support for overloading. This allows the same symbol to be reused at different types. Since ad-hoc polymorphism plays an important role in this manuscript, we will examine this mechanism in more detail in the following subsection.

2.1.1 Ad-hoc polymorphism

Polymorphism is a fundamental feature of functional programming languages. In his lecture notes, Strachey [51] distinguishes between two main forms of polymorphism: parametric polymorphism and ad-hoc polymorphism. Parametric polymorphic functions are *uniform*, meaning that their behavior does not depend on the specific type at which they are instantiated. For example, the function `length`, with type `'a list -> int`, returns `7` when applied to any list of seven elements. In other words, the compiler reuses the same implementation of `length` for every call, regardless of the instantiation of `'a`.

Parametric polymorphism is widely used in functional programming. However, it is not suitable for modeling mathematical function overloading. In mathematics, the symbol `+` has a consistent, intuitive meaning, and no ambiguity arises in expressions such as `1 + 2` or `$\pi + e$` . In the first expression, `+` denotes addition over integers; in the second, it denotes addition over real numbers. The same symbol is consistently used for operations that share common algebraic properties.

In contrast, parametric polymorphism does not allow to have multiple different type-dependent interpretations. This limitation appears in languages such as *OCaml*, where integer and floating-point additions are represented by distinct operators, `+` and `+.` , respectively. Thus, one writes `1 + 2` for integers and `$\pi +. e$` for floating-point numbers. In general, a different symbol must be introduced for each type implementing a given mathematical operation.

To address the need for a shared notation for operations (and properties in proof assistants) across types, Wadler and Blott introduced the notion of type classes in 1989 [59] and were soon implemented in *Haskell* [26] as a pervasive feature of the language. Then other languages, such as *Rocq* [50] also extended their type system to allow this kind of overloading.

Symbol overloading Type classes provide a principled form of ad-hoc polymorphism, enabling a single symbol to be associated with multiple type-specific implementations while preserving static type safety. They provide structure parameterized by one or more

*In *OCaml*, π is `Float.pi` and e is `Float.exp 1.0`

types t and declare the signatures of a set of functions, called methods, whose types may depend on t . An instance of a type class provides a concrete implementation of the type class for a specific choice of t , providing definitions for all of its methods.

Unlike parametric polymorphism, where a unique function is used for the execution of a call, type classes rely on instance resolution. Typeclasses and instances inhabit a database mapping classes to their instances. When compiling a call to a type class method, the elaborator finds the right implementation by selecting the instance of the type class associated with the given type in the given database.

As an example, consider the type class `Add`, parameterized by a type T and equipped with a method `plus` to perform addition between two elements of type T . In *Rocq*, the definition of `Add` is given below.

```
Class Add T := {plus: T → T → T}.
```

The type of the method `plus` is $\forall T, \text{Add } T \rightarrow T \rightarrow T$, where both the type parameter T and the instance argument `Add T` are implicit. The typechecking of the term `plus 3 4` gives:

```
@plus nat ?Add 3 4 : bool
where ?Add : [ |- Add nat]
```

The term has type `nat`, but the instance for `Add nat` cannot be inferred, leaving a metavariable `?Add`, this is because in the type-class database, there is no instance associated to `Add`. To resolve this, we define the instance for natural numbers:

```
Instance addNat : Add nat := {plus := Nat.add}.
```

The declaration `addNat` has type `Add nat` and provides an implementation of the method `plus`. Registering this instance extends the instance database for `Add` with an entry for the type `nat`. Afterwards, the term `plus 3 4` can be elaborated successfully, with `?Add` instantiated to `addNat`.

Similarly, we may define an instance for booleans:

```
Instance addR : Add R := {plus := Rplus}.
```

The same symbol `plus` can then be applied to values of type `nat` or R . Therefore, extending the notation system, we are now allowed to write $1 + 2$ and $\pi + e$ knowing that the elaborator automatically selects the appropriate instance, `addNat` or `addR`.

A key feature of type classes is the ability to define *conditional* instances, whose availability depends on other instances. An example is the instance for the product type.

```
Instance addProd : ∀ T1 T2, Add T1 → Add T2 → Add (T1 * T2) :=
  {plus '(x1,y1) '(x2, y2) := (plus x1 x2, plus y1 y2)}.
```

The type of the instance `addProd` states that it implements `Add (T1 * T2)` provided that instances of `Add T1` and `Add T2` are available. In this way, addition on pairs is derived from addition on their components, that is, we can add $(\pi, 3)$ with $(e, 4)$, $(3, (\pi, 4))$ with $(7, (e, 0))$...

The type class `Add` can be defined in a similar style in *Haskell*, up to minor syntactic differences. In a proof assistant such as *Rocq*, however, type classes support not only function overloading but also *theorem overloading*.

219 For instance, one may enrich `Add` with the associativity specification:

```
Class Add T := { plus: T → T → T;
  assoc: ∀ a b c, plus a (plus b c) = plus (plus a b) c }.
```

220 An instance of this refined class must now provide both an implementation of the
 221 function `plus` and a proof that `plus` is associative. Then, if `addNat`, `addR` and `addProd` are
 222 implemented according the new definition of `Add`, the `assoc` symbol can be used as a proof
 223 that addition is associative over natural and real numbers and pairs.

224 In order to better structure type classes, the system can be extended so that type
 225 classes can inherit properties of existing classes. In group theory this feature consist, for
 226 example, in defining an object called `Magma` with an operation `op`, and a second object
 227 `Semigroup` which extends `Magma` by postulating the associative property on `op`.

228 This way for example, the `Add` class, in some developments is formalized as:

```
Class Magma T := {op : T → T → T}.
Class Semigroup T `{Magma T} :=
  {assoc a b c : op a (op b c) = op (op a b) c}.

Instance addNatM : Magma nat := {op := Nat.add}.
Instance addNatS : Semigroup nat := {assoc := Nat.add_assoc}.
```

229 **Search strategy** Type classes are used in several major *Rocq* libraries, such as *tlc*, *stdpp*,
 230 and *iris* [30]. They provide access to the type-class solver, which automatically infers
 231 instances by searching in a database of declared instances. This database is organized
 232 as an ordered list. Given a type-class goal, the solver tries to unify the goal with the
 233 conclusion of each instance. When unification succeeds, it then tries to solve the premises
 234 of that instance.

235 As an example, consider the instances defined previously for the type class `Add`. To solve
 236 the goal `Add (nat * R)`, the solver first tries `addNat`, which fails because its conclusion does
 237 not match the goal. The same happens with `addR`. The instance `addProd` can be applied
 238 if its premises, namely `Add nat` and `Add R`, can be solved. Since both are derivable, the
 239 resulting instance is `addProd nat R addNat addR`. By default, the solver follows a *depth-*
 240 *first search* (DFS): it explores the database deeply and backtracks when a branch fails.

241 The order of instances in the database is important, since it may affect both perfor-
 242 mance and termination. For instance, the goal `Add ?X`, where `?X` is an existential variable,
 243 can lead to non-termination if `addProd` has the highest priority. In such a case, the variable
 244 may be repeatedly instantiated with terms of the form $((\dots * ?X_{n-2}) * ?X_{n-1}) * ?X_n$, due
 245 to the recursive use of its first premise.

246 The priority of an instance can be specified at the end of its declaration using `| N`,
 247 where `N` is an integer. The instance is then placed at position `N` in the database. A smaller
 248 value of `N` means higher priority. If no priority is given, *Rocq* computes it automatically
 249 by assigning weight 1 to each premise that is itself a type-class constraint.

250 Several mechanisms are available to guide the solver. One common tool is *modes*,
 251 which enforce conditions on the shape of class arguments during search. There are three
 252 kinds of modes in *Rocq*. Informally, a mode is a list of symbols `+`, `!`, or `-`, indicating

whether an argument is treated as an input, partially known (its head is fixed), or as an output.

Modes can be declared in two ways. One can write `#[mode="+"] Add T.`, which specifies that the first argument of `Add` must be instantiated in goals. Alternatively, one can declare it afterwards using `Hint Mode Add +.`

A type class may have several modes. During instance search, these modes are checked dynamically: the search proceeds if at least one mode is satisfied by the current goal.

A characteristic of type classes is that several instances may match the same goal. This requires backtracking in the solver. In some libraries, such as *stdpp*, this behavior is exploited to mimic logic programming. For example, the type class `SetUnfold` is used to transform set expressions into logical properties. For instance, an inclusion $X \subseteq Y$ can be turned into a statement of the form $\forall x, P \ x \rightarrow Q \ x$, under suitable assumptions on X and Y . If no transformation applies, the default instance `set_unfold_default P : SetUnfold P P` is used, leaving the goal unchanged.

This default instance serves as a catch-all rule, as it applies to any goal. Although backtracking can be useful in such patterns, it may introduce ambiguities and lead to unexpected behavior that is difficult to control in practice.

In *Rocq*, backtracking can be disabled using the flag `Typeclasses Unique Instances`. This forces the system to commit to the first solution found during the recursive search. However, this mechanism is not class-specific: once enabled, non-backtracking is applied to all type classes involved in the search. In practice, a user may prefer a more precise behavior, where only certain type classes avoid backtracking, while others retain it. Therefore, at least in the libraries we have analyzed, this flag is not used.

2.2 The *Elpi* language

Elpi [14] stands for *Embeddable Lambda-Prolog Interpreter*. It is a dialect of *λ-Prolog* [40, 41] that supports higher-order logic programming and integrates a goal suspension mechanism based on CHR (Constraint Handling Rules [19, 25]).

The term “embeddable” highlights that *Elpi* is designed to be used as an extension language. Its main application is the *Rocq-Elpi* plugin (see section 2.3), which enables communication between *Elpi* as the meta-language and the *Rocq* proof assistant as the object language.

A comprehensive presentation of the *Elpi* language and its applications can be found in Tassi’s “habilitation à diriger les recherches” [55]. Here we briefly introduce the aspects of *Elpi* that are relevant to this document.

Sections 2.2.1 and 2.2.2 describe the syntax and operational semantics of *Elpi*. Our presentation slightly differs from [55], as it follows the conventions adopted later in chapters 5 and 6. In particular, the syntax and semantics presented here exclude the *CHR* component of *Elpi*; we discuss it informally in section 2.2.5.

In sections 2.2.3 and 2.2.4, we first present an example of execution, highlighting the role of the `Cut` operator in the *Elpi* language. We then discuss two main programming styles in *Elpi*, namely higher-order programming and dynamic programming, and explain how they interact with the *Elpi* semantics.

Program	{	$\mathbb{K} ::= f, g, \dots$	datum
		$\mathbb{P} ::= p, q, \dots$	predicate
		$\mathbb{U} ::= X, Y, \dots$	unification variable
		$\mathbb{N} ::= x, y, \dots$	nominal variable
		$\mathbb{Tm} ::= \mathbb{K} \mid \mathbb{P} \mid \mathbb{U} \mid \mathbb{N} \mid \mathbb{Tm} \ \mathbb{L}$	term
		$\mathbb{L} ::= \lambda \mathbb{N}. \mathbb{L} \mid \mathbb{Tm}$	λ -term
		$\mathbb{A} ::= \text{Cut} \mid \mathbb{Tm} \mid \text{pi } \mathbb{N}. \mathbb{A} \mid \mathbb{R} \Rightarrow \mathbb{A}$	atom
		$\mathbb{R} ::= \mathbb{Tm} :- \mathbb{A}^*$	rule
Types	{	$\mathbb{Kd} ::= \text{kind } \mathbb{K} \ \mathbb{T}_a$	type declaration
		$\mathbb{T}_a ::= \text{type} \mid \mathbb{T}_a \rightarrow \mathbb{T}_a$	type arity
		$\mathbb{Ks} ::= \text{type } \mathbb{K} \ \mathbb{T}_y \mid \text{type } \mathbb{P} \ \mathbb{T}_y$	constructor declaration
		$\mathbb{T}_b ::= \mathbb{K} \mid \mathbb{N} \mid \mathbb{T}_b \ \mathbb{T}_y$	type
		$\mathbb{T}_y ::= \mathbb{T}_a \xrightarrow{m} \mathbb{T}_a \mid \mathbb{T}_b$	λ -type
		$m ::= \text{input} \mid \text{output}$	mode

Figure 2.2: Abstract syntax of *Elpi* programs

295 Finally, in sections 2.2.5 and 2.2.6, we outline two key features of the *Elpi* language:
 296 the constraint handling rules system, and rule naming and grafting.

297 2.2.1 Abstract syntax of *Elpi* (without *CHR*)

298 **Syntax of *Elpi* programs** Figure 2.2 presents the abstract syntax of *Elpi* programs.

299 The basic syntactic categories are data $\mathbb{K}$, predicates $\mathbb{P}$, unification variables $\mathbb{U}$,
 300 and nominal variables $\mathbb{N}$. Nominal variables represent binders introduced either by λ -
 301 abstractions or by **pi**-quantification.

302 A term $\mathbb{Tm}$ is either one of these basic constructs or an application. More precisely,
 303 terms are built by the binary application of a term to a λ -term $\mathbb{L}$. Lambda terms intro-
 304 duce abstractions; in the concrete *Elpi* syntax an abstraction is written **x** \ **t**. Function
 305 application follows a curried style: unlike *Prolog*, *Elpi* does not use parentheses between
 306 the functor and its arguments.

307 An atom $\mathbb{A}$ represents an executable unit. It can be:

- 308 1. the cut operator (**!** in the concrete syntax),
- 309 2. a predicate call,
- 310 3. a **pi**-quantification, written **pi** **x** \ **t** in the concrete syntax, which introduces a fresh
 311 nominal variable **x** visible in **t**, or
- 312 4. the dynamic insertion of a rule $\mathbb{R}$, written **r** **=>** **t**, which makes the rule **r** locally
 313 available while solving **t**.

314 A rule $\mathbb{R}$ consists of a head and a (possibly empty) body, which is a list of atoms. The
 315 atoms in the body are called the premises of the rule.

316 For simplicity, we omit constraints from the syntax, since they do not play an imme-
 317 diate role in the presentation. They are briefly described in section 2.2.5.

The syntax we propose allows writing hereditary Harrop formulas. In this abstract syntax, logical connectives such as negation and disjunction can be treated as first-class objects and implemented directly as *Elpi* rules.

Definition 2.2.1 (Atom polarity). Polarity indicates whether a subformula appears in a *positive* or *negative* position within a formula.

The intuition behind polarity is simple: a formula or atom at the top level is in positive position. When crossing an implication ($\Rightarrow$), the left-hand side flips its polarity, while the right-hand side keeps the polarity of the original formula.

In the example on the right we explicitly annotate the polarity of each subformula.

$$\underbrace{\underbrace{\underbrace{x}_{pos} \Rightarrow \underbrace{y}_{neg}}_{neg} \Rightarrow \underbrace{z}_{pos}}_{pos}$$

This notion of polarity will be used again in section 3.4 when discussing the encoding of instances.

Syntax of *Elpi* types In *Elpi*, users can define the algebraic data types required by their development as well as the signatures of predicates. A new algebraic data type is declared with the `kind` keyword, while constructors of that type are introduced using the `type` keyword. A type is either a definition is either a type declaration, a variable, or the application of a type to another in the case of parametric types. The arrow symbol is used to type function space and in *Elpi* it take an optional mode information. The mode, `input` or `output`, tells the runtime which unification algorithm to use when unifying the arguments of two terms, we dealy this discussion in the next paragraph. By default, no mode information is compiled to `output`.

For example, the following declarations define Church naturals and polymorphic lists.

```
kind nat type.                kind list type → type.
type z nat.                   type nil list A.
type s nat → nat.             type cons A → list A → list A.
```

The type `nat` contains the constant `z` representing zero and the constructor `s` representing the successor of another natural number. A `list` is either empty `nil` or the constructor `cons` that combines a head element with a tail list. Lists are polymorphic: the type of the head element must coincide with the type of the elements stored in the tail.

A predicate is a symbol whose type ends in `prop`. The type `o` is an equivalent notation following the convention of [41]. *Elpi* provides a dedicated syntax for declaring predicates, which also allows the programmer to specify the modes of their arguments.

The declaration `pred p t1 -> t2`, where `t1` and `t2` are types, is equivalent to `type p t1 -> t2 -> prop`. However, it also defines the modes of the two arguments.

Arguments placed before the arrow symbol in a `pred` declaration are considered inputs, while those after the arrow are outputs. These modes are used at runtime to decide which unification strategy must be applied when matching the head of a rule with the current goal.

$$\begin{array}{c}
\frac{}{\text{runE } ((\sigma, \emptyset) :: a) \rightsquigarrow (\sigma, a)} \text{runE}_\top \\
\\
\frac{}{\text{runE } \emptyset \rightsquigarrow \square} \text{runE}_\perp \\
\\
\frac{\text{runE } ((\sigma, g) :: a) \rightsquigarrow r}{\text{runE } ((\sigma, (_, \text{Cut}, a) :: g) :: _) \rightsquigarrow r} \text{runE}_! \\
\\
\frac{\mathcal{B}_E(\mathcal{P}, (\sigma \ t), \sigma, a, g) = a' \quad \text{runE } (a' \ ++ \ a) \rightsquigarrow r}{\text{runE } ((\sigma, (\mathcal{P}, t, _) :: g) :: a) \rightsquigarrow r} \text{runE}_\mathcal{B} \\
\\
\frac{y \# \mathcal{P} \quad \text{runE } ((\sigma, (y +_{\vee} \mathcal{P}, t[x/y], a) :: g) :: a') \rightsquigarrow r}{\text{runE } ((\sigma, (_ \ \mathcal{P} _, \text{pi } x. t, a) :: g) :: a') \rightsquigarrow r} \text{runE}_\vee \\
\\
\frac{\text{runE } ((\sigma, (h +_{\Rightarrow} \mathcal{P}, _ \ t _, a) :: g) :: a') \rightsquigarrow r}{\text{runE } ((\sigma, (_ \ \mathcal{P} _, h \Rightarrow t, a) :: g) :: a') \rightsquigarrow r} \text{runE}_{\Rightarrow}
\end{array}$$

Figure 2.3: Operational semantics of *Elpi*

351 **Program type checking** *Elpi* is an (almost) strongly typed language. When a program
352 is interpreted, the type checker verifies that it is well typed. Any type inconsistency is
353 reported as an error, which typically helps identify and correct implementation bugs early.

354 2.2.2 Operational semantics of *Elpi* (without *CHR*)

355 A substitution σ of type Σ is a partial mapping from unification variables $\mathbb{U}$ to terms
356 $\mathbb{Tm}$. The empty substitution is denoted by ε . The notation $\sigma = \{X \mapsto t\}$ represents a
357 substitution σ where the variable X is assigned to the term t . The domain $\text{dom } \sigma$ of a
358 substitution σ is the set of variables that are assigned in σ .

Definition 2.2.2 (Substitution extension). A substitution σ' is an extension of σ , noted $\sigma \subseteq \sigma'$, iff

$$\exists \sigma'', \sigma + \sigma'' = \sigma' \wedge \text{dom } \sigma'' \cap \text{dom } \sigma = \emptyset$$

360 **Definition 2.2.3** (Substitution application (or term dereferencing)). A substitution σ
361 applied to a term t , noted σt , is the operation allowing to recursively replace the variables
362 in t by their assignment in σ .

363 A typing environment $\mathbb{E}$ maps predicates $\mathbb{P}$ to their signatures.

364 A program $\mathcal{P}$ of type $\mathbb{P}$ is represented as a record with the following structure:

365 $\mathbb{P} := \{\text{index: list } \mathbb{R}; \text{nvar: set } \mathbb{N}; \text{sig: } \mathbb{E} \}$

366 The field **index** is an ordered list of rules representing the rule database. The field
367 **nvar** contains the set of nominal variables that have already been introduced, and **sig** is
368 the signature environment.

We write $x\#\mathcal{P}$ to denote that the nominal variable x is fresh with respect to $\mathcal{P}.\text{nvar}$. The notation $x +_{\forall} \mathcal{P}$ denotes the program obtained by adding x to $\mathcal{P}.\text{nvar}$. Similarly, $h +_{\Rightarrow} \mathcal{P}$ denotes the program obtained by prepending the rule h to $\mathcal{P}.\text{index}$.

A goal g of type $\mathcal{G} = \mathbb{P} \times \mathbb{A} \times \vec{\mathcal{A}}$ is a triple consisting of a program, an atom, and a list of *cut-to alternatives*. An alternative a of type $\mathcal{A}$ is a pair $\Sigma \times \vec{\mathcal{G}}$ composed of a substitution and a list of goals.

Figure 2.3 presents the derivation rules of the *Elpi* interpreter, defined in terms of the function

$$\text{runE} : \vec{\mathcal{A}} \rightarrow (\Sigma, \vec{\mathcal{A}}) + \square.$$

This semantics is inspired by [11, Sec. 4.3] and [55]. Execution is modeled as the evaluation of a stack of alternatives $\mathcal{A}$. The interpreter processes alternatives sequentially. The evaluation of the alternative list either succeeds, producing a substitution and a new list of alternatives, or fails, returning $\square$.

In figure 2.3, rule $\text{runE}_{\top}$ captures success: if the first alternative contains an empty list of goals, the interpreter returns its associated substitution together with the remaining alternatives. Conversely, rule $\text{runE}_{\perp}$ states that evaluation fails when the list of alternatives is empty.

The remaining rules of runE , namely runE_i , runE_B , $\text{runE}_{\forall}$, and $\text{runE}_{\Rightarrow}$, apply when the first alternative contains a non-empty list of goals. Such alternative list has the form $((q :: g) :: a)$, where q is the current goal atom (or query). The derivation rule to use depends on the syntactic form of q , which can be one of the four forms of $\mathbb{A}$ listed in figure 2.2.

If q is a *Cut*, rule runE_i discards the current alternatives a and replaces them with the cut-to alternatives stored in q . Evaluation then continues with the remaining goals g .

If q is a *pi*-quantification $\text{pi}x.t$, the interpreter generates a fresh nominal variable y not occurring in $\mathcal{P}$, replaces all free occurrences of x in t with y , and continues evaluation with the resulting atom.

If q corresponds to the dynamic insertion of a rule h , evaluation continues with atom t in the program extended with the local rule h . Intuitively $h \Rightarrow t$ amounts to a well-bracketed use of *Prolog*'s `assert/1` and `retract/1`.

Finally, if q is a predicate call, the interpreter performs *backchaining* over the rules stored in $\mathcal{P}.\text{index}$. Backchaining retrieves the rules that can be used on the query q . In standard *Prolog* this is done by selecting rules whose head unifies with q . In *Elpi* the process depends on predicate modes, which determine if a argument should be “matched” or “unified”.

Elpi provides two procedures, `unify` and `matching`. Both have type

$$\mathbb{Tm} \rightarrow \mathbb{Tm} \rightarrow \Sigma \rightarrow \Sigma + \square,$$

taking two terms and an initial substitution σ . They either fail (returning $\square$) or succeed with an updated substitution σ' that is an extension of σ .

Definition 2.2.4 (Pattern fragment). A λ -term belongs to Miller’s pattern fragment if in every subterm of the form $F t_1 t_2 \dots t_n$, where F is a higher-order unification variable, each term t_i is a bound variable distinct from any t_j (with $i \neq j$).

The procedure `unify` implements higher-order unification restricted to the pattern fragment [40] and is used for arguments in `output` position. The procedure `matching`, inspired by *B-Prolog*'s “matching clauses” [62], is used for arguments in `input` position.

Definition 2.2.5 (*unify*). Two terms q and u *unify* under a substitution σ if there exists a substitution σ' extending σ such that $\sigma'q = \sigma'u$, and σ' is a most general unifier for the two terms.

Definition 2.2.6 (*matching*). A query q *matches* a pattern u under a substitution σ if there exists a substitution σ' extending σ such that $\sigma'u = \sigma q$, where σ' assigns at most the free variables occurring in u .

Backchaining is defined by the function $\mathcal{B}$:

Definition 2.2.7 (*Backchain*, $\mathcal{B} : \mathbb{P} \times \text{set } \mathbb{U} \times \mathbb{Tm} \times \Sigma \rightarrow \overrightarrow{\Sigma \times \mathbb{A}}$).

$$\mathcal{B}(\mathcal{P}, V, q, \sigma) = \left[(\sigma', b) \mid \mathcal{U}(\mathcal{S}(\mathcal{P}.\text{sig}, q), q, h, \sigma) = \sigma' \neq \square \mid (h : -b) \in \text{fresh}(V, \mathcal{P}.\text{index}) \right].$$

Intuitively, $\mathcal{B}$ selects the rules that can be applied to the current query q under a substitution σ . The variables occurring in the program rules are assumed to be disjoint from those in the current execution state. To guarantee this, the program is refreshed using the function `fresh`, which renames its variables with respect to the set V provided as input to $\mathcal{B}$.

For each applicable rule, $\mathcal{B}$ returns the substitution obtained by matching/unifying the rule head with q , together with the corresponding rule body.

The definition relies on two auxiliary functions:

$$\mathcal{S} : \mathbb{E} \rightarrow \mathbb{Tm} \rightarrow \mathbb{T}\mathbb{y} + \square \tag{2.1}$$

$$\mathcal{U} : (\mathbb{T}\mathbb{y} + \square) \rightarrow \mathbb{Tm} \rightarrow \mathbb{Tm} \rightarrow \Sigma \rightarrow \Sigma + \square. \tag{2.2}$$

The function $\mathcal{S}$ retrieves the signature of the predicate occurring in a term. It returns the signature if the predicate is present in the environment, and $\square$ otherwise.

The function $\mathcal{U}$ tests whether the query q matches/unifies the head h of a rule under an initial substitution σ . The predicate signature determines the mode of each argument, which in turn selects the appropriate procedure: `matching` for `input` arguments and `unify` for `output` arguments. If all arguments satisfy the unification condition, $\mathcal{U}$ returns the resulting substitution σ' ; otherwise it returns $\square$.

Definition 2.2.8 (*Elpi backchain*, $\mathcal{B}_E : \mathbb{P} \times \mathbb{Tm} \times \Sigma \times \overrightarrow{\mathcal{A}} \times \overrightarrow{\mathcal{G}} \rightarrow \overrightarrow{\mathcal{A}}$).

$$\mathcal{B}_E(\mathcal{P}, t, \sigma, a, g) = \left[\sigma', ([(\mathcal{P}, q, a) \mid q \in b] ++ g) \mid (\sigma', b) \in \mathcal{B}(\mathcal{P}, \text{get_fv}(t, g :: a, \sigma), t, \sigma) \right].$$

The function $\mathcal{B}_E$, used by rule `runEB` in figure 2.3, builds a new list of alternatives from the result of $\mathcal{B}$ (`get_fv` is the function retrieving the list of free variables in the term t , the list of alternatives $g :: a$ and the substitution σ). For each filtered rule, the atoms in the rule body become new goals whose cut-to alternatives are a and whose continuation is g . The substitution associated with each alternative is the one returned by $\mathcal{B}$.

runE rules	run rules	run rules	runE rules
$\text{runE}_\top$	$\text{run}_\top$	$\text{run}_\top$	$\text{runE}_\top$
$\text{runE}_\perp$	unreachable case	$\text{run}_\perp$	$\text{runE}_\mathcal{B}$ and $\text{runE}_\perp$
$\text{runE}_\mathcal{B}$	$\text{run}_{\circlearrowleft, \perp, \sigma, @} + \text{IH}$	$\text{run}_{\circlearrowleft}$	$\text{runE}_\mathcal{B}$ and IH
$\text{runE}_!$	$\text{run}_!$ and IH	$\text{run}_@$	$\text{runE}_\mathcal{B}$ and IH
$\text{runE}_\forall$	$\text{run}_\forall + \text{IH}$	run_σ	$\text{runE}_\mathcal{B}$ and IH
$\text{runE}_\Rightarrow$	$\text{run}_\Rightarrow + \text{IH}$	$\text{run}_!$	$\text{runE}_!$ and IH
		$\text{run}_\forall$	$\text{runE}_\forall$ and IH
		$\text{run}_\Rightarrow$	$\text{runE}_\Rightarrow$ and IH

(a) $\text{runE} \rightarrow \text{run}$ (b) $\text{run} \rightarrow \text{runE}$

Table 2.1: High-level proof schema of theorem 2.2.1

Comparison with Tassi’s *hdr* In the previous subsections, we presented the *Elpi* language, including its syntax and semantics. This formalization has been developed in collaboration with Enrico Tassi. A comprehensive description of *Elpi* can be found in his work *hdr* [55].

We highlight here the main differences between his presentation and the one adopted in this section. First, in the syntax, we use binary application for terms, instead of the n -ary application used in *hdr*. Second, in the semantics, programs carry not only the list of rules and nominal variables, but also the signature of declared predicates. We also omit the formal treatment of constraints, as it is not required for the remainder of this manuscript; a brief informal explanation is sufficient for our purposes (see section 2.2.5).

Another difference concerns the definition of **runE**. In Tassi’s presentation, **runE**, called **run**, is defined as a procedure that takes two arguments: the current alternative and the list of future alternatives. This choice reflects closely the concrete implementation in *Elpi*, where execution starts from a query, and a query is represented as an alternative. In contrast, we define **runE** with a single input, namely a list of alternatives, from which the first element is extracted by pattern matching when needed. This design choice is motivated by the mechanization in *Rocq* (see chapter 6), where we need to prove the equivalence between two semantics and to represent the execution of **runE** on an empty list of goals. Despite these differences, as the theorem below claims, the two semantics are equivalent when executed on non-empty lists of alternatives, modulo the absence of constraints, which is forced to the empty list in the **run** semantic (this is performed by the mapping of one list to the other and putting the empty set, i.e. $\emptyset$, of constraints).

Theorem 2.2.1.

$$\begin{aligned}
&\forall a_0 a_s r \sigma, \text{ let } (a'_0, a'_s) := \text{map } (\lambda x. (\emptyset, x)) (a_0 :: a_s) \text{ in} \\
&\quad \text{let } r' := \text{map } (\lambda x. (\emptyset, x)) r \text{ in} \\
&\quad \text{runE } (a_0 :: a_s) \rightsquigarrow (\sigma, r') \leftrightarrow \text{run } a'_0 a'_s \rightsquigarrow (\emptyset, \sigma, r')
\end{aligned}$$

Proof. The proof is relatively simple. It is based on two inductions, one for each direction of the $\leftrightarrow$ operator, respectively on **runE** and **run**. The intuition is illustrated in table 2.1, where IH denotes the induction hypothesis.

```

pred f int → int.
f 1 2.
f 2 3.
pred r int → int.
r 0 0.
r 0 1.
r 0 2 :- !.
pred g int → int.
g A C :- r A B, f B C, !.
g D F :- f D E, f E F.

```

Figure 2.4: Running example of a *Elpi* program

465 The first subtable corresponds to the left-hand side of the implication, while the second
 466 one corresponds to the right-hand side. In both cases, the goal is to relate the constructors
 467 of the two semantics.

468 More precisely, we match each constructor of one semantics with the corresponding
 469 constructors of the other, and, since our semantics has fewer constructors, depending on
 470 the considered case, several constructors of **run** may be used on a single constructor of
 471 **runE**. We also point out, that the **runE**_⊥ case in 2.1a, is unreachable since we start from
 472 an non-empty list of goals. $\square$

473 2.2.3 Example of program execution in *Elpi* with Cut

474 We believe that the main difficulty in the semantics presented in the previous section lies
 475 in the treatment of choice points and their interaction with the **Cut** operator.

476 Several variants of the cut operator exist in the literature. The variant we consider,
 477 used in *Elpi*, is the so-called *hard cut* (see, for example, the documentation of *SWI-Prolog*
 478 [52]). When the **backchain** function returns a non-empty list of alternatives, the
 479 first alternative is executed immediately, while the remaining ones are stored as choice
 480 points.

481 Within a selected alternative, premises are evaluated from left to right, each atom
 482 being executed in sequence. When a **Cut** is encountered, it removes two kinds of choice
 483 points: (i) those generated by **backchain** for the current predicate call, and (ii) those
 484 created during the evaluation of premises to the left of the **Cut**.

485 As a running example, we consider the program shown in figure 2.4. For clarity,
 486 figure 2.5 illustrates its execution, explicitly showing the pointers to the cut-to alternatives.
 487 We only highlight the alternatives targeted by the **Cut** operator, since the alternatives
 488 stored for predicate calls do not play a role in the semantics discussed here.

489 Consider the program in figure 2.4 and the query **g 0 R**. Both rules for **g** apply to this
 490 query. Backchaining therefore returns the following two alternatives:

$$[(\sigma_1, [\mathbf{r} \ A \ B, \mathbf{f} \ B \ C, !]), (\sigma_2, [\mathbf{f} \ D \ E, \mathbf{f} \ E \ F])]$$

491 with $\sigma_1 := \{A \mapsto 0, C \mapsto R\}$ and $\sigma_2 := \{D \mapsto 0, F \mapsto R\}$. For simplicity, we choose an example
 492 where variable refreshing plays no role, so we do not discuss the set $\mathcal{F}_v$ any further.

493 Execution continues with the first premise of the first alternative, namely **r A B** under
 494 σ_1 , i.e. **r 0 B**. The remaining alternatives are stored as choice points. Backchaining

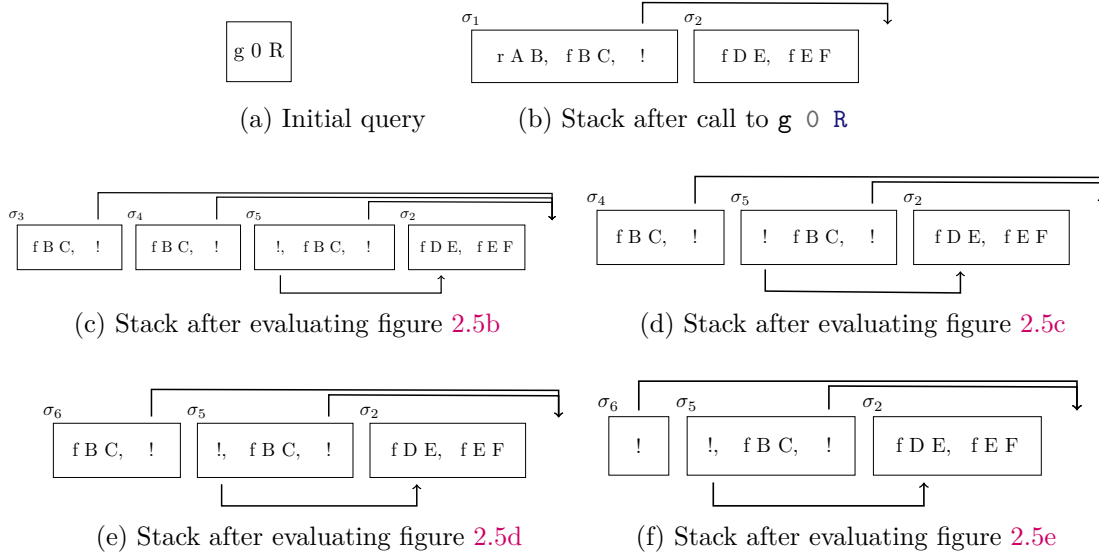


Figure 2.5: Example of execution

495 `r 0 B` returns $[(\sigma_3, []), (\sigma_4, []), (\sigma_5, [!])]$ where $\sigma_3 := \sigma_1 + \{B \mapsto 0\}$; $\sigma_4 := \sigma_1 + \{B \mapsto 1\}$ and
 496 $\sigma_5 := \sigma_1 + \{B \mapsto 2\}$.

497 The next step of interpretation continues with `f B C` under σ_3 , that is, `f 0 R`, and
 498 leaves $[(\sigma_4, []), (\sigma_5, [!])]$ as new choice points. This time, backchaining fails (no rule for
 499 `f` matches input 0). We therefore backtrack to the most recently introduced choice point
 500 and retry `f B C` under σ_4 , i.e. `f 1 R`. This succeeds with $\sigma_6 := \sigma_4 + \{R \mapsto 2\}$.

501 We now reach the cut. At this point there are still two unexplored alternatives: one
 502 from the third rule for `r` and one from the second rule for `g` (respectively, σ_5 and σ_2). Both
 503 are discarded, because they were created when, or after, the first rule for `g` was selected.
 504 In contrast, a cut does not discard choice points created earlier; in this example, there are
 505 none. The final solution is: $\{A \mapsto 0, B \mapsto 1, C \mapsto R, R \mapsto 2\}$.

506 2.2.4 Higher-order, dynamic and meta- programs in *Elpi*

507 One of the main features of *Elpi* is the possibility to work with higher-order and dynamic
 508 programs. In this section, we present some simple examples illustrating these two aspects.
 509 These examples will be used later as a reference when discussing determinacy checking in
 510 *Elpi* (see chapter 5).

511 **Higher-order programming** A well-known higher-order predicate in *Elpi* is `map`, whose
 512 implementation is shown below.

```
pred map (pred A -> B), list A -> list B.
map _ [] [].
map F [X|XS] [Y|YS] :- F X Y, map F XS YS.
```

513 This is a standard Prolog-style definition. The predicate takes a higher-order argument
 514 `F`, which is applied to each element of the input list. In the base case, the empty list is

mapped to the empty list. In the recursive case, the head X is transformed into Y using F , and the predicate is then applied recursively to the tail XS .

Another common higher-order construct in *Elpi* is the predicate `once`, defined as follows.

```
pred once (pred) ->.
once P :- P, !.
```

The predicate `once` acts as a wrapper around a goal. It executes the goal P and then removes all remaining choice points. As a consequence, only the first solution is returned. For example, the query `once (r 0 X)`, where r is defined in figure 2.4, produces a single result, binding X to 0.

The behavior of the `Cut` operator depends on where it is placed. The following snippets illustrate similar programs with different uses of `Cut`.

```
pred p1 int -> int.      pred p2 int -> int.      pred p3 int -> int.
p1 1 X :- r 0 X.         p2 1 X :- r 0 X, !.         p3 1 X :- once (r 0 X).
p1 1 3.                  p2 1 3.                  p3 1 3.
```

The query `p1 1 X` returns four solutions for X , namely 0, 1, 2, and 3. This is because no `Cut` is used, so all possible solutions are explored by backtracking.

The query `p2 1 X` returns only one solution, $X = 0$. In this case, after the first solution of `r 0 X` is found, the `Cut` removes all choice points, both for r and for $p2$.

Finally, the query `p3 1 X` returns two solutions, 0 and 3. Here, the use of `once` removes the choice points generated by r , but it does not affect the second rule for $p3$.

Dynamic programming The domain where *Elpi* really shines is the manipulation of higher-order data represented via the so-called λ -tree syntax [42] also called HOAS [47]. Such data is pervasive in interactive provers based on Type Theory such as *Rocq*. In the paradigmatic example below `tm` is the data type for λ -terms.

```
kind tm type.           % datatype of  $\lambda$ -terms in HOAS form
type app tm -> tm -> tm. % - binary application
type lam (tm -> tm) -> tm. % -  $\lambda$ -abstraction
```

In the HOAS encoding of syntax with binders we do not find a node for variables: we re-use the notion of bound variable of the programming language to encode the one of the syntax tree. At the level of types we see that the `lam` node carries an *Elpi* function over terms. It is by applying this function that one can recover the body of the λ -abstraction. A simple program manipulating higher-order data is the function performing a deep copy of a term:

```
pred copy tm -> tm.
copy (app A B) (app C D) :- copy A C, copy B D.
copy (lam F) (lam G)      :- pi x\ copy x x => copy (F x) (G x).
```

The copy of an application is built from the copies of the subterms. What is more interesting is how one copies under a binder. The “`pi x\ copy x x =>`” instruction explains how to copy the new constant x . Finally the code performs a recursive call over $F x$: by applying the *Elpi* function F to the fresh symbol x one obtains the body of the λ -abstraction where the bound variable is replaced by the new symbol x .

In *Elpi*, all predicates are *dynamic*, or open, hence the code of `copy` can be augmented with rules at runtime.

The `copy` predicate, as given, implements the identity and may seem rather useless, but it is not! Since the program is open, one can add new rules for `copy` and change its behavior. For example the code below computes the weak head normal form of a λ -term by contracting the application of a λ -abstraction to an argument and uses `copy` to perform the substitution:

```
pred whd tm -> tm.
whd (app H A) R :- whd H (lam F), !,           % we fire the redex
    pi x\ copy x A => copy (F x) S, whd S R.    % and we continue
whd X X.                                         % or we stop
```

The argument `A` is put in place of the (previously) bound variable in `F` by replacing that variable with `x` and by running `copy` with the addition of a special rule copying `x` to `A`. Our static analysis understands this programming pattern and is able to recognize that `whd` is also a function. In particular the first rule, if taken, discard the second rule (due to the cut) and only calls functions.

Meta-programming In *Elpi*, code is a first-class object, meaning that it can be inspected and generated within the language itself. As an example, consider a program that performs the fusion of two calls to `map` [58].

```
pred comp (pred A -> B), (pred B -> C), A -> C.
comp F G X Z :- F X Y, G Y Z.

pred fuse (pred list A -> list B), (pred list B -> list C) ->
    (pred list A -> list C).
fuse (map F) (map G) (map (comp F G)).
```

The predicate `fuse` analyzes two programs. If both correspond to mapping a list, with functions `F` and `G`, it produces a new program of the form `map (comp F G)`, where `comp` denotes function composition.

This approach can improve efficiency. For instance, instead of executing the sequence `map F L1 L2, map G L2 L3`, which traverses the list twice, one can combine the two computations. By calling `fuse (map F) (map G) H`, a new program `H` is obtained. Then, `H L1 L3` computes the same result with a single traversal of the list.

More generally, meta-programming is widely used in *Rocq-Elpi*. It allows the generation of new rules that can be stored on disk and reused in later executions of the program.

2.2.5 CHR: goal suspension

Constraint Handling Rules (*CHR*, [19, 18]) is a declarative, rule-based language designed to manipulate a multiset of constraints. A *CHR* program consists of rules that match patterns of constraints, optionally guarded by conditions, and transform the current constraint store by adding or removing elements. Operationally, execution proceeds by repeatedly selecting applicable rules and applying them to the constraint store. This model offers a flexible way to express simplification, propagation, and their combination, known as *simpagation*.

From a conceptual point of view, *CHR* can be seen as a rewriting system over constraints, where rules describe how local configurations evolve. In contrast to standard logic programming rules, which operate on a single goal, *CHR* rules may inspect and rewrite several constraints at once. This aspect is important when correctness depends on the interaction between constraints, rather than on their independent resolution. For this reason, *CHR* naturally supports a more global view of computation.

In *Elpi*, *CHR* is integrated as an extension of the underlying λ -*Prolog* engine. The motivation for this integration comes from the need to combine and refine suspended constraints generated during proof search.

We do not enter into the details of the syntax or the implementation of *CHR*, since they play only a minor role in this manuscript. A more complete presentation can be found in [25, Sec. 4.3] and [55, Sec. 2.4].

2.2.6 Rule names and insertion

In the *Elpi* database, the rules used during backchaining are stored in an ordered data structure, together with their implementation. Each rule can optionally be given a name by the user.

These names are useful when adding new rules that should have higher or lower priority than existing ones. In this way, the user can control the position of rules in the search order.

In *Elpi*, inserting a rule before or after an existing one is called *grafting*.

```
:name "nth-fail"
nth _ _ :- error "The list is too short".
```

The definition of `nth` above is a catch-all rule, since it applies to any query. This rule is named `nth-fail`. When defining additional rules for `nth`, one can write:

```
:before "nth-fail"
nth 0 [X|_] :- X.
```

This declaration places the new rule just before the catch-all rule. A similar syntax can be used with `:after` to insert a rule after an existing one.

The usefulness of grafting becomes clearer in section 3.4.1, where typeclass instances must be inserted at specific positions in the database to respect the intended priority.

2.3 *Rocq-Elpi*: A Logical Framework for *Rocq*

The *Rocq-Elpi* language is a plugin for *Rocq* that enables the use of *Elpi* within the *Rocq* environment. In the following sections, we first explain the idea behind encoding *Rocq* terms in *Elpi* (see section 2.3.1) using higher-order abstract syntax (HOAS) [47]. In section 2.3.2, we provide some important API that are widely used in *Rocq-Elpi*. Finally, in section 2.3.4, we provide a brief description of how commands and tactics are written in *Rocq-Elpi* and how they integrate with *Rocq*.

```

kind term type.

type sort    sort → term.           % Prop, Type@{i}
type global  gref → term.

type app     list term → term.       % app [hd/args]
type fun     name → term → (term → term) → term.
type prod    name → term → (term → term) → term.

...                               % Other constructors omitted

```

Figure 2.6: HOAS of *Rocq* terms in *Elpi*

<i>Rocq</i> terms	<i>Elpi</i> HOAS	description
<code>plus 0</code>	<code>app [<<plus>>, <<0>>]</code>	term application
<code>fun x => ?F a</code>	<code>fun `x` y\ app [G, <<a>>]</code>	λ -abstraction
<code>$\forall$ x, p x</code>	<code>prod `x` z\ app [<<p>>, z]</code>	$\forall$ -quantif and binding

Table 2.2: Examples of *Rocq* terms encoded in *Elpi* using HOAS

2.3.1 HOAS: λ -terms representation

In section 2.2.4, we sketched how to encode the syntax of a language using the λ -tree syntax of *Elpi*, also known as higher-order abstract syntax (HOAS) [47]. This syntax is inspired by the λ -calculus, which forms the core of the *Rocq* system.

The encoding of *Rocq* terms is performed using a special *Elpi* type called `term`. The snippet depicted in figure 2.6 shows the main constructors for terms.

The base constructors are `sort`, which encodes the **Type** and **Prop** terms of *Rocq*, and `global`. The `global` constructor represents user-defined symbols such as types, inductives, and constants.

These symbols are identified by a `gref`, which is an opaque object containing the unique identifier of the corresponding *Rocq* symbol. For better space management we use a more compact notation: opaque *Rocq* symbols are written using guillemets, avoiding the explicit use of the `global` constructor. For instance, the *Rocq* constant `nat` is represented in HOAS form as `<<nat>>`.

Term application in *Rocq* is represented by the `app` constructor, which takes a list of terms. λ -abstractions and $\forall$ -quantifications are encoded using the `fun` and `prod` constructors, respectively. Each of these constructors carries the `name` of the *Rocq* binder (used only for pretty printing), the type of the binder, and an *Elpi* higher-order object of type `term → term`, which represents the abstraction of the bound variable in the term.

The *Rocq-Elpi* backend translates *Rocq* terms into HOAS *Elpi* terms and conversely when the two systems interact. In table 2.2, we show some examples of this encoding: the first column contains *Rocq* terms, and the second column their HOAS representation in *Elpi*.

The second and third rows illustrate how binders are handled for λ -abstraction and $\forall$ -quantification. The original binder name from *Rocq* is stored in the `fun` and `prod` constructors, while the name of the bound variable in *Elpi* is not relevant. In the third row,

the *Elpi* binder (called *z*) is reused to represent the abstraction of the object language when applied to the predicate *p*.

The second row shows a *Rocq* unification variable *?F*. It is translated into an *Elpi* unification variable *G*, which internally corresponds to *?F*. When *G* is instantiated during the execution of the *Elpi* program, the result is translated back into a *Rocq* term and assigned to *?F*.

Quotations and Antiquotations Writing *Rocq* terms can be tedious, as it requires explicitly constructing their full HOAS representation. As shown in table 2.2, the *Elpi* HOAS syntax is significantly more verbose than standard *Rocq* syntax. Moreover, *Rocq* often includes implicit subterms, whereas *Elpi* does not provide such a mechanism.

To simplify development, *Rocq-Elpi* offers a system of quotations and antiquotations. This mechanism allows embedding *Rocq* syntax within *Elpi* code, and conversely *Elpi* syntax within *Rocq* code, in a recursive manner.

An illustrative example, adapted from [55], is the following:

$$\underbrace{\text{print}}_{Elpi} \left\{ \underbrace{\{1 + \}}_{Rocq} \text{lp} : \underbrace{\{\{\text{app}[\langle\langle S \rangle\rangle, \underbrace{\{\{0\}\}}_{Rocq} \underbrace{\{\}}_{Elpi}\}}_{Elpi} \underbrace{\{\}}_{Rocq} \right\}$$

Within an *Elpi* code block, the syntax $\{\{ \dots \}\}$ denotes a quotation that allows writing *Rocq* code. Conversely, within *Rocq* code, the syntax $\text{lp} : \{\{ \dots \}\}$ denotes a quotation for embedding *Elpi* code. These constructs can be nested arbitrarily.

2.3.2 API in *Rocq-Elpi*

Thanks to the HOAS representation of *Rocq* terms in *Elpi*, the two languages become closer and can share a common representation. This makes their interaction easier and allows interleaving computations between them. The *Rocq* language exposes several APIs that can be used directly from *Elpi*.

```
pred coq.unify-eq term, term → diagnostic.
```

The predicate above is widely used in *Elpi* developments. It takes two terms and attempts to unify them. If unification fails, it returns a diagnostic explaining the error. Internally, this predicate relies on the *Rocq* unification algorithm. When *Elpi* `coq.unify-eq` is called, the HOAS representation of the terms is translated into *Rocq* terms before performing the unification.

Other predicates allow extending the *Rocq* system with new definitions, modules, modes, and more.

```
pred coq.TC.declare-class gref →.
```

For example, the predicate above allows declaring a type class from a global reference in *Rocq* directly within *Rocq-Elpi*.

2.3.3 Declaring Databases in *Rocq-Elpi*

In [55, Sec. 4.4.3], a database is defined as a non-executable *Elpi* program. It contains a collection of code that can be included into commands or tactics by referring to its name.

When a program is executed, it does not use the state of the database at the moment when the command was defined. Rather, it accesses the current content of the database at execution time. This means that any update to the database is visible to all programs that include it.

An *Elpi* database can be dynamically modified by adding new predicate signatures and rules.

```
kind clause type.
%           Name      Grafting      Rule
type clause string → grafting → prop → clause.

%
%           DB name Indexing PName
pred coq.elpi.add-predicate string, string, string,
                             list (pair argument_mode string) →.

%
%           Scope      DB name      Clause
pred coq.elpi.accumulate scope, string, clause →.
```

The `coq.elpi.add-predicate` predicate takes as arguments the name of the database where the predicate is inserted, the indexing strategy[†], the predicate name, and its specification (argument modes and types).

The `coq.elpi.accumulate` predicate adds a new clause to a database within a given scope. A clause is a specific type with a single constructor, which contains the clause name, its grafting with respect to the existing database, and the rule itself.

These last two predicates are particularly useful in the context of sections 3.3 and 3.4.

2.3.4 Writing Commands and Tactics in *Rocq-Elpi*

We now describe how to implement commands and tactics in *Rocq-Elpi*. In table 2.3, we summarize how to define commands and tactics, how to invoke them, and what their entry points are. Both commands and tactics are typically built on top of an *Elpi* database, which they can read and, if needed, modify. A database is added using the command `Elpi Accumulate Db db_name`. The code of commands and tactics can be extended as follows:

```
Elpi Accumulate [command_name|tactic_name] lp:{{
```

[†]We do not discuss indexing in detail, as it is not central for this manuscript. It determines the data structure used to store rules in order to improve search efficiency.

Commands	Tactic	Description
<code>Elpi Command command_name.</code>	<code>Elpi Tactic tactic_name.</code>	Declaration
<code>Elpi command_name.</code>	<code>elpi tactic_name.</code>	Invocation
<code>main</code>	<code>msolve</code> and <code>solve</code>	Entry point

Table 2.3: Command and Tactic main commands

```

kind goal type.
kind sealed-goal type.
type nabla (term → sealed-goal) → sealed-goal.
type seal goal → sealed-goal.

typeabbrev goal-ctx (list prop).
%      Context    RS      Ty      Sol      Arguments
type goal goal-ctx → term → term → term →
      list argument → goal.

```

Figure 2.7: Representation of goals and sealed goals in *Rocq-Elpi*

```

% New rules added here
}}.

```

As mentioned at the beginning of this chapter, commands are used to extend the *Rocq* language with new definitions and theorems. In other words, a command does not return a value: it is mainly used for printing or for modifying the internal state of the system. The entry point of a command is a predicate named `main`, with signature `pred main list argument →`. An argument can be a string, an integer, a term, or an open term.

Tactics in *Elpi* are used to solve *Rocq* goals through the *Elpi* interpreter. Their declaration is similar to that of commands. The entry point of a tactic is the predicate `msolve`, with the following type:

```
pred msolve list sealed-goal → list sealed-goal.
```

The `msolve` predicate takes as input a list of *Rocq* sealed goals and returns an updated list of sealed goals. The type `sealed-goal`, shown in figure 2.7, represents goals that may contain abstractions. These abstractions are encoded using the `nabla` constructor.

Under a spine of `nabla` abstractions, the `seal` constructor contains a goal. A goal is a tuple with five components: a context C , a raw solution S_r , a type T_y , a final solution S , and a list of arguments L .

The context C maps free variables to their types. The term S_r is a trigger: when it is instantiated, it is elaborated against the type T_y . The type T_y is the type of the *Rocq* goal to be solved. The term S is the final solution, which must have type T_y . The list L contains arguments that control the behavior of the resolution process. In this work, we mainly use two projections of goals, namely the context and the goal type.

If the predicate `msolve L N` is not defined, or if it fails when applied to the list of goals L , then the system attempts to solve each goal in L individually using the predicate `solve`. In this case, `solve` is called on the content of the `seal` constructor, meaning that all `nablas` have already been traversed.

```
pred solve goal → list sealed-goal.
```

The final point addressed in this section concerns the `refine` predicate.

Given a goal G and a proof term P , one may wish to typecheck P against the G and obtain the corresponding list of subgoals that must be solved recursively. The *Rocq* tactics performing this task is called `refine`.

In *Elpi*, the `refine` predicate exists and has the same behavior: it takes a proof term and a goal, it typechecks the proof against its type and returns a list of subgoals on the for of a list of sealed-goals. It is customary for *Elpi* tactics to invoke `refine` in order to produce the output list the `solve` predicate.

For performance reasons, *Elpi* also provides a variant, `refine.no_check`, which skips typechecking and directly produces the list of sealed subgoals from the given term and goal. This results in faster, although less reliable, computation.

CHAPTER 3

Architecture of a Type-Class Solver in *Rocq-Elpi*

Type classes provide an elegant mechanism for implementing ad-hoc polymorphism [58]. As explained in section 2.1.1, they allow the same symbol to be reused in different contexts, and in a proof assistant like *Rocq*, instances of a type class preserve the same mathematical properties. From a Prolog-like perspective, a type-class goal of *Rocq* corresponds to resolving a query to a predicate that associates its inputs with an instance solving the type-class constraint. In *Rocq*, instances act as rules stored in a database, which are used to backchain and resolve type-class queries.

A meta-language integration for type-class resolution must provide two components: (1) a compiler that translates *Rocq* class and instance declarations into the meta-language representation, and (2) a type-class solver that animates the search by interacting with the resulting database.

In our setting, the meta-language used to implement the meta-solver is *Elpi*. Consequently, the main task is to implement the class and instance compiler, which incrementally builds the database on which the *Elpi* program operates.

As explained in section 2.1.1, the type-class solver itself typically follows a *Prolog*-like resolution strategy: instances are tried according to their priority, and backtracking may reconsider previous choices when a branch fails. In *Elpi*, this behavior comes for free, since the resolution mechanism of the type-class solver strongly imitates the standard execution model of the *Elpi* runtime.

The remainder of this work is organized as follows. We first demonstrate how *Rocq* terms can be compiled into an *Elpi* database and how type-class goals can be translated into *Elpi* queries, enabling proof automation through *Elpi*'s search mechanism. In section 3.2, we present a minimal instance compiler, showing how a basic type-class solver can be implemented with few lines of code. While simple, this approach lacks important features required for handling complex instances. We therefore introduce a more powerful compiler in sections 3.3 and 3.4.

Next, in sections 3.4.1 to 3.4.3, we discuss how rule priorities influence database construction, how *Rocq* goals are encoded as *Elpi* queries, and how users can extend the database with custom rules to tailor the solver's behavior.

We then report on our experimental evaluation in section 3.5, where the solver is applied to real-world libraries such as *stdpp* and *tlc*. Benchmarks presented in section 3.5.1 compare the performance of *Rocq* and *Elpi*, analyzing how execution time in *Elpi* is distributed across different phases of the search. In section 3.5.2, we examine the differences between *Elpi* and *Rocq* modes and their impact on resolution. Finally, in section 3.5.3, we discuss the unification challenges that arise when using *Elpi*'s unification algorithm on *Rocq* terms. The discussion concludes in section 3.6.

3.1 Interfacing *Rocq* with *Elpi* for Type-Class Resolution

To address type-class unification problems, we extend *Rocq* with the following APIs (simplified here for clarity):

```
(* API for Compiler *)
val register_observer : string → (event → unit) → unit
val activate_observer : string → unit
val deactivate_observer : string → unit

(* API for Solver *)
val register_solver : string → tc_solver → condition → unit
val activate_solver : string → unit
val deactivate_solver : string → unit
```

This APIs allow one to declare new compilers and new solvers inside the *Rocq* system.

A compiler is implemented as an observer, which is triggered whenever a new class or instance is declared. To define a new observer, one provides a name and an event handler to `register_observer`. The observer is stored in a table mapping names to handlers. The handler is a higher-order function, invoked each time an event occurs. Its role is to process the received information and update its internal database. Observers can be activated or deactivated when needed.

A type-class solver is registered via `register_solver`, which takes a name, a value of type `tc_solver`, and a condition. The `tc_solver` is a higher-order function that receives a *Rocq* context and a list of goals, and returns an updated environment where some type-class goals may be solved. The condition explains if the current solved is meant to solve the current list of type-class goals. This is useful, for example, if the aim of a solver is meant to be used on particular type-class goals.

Solvers are also stored in a table indexed by their name. During resolution, the *Rocq* type-class mechanism scans the registered solvers and selects the first active one that satisfies its condition. If the condition does not hold, the next solver is considered. As for observers, solvers can be activated or deactivated by the user.

3.2 Simple instance compiler

Conceptually, a type-class goal can be translated into a query with two arguments: the goal itself and a concrete instance witnessing it. The search is expressed using the predicate `pred tc term -> term`.

By invariant, a query of the form `tc T I` ensures that the term `I` has type `T`.

```

Class Add T := {plus: T → T → T}.

Instance addNat : Add nat := {plus := Nat.add}.
Instance addR : Add R := {plus := Rplus}.

Instance addProd : ∀ T1 T2, Add T1 → Add T2 → Add (T1 * T2) :=
  {plus '(x1,y1) '(x2, y2) := (plus x1 x2, plus y1 y2)}.

```

Figure 3.1: Add class and its instances

```

% Inst Ty Args Prems Res
pred comp term, term, list term, list prop → prop.
comp I {{lp:T → lp:Bo}} Ag P (pi y\ R y) :- !, % r→
  pi x\ comp I Bo [x|Ag] [tc T x | P] (R x).
comp I {{∀ x, lp:(Bo x)}} Ag P (pi y\ R y) :- !, % r∀
  pi x\ comp I (Bo x) [x|Ag] P (R x).
comp I T Ag P (tc T Proof :- [true | Body]) :- % rB
  mk-app I {rev Ag} Proof,
  rev P Body.

pred compile gref →.
compile G :- coq.env.typeof G Ty,
  comp (global G) Ty [] [] R,
  coq.elpi.accumulate _ "tc.db" (clause _ _ R).

```

Figure 3.2: Simple *Elpi* compiler

As an example, consider the `Add` type-class introduced in section 2.1.1. For convenience, the full definition of the class and its instances is reproduced in figure 3.1.

After an instance is declared, an observer triggers the instance compiler shown in figure 3.2. The compiler is similar to the example presented in [55, Sec. 4.5].

The entry point of the compiler is the predicate `compile`, which takes the global reference `G` of the instance as input. Using the `coq.env.typeof` API, the compiler retrieves the type `Ty` of `G`. Compilation then proceeds by invoking the auxiliary predicate `comp` to produce the rule `R`. Finally, `coq.elpi.accumulate` adds the clause `R` to the database, here called `tc.db`.

The predicate `comp` performs the core of the compilation. It takes four arguments: the instance term; its type, which is structurally consumed during recursion; and two accumulators storing the list of instance arguments and the list of premises that will form the body of the resulting rule. From these inputs, `comp` produces the *Elpi* rule corresponding to the compiled instance.

Compilation proceeds by analyzing the structure of the instance type. Three cases are relevant. Rules $r\rightarrow$ and $r\forall$ handle quantifications, while rule rB is the base case that constructs the final *Elpi* rule from the information accumulated during the recursion.

The base case is reached when the current type `Ty` is not a quantification. The compiler then constructs the final rule by assembling its head and body.

The head is the predicate `tc` applied to the instance type `T` and to a proof term `Proof`. The proof is obtained by applying the instance `I` to its arguments `rev Ag`. Since the arguments are accumulated in reverse order during the traversal, the list `Ag` is reversed before building the application.

The premises of the clause are stored in the accumulator `P`. As for the arguments, the list is reversed before constructing the final rule.

For example, compiling the instance `addNat` produces the rule

```
tc {{Add nat}} {{addNat}}.
```

When the current type `Ty` is a quantification, the compiler introduces a fresh nominal variable representing the bound variable in the recursive call. Two situations must be distinguished depending on the shape of the quantification.

If the quantification corresponds to a premise of the instance, rule $\mathcal{r}\rightarrow$ is used. This rule introduces a fresh variable `x` representing the conditional instance with type `T`. The premise `tc T x` is added to the list of premises `P`, and the variable `x` is added to the list of instance arguments `Ag`.

Rule $\mathcal{r}\forall$ handles dependent quantification corresponding to parameters of the instance, that are supposed not to have a class as type. In this case the abstraction is simply consumed: the list of premises `P` remains unchanged, while the variable `x` is added to the list of instance arguments.

Both rules $\mathcal{r}\rightarrow$ and $\mathcal{r}\forall$ return a term of the form `(pi y \ R y)`, where `R` is the result of the recursive call to `comp`. Since `R` has type `term -> prop` and abstracts the local variable `x`, the compiler introduces a fresh variable `y` to quantify over the corresponding argument. This ensures that the resulting rule is closed (ground), as required by the `coq.elpi.accumulate` predicate.

As an example, consider the instance `addProd`, whose type is

$$\forall \underbrace{x\ y}_{\mathcal{r}\forall}, \underbrace{\text{Add } x \rightarrow \text{Add } y}_{\mathcal{r}\rightarrow} \rightarrow \underbrace{\text{Add } (x * y)}_{\mathcal{r}B}.$$

Compiling this instance yields the following *Elpi* rule:

```
pi L R PL PR \
  tc {{Add (lp:L * lp:R)}} {{addNat lp:L lp:R lp:PL lp:PR}} :-
    tc {{Add lp:L}} PL, tc {{Add lp:R}} PR.
```

This rule provides a proof for `Add (x * y)` provided that the two premises `Add x` and `Add y` can be solved.

To also solve type-class goals, we define an *Elpi* tactic containing the following rule:

```
solve (goal _ _ Ty _ _ as G) S :- tc Ty P, refine P G S.
```

Given a *Rocq* goal `G`, the tactic retrieves its type `Ty`, performs a backchaining step on `tc Ty P` to obtain a proof `P`, and refines the goal `G` with `P`. The refinement produces a list of subgoals `S`, which are returned to *Rocq*.

The full code, together with a working example, can be found in appendix [A](#).

The implementation presented in this section illustrates the expressive power of the *Elpi* language: the core of the compiler fits in roughly eight lines of code, while a similar number of lines is required to complete the *Elpi* command for the compiler. The solver itself consists of a single rule. However, this simplified implementation has some limitations it is worth to point out.

The predicate `tc` does not expose the class arguments as *Elpi* arguments. Consequently, the *Rocq* modes (see section 2.1.1) declared on a class are not preserved in the *Elpi* solver. The compiler should therefore generate a specialized predicate for each class, with arguments and modes matching those of the class so that rule backchaining can be performed correctly. This problem is addressed in section 3.3.

Rule $r\forall$ does not account for the possibility that the quantified variable x is itself a premise of the class. In complex instances, x may have a class as its type and should therefore appear as a premise in the generated rule. This problem is addressed in section 3.4.

The compiler does not handle instances of the form $(c_1 \rightarrow c_2) \rightarrow c_3$, where the implementation of c_3 requires a proof of c_2 in a context extended with a local instance of c_1 . Supporting such cases requires generating hereditary Harrop formulas and therefore tracking the polarity (see section 2.2.1) of the instance being compiled. This problem is addressed in section 3.4.

The `solve` rule of the solver ignores the context of the goal. As discussed in section 2.3.4, the first projection of a goal in *Rocq-Elpi* is its context, which may contain local instances that should be compiled and used during resolution. This problem is addressed in section 3.4.2.

3.3 Class compilation

When a class c is declared in *Rocq*, the compiler generates a corresponding *Elpi* predicate. If the class has n parameters, the generated predicate has $n + 1$ arguments: the first n arguments correspond to the parameters of c , while the final argument represents the instance witnessing the class in the current rule.

From the perspective of the *Elpi* solver, two aspects of a class declaration in *Rocq* are particularly relevant: the global reference identifying the class and the modes associated with its parameters. In *Rocq*, users may explicitly declare modes using the attribute `#[mode="..."]`. If no mode declaration is provided, the parameters of the class are considered outputs by default.

The compiler responsible for generating the corresponding *Elpi* predicate is shown in figure 3.3.

The entry point of the compiler is the predicate `add-class-pred`, which takes two arguments: the global reference `C` of the class and a string `S` describing its modes. The string `S` encodes the modes using the three *Rocq* symbols `-`, `+`, `!`, separated by spaces. If the user does not provide an explicit mode declaration, `S` is the empty string.

Predicates can be created dynamically in *Elpi* using the API `coq.elpi.add-predicate`. This primitive takes four arguments: the database in which the predicate will be stored

```

pred dft-class-mode term → list mode-type.
dft-class-mode (prod _ _ B) [pr out "term" | L] :- !,
  pi x\ dft-class-mode (B x) L.
dft-class-mode _ [pr out "term"].

pred str→mode string → mode-type.
str→mode "-" (pr out "term").
str→mode "+" (pr in "term").
str→mode "!" (pr in "term").

pred str→modes gref, string → list mode-type.
str→modes C "" M :- !, dft-class-mode {coq.env.typeof C} M.
str→modes _ S M' :-
  map {rex.split " " S} str→mode M,
  append M [pr out "term"] M'.

pred add-class-pred gref, string →.
add-class-pred C S :-
  gref→pred-name C N,
  str→modes C S M,
  coq.elpi.add-predicate "tc.db" _ N M.

```

Figure 3.3: Class compiler

(in our case `tc.db`), the indexing strategy, the predicate name `N`, and a list `M` of values of type `mode-type`. Each element of `M` specifies the mode and the type of one argument of the predicate. The predicate name is constructed from the global reference of the class using the predicate `gref→pred-name`. The list of modes `M` is obtained by processing the string `S` with the predicate `str→modes`.

Two cases must be distinguished depending on whether the string `S` is empty. If `S` is empty, the compiler derives the modes from the type of the class. It retrieves the type of `C` and creates a pair `pr out "term"` for each quantification in that type. In the base case, an additional output argument corresponding to the instance is added, again represented by the pair `pr out "term"`.

As an example, consider the declaration of the class `Add` (cf. figure 3.1). When this class is declared, the *Elpi* class compiler is triggered and receives the *Rocq* symbol `indt «Add»` as input. Since no mode attribute is provided, the compiler receives the empty string.

From the symbol `Add` the compiler retrieves its type and determines the number of parameters. In this example, the HOAS representation of the type of `Add` in *Elpi* is `{{Type → Type}}` which indicates that the class has a single parameter.

From this information the compiler generates the predicate associated with the class:

```
pred tc-Add → term, term.
```

If `S` is not empty, the user has provided explicit modes for the class. In this case, each *Rocq* mode is translated into the corresponding *Elpi* mode. This translation is performed by the predicate `str→mode`, which maps `-` to the *Elpi* output mode, and both `+` and `!` to the *Elpi* input mode.

3.3.1 *Rocq* and *Elpi* mode discrepancies

Our typeclass solver is not intended to replicate the *Rocq* solver. Instead, it aims to exploit, whenever possible, the features of the *Elpi* language. A natural claim is that mapping two *Rocq* mode to only one in *Elpi* is that the *Rocq* and *Elpi* type-class solvers will have different runtime behaviors.

We can note that there is no direct correspondence between *Rocq* modes and *Elpi* modes. While the *Rocq* mode `-` corresponds naturally to the *Elpi* output mode, neither `+` nor `!` has a precise counterpart among the *Elpi* input mode. Furthermore, a *Rocq* class may declare multiple mode specifications, whereas an *Elpi* predicate admits only a single mode declaration.

We recall that the *Rocq* modes `+` and `!` impose conditions on the shape of the argument: `+` requires a ground term, while `!` requires a term with a rigid applicative head. During the resolution phase, if a goal does not satisfy the mode constraint imposed by the class declaration, a failure immediately stops the instance search. In contrast, *Elpi* modes do not check the shape of the argument before backchain the query into the database, it, instead, modifies the unification strategy by using the `matching` procedure (see section 2.2.2).

In order to see the different mode behavior of the *Rocq* and *Elpi* modes, we rework the implementation of the `Add` type class.

```
#[mode="+"] Class Add T :=

Instance addNat : Add nat.
Instance addDummy T : Add T.
```

Consider a goal $g = \text{Add } ?X$ where $?X$ is a meta-variable (evar in *Rocq* jargon). With the *Rocq* mode system, this goal has no solution because the argument of the class is a non-ground term and therefore does not satisfy the mode constraint. In contrast, under the *Elpi* mode interpretation the instance `addDummy` is a valid solution.

The instance `addDummy` is considered as a catchall: the first argument of the `Add` class is a variable, i.e a pattern that can be used on any goal for `Add`. It means that `addDummy` can be used as a solution for g .

3.4 Instance compilation

As explained in section 3.1, instance compilation is triggered immediately after an instance declaration. Figure 3.4 shows an instance compiler implemented in *Elpi* where we take into account the type of dependent quantification and the polarity of the instance being compiled is an essential ingredient for compiling deep hereditary Harrop formulas.

Similarly to figure 3.2, we define a `compile` predicate which calls the `comp` auxiliary function. `comp` this time takes one more argument which is a boolean and tells the polarity of the term being compiled.

The compilation proceeds by analyzing the structure of the instance type. There are three relevant cases. Rules `r1` and `r2` handle quantifications, while rule `r3` is the base case that constructs the final *Elpi* rule from the information accumulated during the recursive traversal.

When the current type has the form $\forall x : \text{Ty}, \text{Bo } x$, the abstraction binds a variable of type `Ty` in the body `Bo`. The test `is-class? Ty` checks whether `Ty` has the shape

```

pred build-rule bool, prop, list prop → prop.
build-rule tt Head Prefs (Head :- Prefs).
build-rule ff Head Prefs (Prefs => Head).

%          Pol  Inst Ty  Args      Prefs      Res
pred comp bool, term, term, list term, list prop → prop.
comp B I {{∀ x: lp:Ty, lp:(Bo x)}} Ag P (pi y\ R y) :- % r→
  is-class? Ty, !,
  pi x\
    comp {¬ B} x Ty [] [] (M x),
    comp B I (Bo x) [x|Ag] [M x | P] (R x).
comp B I {{∀ x, lp:(Bo x)}} Ag P (pi y\ R y) :- !,      % r∀
  pi x\ comp B I (Bo x) [x|Ag] P (R x).
comp B I Ty Ag P R :-                                     % rB
  safe-dest-app Ty C CAg,
  mk-app I {rev Ag} Proof,
  make-rule-head C {append CAg [Proof]} Head,
  build-rule B Head {rev P} R.

pred compile gref →.
compile G :- coq.env.typeof G Ty,
  comp tt (global G) Ty [] [] R,
  coq.elpi.accumulate _ "tc.db" (clause _ _ R).

```

Figure 3.4: Instance compiler

949 $\forall x_1 \dots x_n, T$ (possibly with zero quantifiers), where T is a term whose applicative head
 950 is a type class.

951 If this condition holds, we are compiling a conditional instance and rule $r \rightarrow$ is applied.
 952 The rule introduces a fresh nominal variable x representing both the abstracted variable
 953 in Bo and the local instance with type Ty . To construct the premise corresponding to x ,
 954 the compiler recursively invokes `comp` on x while switching the polarity of the rule. This
 955 recursive call produces the premise M , which abstracts x . Compilation then continues on
 956 $Bo\ x$, while extending the accumulators with the new instance argument x and the new
 957 premise M .

958 Rule $r\forall$ handles the case where the quantified type is not a type class. In this situa-
 959 tion, the compiler behaves exactly as rule $r\forall$ in figure 3.2: it simply consumes under the
 960 abstraction, leaving the list of premises P unchanged and adding the variable x to the list
 961 of instance arguments.

962 As in figure 3.2, the base case of `comp` is reached when the current type Ty is not a
 963 quantification. In this case, Ty is a type class C applied to a list of arguments CAg . The
 964 compiler then constructs the final rule by building its head and premises.

965 To build the head $Head$, the compiler first constructs the proof term $Proof$, obtained
 966 by applying the instance I to its arguments `rev Ag`. The list Ag is stored in reverse
 967 order during traversal, so it is reversed before constructing the application. The predicate
 968 `make-rule-head` takes the class name as its first argument in order to generate the predicate
 969 corresponding to the class. The arguments of this predicate are obtained by appending
 970 the proof term to the list of class arguments.

```

Inductive a := p | q | r.
Variable to_prop : a → Prop.

Inductive mlogic :=
| atom : a → mlogic
| and : mlogic → mlogic → mlogic
| impl : mlogic → mlogic → mlogic.

Fixpoint interp (p: mlogic) : Prop :=
  match p with
  | atom a => to_prop a
  | impl a b => interp a → interp b
  | and a b => interp a /\ interp b
  end.

Class Provable (T : mlogic) := { proof : interp T }.

Instance Pand F1 F2 :
  Provable F1 → Provable F2 → Provable (and F1 F2).

Instance Pimpl F1 F2 :
  (Provable F1 → Provable F2) → Provable (impl F1 F2).

```

Figure 3.5: Example with simple logic

971 The premises of the clause are stored in the accumulator `P`. As with the instance
 972 arguments, they are reversed before constructing the final rule.

973 Finally, the predicate `build-rule` assembles the rule according to the current polar-
 974 ity. With positive polarity, it produces a standard clause `Head :- Prem`s. With negative
 975 polarity, it produces the implication `Prem`s $\Rightarrow$ `Head`.

976 Given the instance in figure 3.1, this version of the compiler produces the following
 977 rules:

```

tc-Add {{nat}} {{addNat}}.
tc-Add {{R}} {{addR}}.
pi L \ pi R \ pi PL \ pi PR \
  tc-Add {{lp:L * lp:R}} {{addProd lp:L lp:R lp:PL lp:PR}} :-
  tc-Add L PL, tc-Add R PR.

```

978 These rules are similar to those produced by the compiler in section 3.2, except for the
 979 specialization of the predicate `tc-Add` for the `Add` class.

980 An interesting example of compilation, where polarities play a central role (and are
 981 incorrectly handled by the compiler in section 3.2), is shown in figure 3.5. The code illus-
 982 trates an encoding of a fragment of propositional logic using the *Rocq* typeclass mechanism
 983 as a proof search engine.

984 We define an inductive type `mlogic` representing a restricted language of formulas
 985 consisting of atoms, conjunctions, and implications. The constructor `atom` corresponds to
 986 logic atoms, while `and` and `impl` correspond to conjunction and implication, respectively.

The function `interp` assigns a semantics to `mlogic` formulas by translating them into *Rocq* propositions. This interpretation is compositional: an atom evaluates to a proposition using the `to_prop` function, a conjunction is interpreted as a logical conjunction, and an implication as a function type.

Automatic reasoning is expressed via the typeclass `Provable T`, which packages a proof of `interp T`. This allows *Rocq*'s typeclass resolution mechanism to act as a proof search procedure.

The inference rules of the logic are encoded as typeclass instances. The instance `Pand` corresponds to conjunction introduction: if `F1` and `F2` are provable, then `and F1 F2` is also provable.

Similarly, the instance `Pimpl` captures implication introduction. It states that to prove `impl F1 F2`, it suffices to show that `F2` is provable under the assumption `F1`, represented as `fact F1`.

The instance `Pand` is compiled into:

```

pi L \ pi R \ pi PL \ pi PR \
  tc-Provable {{and lp:L lp:R}} {{Pand lp:L lp:R lp:PL lp:PR}} :-
    tc-Provable L PL, tc-Provable R PR.

```

This rule closely mirrors the rule for `addProd`, since both rely on the same search principle: constructing a proof requires recursively constructing proofs for the subcomponents (the arguments of `and`, respectively `*`).

The rule corresponding to the instance `Pimpl` is more subtle, since it involves a higher-order hypothesis. Its type is:

$$\forall F1\ F2, \underbrace{(\underbrace{\text{Provable } (\text{fact } F1)}_{\text{positive}} \rightarrow \underbrace{\text{Provable } F2}_{\text{negative}})}_{\text{negative}} \rightarrow \underbrace{\text{Provable } (\text{impl } F1\ F2)}_{\text{positive}}$$

The compiled rule is:

```

pi F B Q \
  tc-Provable {{impl lp:F lp:B}} {{Pimpl lp:F lp:B lp:Q}} :-
    pi p \ tc-Provable {{fact lp:F}} p => tc-Provable B {{lp:Q lp:p}}.

```

The structure of this rule matches the intended semantics: the head targets `impl F B`, while the body attempts to construct a proof of `B` in a context extended with the assumption `F`.

3.4.1 Rule Priorities

When declaring instances, the priority assigned to each rule plays a fundamental role in the performance of the resolution phase. In *Rocq*, every instance is associated with a priority at declaration time.

To support rule priorities, we extend the instance trigger so that it also propagates priority information to the *Elpi* compiler. To reproduce a similar behavior in *Elpi*, we exploit its grafting mechanism (see section 2.2.6), which allows rules in the database to be named and enables new rules to be inserted before or after existing ones.

```

pred get-ITy prop → term, term.
get-ITy (decl I _ Ty) I Ty.
get-ITy (def I _ Ty _) I Ty.

pred compile-ctx prop → prop.
compile-ctx C R :- get-ITy C I Ty, is-class? Ty, comp tt I Ty [] [] R.

solve (goal C _ Ty _ _ as G) S :-
  map-filter C compile-ctx H,
  comp ff P Ty [] H R, R,
  refine P G S.

```

Figure 3.6: Goal compiler

1018 In the *Elpi* compiler, we predeclare a fixed number of hooks, each associated with an
 1019 integer n representing a grafting position. We account for this mechanism by modifying
 1020 the `compile` rule in figure 3.4 so that it takes this integer as input. When a rule is added
 1021 to the database, we explicitly specify that it must be grafted `:before` the hook named n .

1022 3.4.2 Goal Encoding

1023 With the database compilation in place, we now define the tactic used to solve type-
 1024 class goals. The implementation of the `solve` predicate is shown in figure 3.6.

1025 A *Rocq* goal in *Elpi* contains, among other information, the context `C` in which the goal
 1026 is defined and a term `Ty` representing the type to be solved. The first step is to traverse
 1027 the context, select the hypotheses that correspond to type classes, and compile them using
 1028 the `comp` procedure. This produces a list of local rules `H`, which are added to the program
 1029 during resolution.

1030 Next, we compile the goal type `Ty`. As in hereditary Harrop formulas, a goal corre-
 1031 sponds to an atom in negative position. Therefore, the compilation is performed with
 1032 polarity `ff`. Concretely,

```
1033 comp ff P Ty [] H R
```

1034 constructs an atom in negative position. Here, `P` is a unification variable representing the
 1035 proof term, `Ty` is the goal to be compiled, the list of instance parameters is empty, and
 1036 `H` provides the premises extracted from the context. The result of the compilation is an
 1037 atom `R` of the form `H => Q`, where `Q` is a query corresponding to a type class whose proof
 1038 argument is `P`.

1039 Evaluating `R` instantiates the proof term `P`, which is then used to refine the original
 1040 *Rocq* goal.

1041 3.4.3 User-Defined Rules

1042 In addition to the automatic compilation of rules, users can extend the database with
 1043 custom rules to implement specialized or more efficient resolution strategies in specific
 1044 situations.

Using the `Elpi Accumulate` command, users can define ad hoc rules, such as the one shown below:

```
tc-Add {{lp:A * lp:A}} P :-
  P = {{let x := lp:A in let p := lp:Pa in addProd x x p}},
tc-Add A Pa.
```

This rule matches products whose two components are identical. When searching for an instance of the class `Add` applied to a type `A`, both components of the pair yield the same result, so the search is performed only once. Moreover, the resulting proof term shares memory between subproofs and their types, improving space and time efficiency.

In the simple rule presented above, we illustrate the expressive power of meta-programming in *Elpi*. Since *Elpi* is a Turing-complete language, users can arbitrarily customize the proof search by defining their own rules. In particular, there are no restrictions on these extensions: users are free to implement any custom rule that suits their needs.

Extending the database with ad-hoc rules is possible in the *Rocq* system thanks to the `Hint` system, that allows to tweak somehow the search engine, for example the `Hint Extern` command allows to apply a list of tactics on goal matching a particular pattern. This approach, however, asks the user to write ad-hoc rules in a language which is not the same of the *Rocq* solver, and which, in some cases, is not the power of expressiveness of *Elpi*.

3.5 Experimental results

3.5.1 Benchmarks

We evaluate our solver on existing libraries, in particular *tlc* [7] and *stdpp* [31]. Both libraries use type-class inference as a main implementation tool. Studying them helps us understand the practical needs of *Rocq* users. This can reveal limitations of our implementation and suggest directions for future work. An important goal of this evaluation is also to compare our solver with the native *Rocq* solver.

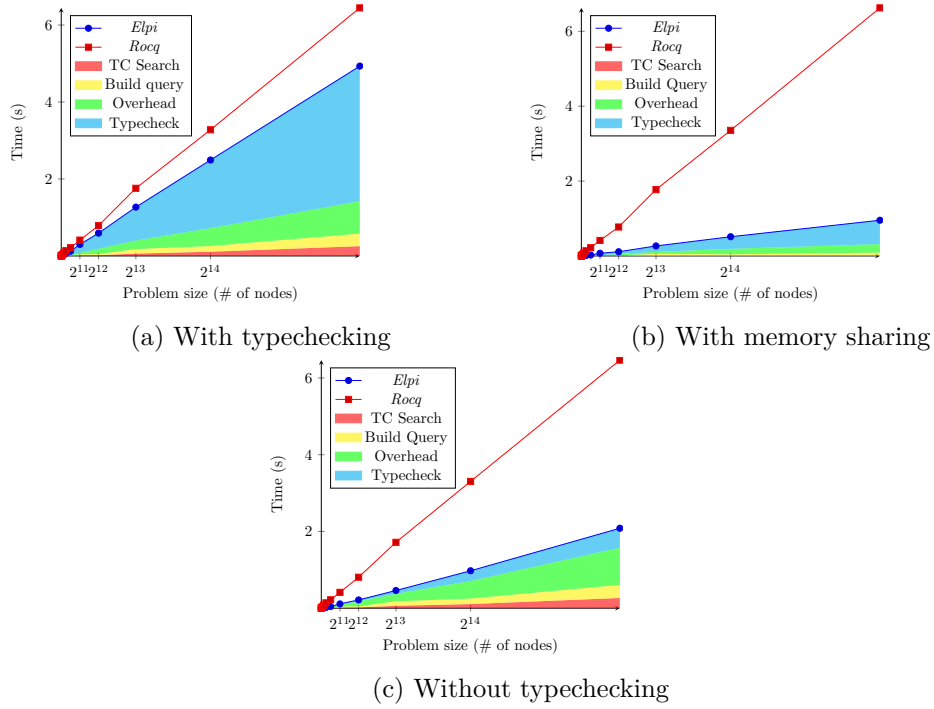
As a first benchmark, we consider a set of classes and instances from *stdpp*. We focus on the `Inj` type class, defined as follows:

```
Class Inj A B (f : A → B) :=
  inj x y : f x = f y → x = y.
```

The type class `Inj` expresses that a function is injective. A function f is injective if, for all x and y , $f x = f y$ implies $x = y$.

The library provides 13 instances of this class. These include the identity function (`id`) and the constructors `inl` and `inr`, which are the injections of a sum type. In this benchmark, we focus on function composition. The composition (written $\circ$) of two functions f and g is injective if both functions are injective.

To evaluate performance, we measure the time needed by the instance solver to solve goals of increasing size. This helps us understand where time is spent and compare *Elpi* with *Rocq*. The benchmark goals are generated by the following function. It builds a balanced tree where internal nodes are compositions and leaves are a fixed injective function f :

Figure 3.7: Benchmarks of *Elpi* solver

$$g_n = \begin{cases} f & \text{if } n = 0, \\ g_{n-1} \circ g_{n-1} & \text{if } n > 0. \end{cases}$$

For a given n , we submit the goal `Inj eq eq  $g_n$` to the solver. The size of this goal grows exponentially with n . Figure 3.7a shows the resolution time for n from 0 to 15. The x -axis is the number of nodes (logarithmic scale), and the y -axis is the time in seconds. The plot compares *Rocq* with a detailed analysis of the *Elpi* solver.

For *Elpi*, we split the total time into four parts: type-class resolution, query construction (translation of *Rocq* terms into *Elpi*), communication between *Elpi* and *Rocq*, and typechecking of the final proof.

For *Rocq*, it is harder to separate these costs. Query construction and communication do not appear. However, instance search and typechecking are mixed together. In fact, partial typechecking is done after each instance selection. For this reason, we cannot easily measure them separately.

The results in figure 3.7a show that instance search is very fast. For goals with 2^{15} nodes, it takes about 0.25 seconds out of a total of 4.9 seconds. On the other hand, query construction and communication grow with the size of the goal. This cost comes from the interaction between *Rocq* and *Elpi* and does not depend on the *Elpi* interpreter itself.

The main cost is typechecking the proof term. It represents about 77% of the total time. We consider two ways to reduce this cost. First, we can introduce ad hoc rules for the specific problem. Second, we can skip typechecking of the produced proof.

The first approach also produces smaller proof terms, which are faster to check. As explained in section 3.4.3, memory sharing can reduce the complexity from exponential

to linear. This effect is shown in figure 3.7b: smaller terms give faster typechecking and faster translation from *Elpi* to *Rocq*. However, this solution is not very portable, since it depends on the specific goal.

The second approach, i.e. skipping typechecking, reduces the time of the proof refinement to the goal by about a factor of 5, as shown in figure 3.7c. In most cases, this does not cause problems. The proof term should be correct, because the rules in the database are compiled in a context where they already typecheck. The main problem comes from universe constraints.

In the *Elpi* solver, universes are removed from the compiled clauses to simplify unification. For example, the *Rocq* term `Type@{i}` corresponds to the HOAS term `sort (typ i)`. During compilation, the level i is replaced by a unification variable. This allows matching terms at different levels at runtime.

However, removing universes from the database can introduce inconsistencies or may forget to state some universe constraint. These are usually detected during typechecking. A possible solution, which we do not implement, is to add universe constraints explicitly in the compiled rules. In this way, part of typechecking would be done during the search. This approach is similar to the strategy used by the *Rocq* solver.

3.5.2 Modes: *Elpi* vs *Rocq* in *stdpp*

We previously discussed discrepancies between *Elpi* and *Rocq* modes in section 3.3.1. Here, we highlight a family of implementation choices in the *stdpp* library that make the *Elpi* mode system inadequate. As a running example, consider the following class.

```
#[mode="- !"] Class Union A M :=
  union: (A → A → option A) → M → M → M.
```

The class is parametrized by two types, A and M , representing respectively the type of keys and the type of collections. The mode declaration requires M to have a rigid applicative head, while no constraint is imposed on A . The class provides the overloadable symbol `union`, which computes the union of two collections. In case of ambiguity, a higher-order function f is used to select between conflicting elements. A typical example arises with dictionaries: when a key appears in both inputs, f determines which value to retain.

We now consider an instance of `Union` from *stdpp*, together with a lemma that relies on it.

```
Instance union {A} : Union A (option A) := ...
```

```
Lemma union_Nr A (f: A → A → option A) mx : union f mx None = mx.
```

In this instance, `union` behaves as follows: it returns `None` if both arguments are `None`; it returns `f x y` if the arguments are `Some x` and `Some y`; otherwise, it returns the non-`None` argument.

In the lemma, an implicit type-class constraint must be resolved. The goal is `Union A (option ?X)`, where `?X` is a unification variable corresponding to the implicit argument of the `None` constructor. At this point, the typechecker lacks sufficient information to instantiate `?X`. The expected behavior is that type-class resolution both selects an instance and determines the value of `?X`.

In *Rocq*, the goal is solved using the instance `union`. In *Elpi*, however, the same instance cannot be applied because of mode restrictions. The input mode is

declared on the second argument of the class, in particular, matching the query `tc-Union {{A}} {{option lp:X}}` against the clause head for `ounion` fails, since *Elpi* does not assign variables in input positions during matching.

Another subtle point arises when additional instances are present. If, before the lemma `union_None_r`, the database is extended with the following instance,

```
Instance ounion {A} : Union A (option (option A)).
```

then, under *Rocq* modes, it becomes unclear which instance is selected to solve the goal. Indeed, both instances match the constraint `Union A (option ?X)`. Depending on their priorities, `?X` may be instantiated differently. As a result, developers may be unaware of which instance has been chosen, potentially leading to type errors later in the development.

In contrast, in *Elpi*, even with the additional instance `ounion`, type-class resolution still fails to make progress. In particular, the query `tc-Union {{A}} {{option lp:X}}` has no solution. To overcome this limitation, the type of the argument `M` must be provided explicitly in the lemma.

```
Lemma union_Nr A (f : A → A → option A) mx : union f mx (@None A) = mx.
```

3.5.3 Unification problems

As explained in section 3.1, the *Elpi* solver can be used as an alternative – not a replacement – to the *Rocq* solver. In a development, one can enable or disable either solver depending on the needs.

When running the *Elpi* solver on existing libraries, one may observe failures in situations where the *Rocq* solver succeeds, and vice versa, although the latter case is rare. In our setting, we focused on making the target library compile with the *Elpi* solver. Note that in this library, all type-class problems are solvable with the *Rocq* solver.

When a type-class resolution fails in *Elpi*, the cause can be investigated by inspecting the execution traces of both *Elpi* (using the *Elpi* trace browser [55, Sec. 3.5]) and *Rocq* (using the `Set Typeclasses Debug` option). Our analysis shows that most discrepancies arise from unification issues. We distinguish two main categories encountered in our study: failures related to $\beta\eta$ -reduction, and failures related to δ -reduction (see [63]).

A typical $\beta\eta$ -unification example is the term `fun x y => ?F y x` unified with `map*`, where `?F` is a unification variable. In *Rocq*, this succeeds by instantiating `?F` with `fun x y => map y x`. In contrast, *Elpi* represents these terms as HOAS expressions, namely `fun _ _ x\ fun _ _ y\ app[F, y, x]` and `global (const «map»)`. Since these terms have different rigid heads, unification fails.

This behavior is expected from the *Elpi* unification algorithm, but it is problematic in the context of instance resolution, where equivalent problems – expressed differently – should be solvable. For example, using *Elpi*-style λ -abstraction and application, the term `(x\ y\ F y x)` unifies with `map`, assigning `F` to `x\ y\ map y x`. To address this, we extend the instance compiler to detect potential $\beta\eta$ -unification issues. Such terms are replaced with a fresh unification variable `v`, simplifying unification in *Elpi*. The variable `v` is then linked back to the original *Rocq* term, so that instantiations in *Elpi* are reflected in *Rocq*.

*The standard list map function

This approach led to our contribution presented in [16], described in detail in chapter 4, which provides an automatic solution to $\beta\eta$ -unification issues.

A concrete example of $\beta\eta$ -unification failure in the *Elpi* solver arises from the compiled rule for the instance `Pimpl`, introduced at the end of section 3.4. Consider the goal `Provable (impl a (fact a))` for some proposition `a`. This goal has the solution `Pimp a (fact a) (fun x => x)`. However, the compiled rule fails to find this solution due to $\beta\eta$ -unification.

After applying the rule for `Pimpl`, the sub-query becomes

```
tc-Provable {{fact a}} {{lp:Q lp:p}}
```

in a context extended with the hypothesis $h = \text{tc-Provable } \{\{\text{fact } a\}\} p$. Solving the sub-query requires unifying it with h . However, the HOAS term corresponding to $\{\{\text{lp:Q lp:p}\}\}$ is `app[Q, p]`, which cannot be unified with `p`, whose HOAS representation is simply `p`.

The second class of issues concerns δ -unification, which is not handled by the *Elpi* unification mechanism. In *Rocq*, definitions can be unfolded during unification. For example, the function `id` unifies with `fun x => x` after unfolding its definition. In contrast, *Elpi* – and more generally *Prolog*-like systems – does not support definitional unfolding during unification.

An example of failure due to missing δ -unification is given by the following code:

```
Class Extens A := ...
Instance Extens_f1 A1 A2: Extens (∀ x1 : A1, A2 x1) := ...
Definition map A B := A -> option B.
```

The *Rocq* solver can solve the goal `Extens (map A B)` using the instance `Extens_f1`. However, *Elpi* fails because `map A B` does not syntactically match a term starting with a quantifier.

Providing a general and automatic solution for δ -unification in *Elpi* is non-trivial. In some cases, the absence of δ -unification is actually desirable, since unfolding definitions can make unification unpredictable and harder to debug. Moreover, it may introduce performance issues [63].

To address δ -unification issues, the user has two options. First, one can declare `map` as a notation for `A -> option B`, since notations are unfolded by the elaborator, creating no ambiguity in the unification algorithm. Alternatively, one can add an ad-hoc rule to the solver:

```
Elpi Accumulate TC.Solver lp:{{
  tc-Extens {{map lp:A lp:B}} R :-
    tc-Extens {{lp:A -> option lp:B}} R.
}}
```

This rule manually implements unfolding: when the first argument of `Extens` is `map`, the solver recursively processes the unfolded definition.

3.6 Discussion

The previous sections highlighted the flexibility of the *Elpi* language and its suitability for compiling *Rocq* terms into *Elpi* rules. In particular, *Elpi* provides a natural framework for proof automation: its *Prolog*-like search engine acts as an inference mechanism, provided that the program contains the derivation rules required to solve a given query. Although the type-class engine in *Rocq* exhibits a similar behavior, our goal was not to faithfully reproduce the *Rocq* solver. Instead, we aimed to leverage the distinctive features of *Elpi* whenever possible.

A key advantage inherited from *Elpi* is the control the user has over the search process. This control is achieved through several mechanisms, including rule priorities, predicate modes, the `unify/matching` unification algorithm, and the `Cut` operator. All these aspects have been discussed in previous sections, with the exception of the `Cut` operator.

We have deliberately postponed the discussion of `Cut`, as it has no direct counterpart in the *Rocq* type-class solver. As explained in section 2.2.2, the `Cut` operator allows the user to commit to a solution once certain conditions (e.g., rule guards) are satisfied. To fully exploit this feature, the *Rocq* language would require a syntax extension capable of indicating where `Cut` should be introduced during compilation to *Elpi* rules. Discussions with other researchers suggest that `Cut` may provide an effective approach to addressing a well-known issue in type-class resolution: determinacy.

In logic programming, determinacy is the property that a predicate produces at most one solution for any given call. The `Cut` operator is a central mechanism for enforcing this property. Transposed to type-class resolution, determinacy would ensure that a class declared as deterministic remains so, regardless of the rules added to the database.

In *Rocq*, the flag `Typeclasses Unique Solutions` enforces a related behavior: it computes all possible solutions to a type-class query and returns a result only if exactly one solution exists; otherwise, it raises an error. However, as noted in the *Rocq* manual [48], this approach is computationally expensive, since it requires exploring the entire search tree.

In contrast, determinacy has been extensively studied in logic programming (e.g., [12, 35, 28]) and can be verified statically using abstract interpretation. This approach incurs no runtime cost during goal execution. Moreover, since determinacy is a general property of logic programs rather than being specific to type-class resolution, we extend *Elpi* with a determinacy checker implemented at the compiler level. As a result, all applications built on *Elpi*, including the type-class solver, can benefit from this analysis.

We presented this work in [17], where we describe the implementation of a determinacy checker for *Elpi* that accounts for both the `Cut` operator and higher-order features such as local rule insertion. A detailed discussion is provided in chapter 5. Finally, we formalize the first-order fragment of the checker within *Elpi* in the *Rocq* system; this formalization is described in chapter 6.

CHAPTER 4

$\beta\eta$ -unification support for meta-programs

In the previous chapter, and in particular in section 3.5.3, we have talked about the unification issues that we encounter in the *Elpi* class solver when unifying certain *Rocq* terms. Unification up to δ is not possible in *Elpi*, but the $\beta\eta$ -conversion rules are part of the *Elpi* unification algorithm. In this chapter we propose a framework allowing to ease the unification of the HOAS representation of the object language terms in the meta language. The sentence which better describe the aim of this chapter: *how to reuse the higher-order unification procedure of the meta language in order to simulate a higher-order logic program for the object language*.

The example we have given in section 3.5.3 providing a source of $\beta\eta$ -unification problem is due to the instance `Pimpl`.

```
tc-Provable {{impl lp:F lp:B}} {{Pimpl lp:F lp:B lp:Q}} :-  
  pi p \ tc-Provable {{fact lp:F}} p => tc-Provable B {{lp:Q lp:p}}.
```

The variable `Q` in *Rocq* is an object living in the function space, while in the HOAS representation of *Elpi* `Q` is a first-order variable of type `term`. As shown in section 3.5.3, the *Rocq* statement `Provable (impl a (fact a))` can be solved by the *Rocq* type-class solver using the instance `Pimpl`. On the contrary, *Elpi* fails: the rule above would be used on the query `tc-Provable {{impl a (fact a)}} S` but the sub-query `tc-Provable {{fact a}} {{lp:(Q p)}}` will fail since not unifying with the local rule `tc-Provable {{fact a}} p`.

The root cause of the problems we outlined in this example is a mismatch between the equational theories of the meta language and the object language, which in turn makes the unification procedures of the meta language weak. The equational theory of the meta-language *Elpi* encompasses $\beta\eta$ -equivalence and its unification procedure can solve higher-order problems in the pattern fragment (definition 2.2.4). Although the equational theory of *Rocq* is much richer, for efficiency and predictability reasons automatic proof search procedures typically employ a unification procedure that only captures a $\beta\eta$ -equivalence and only operates in the pattern fragment.

A more sophisticated encoding of *Rocq* terms that, on a first approximation, would reshape the compiled rule for `Pimpl` as follows:

```
tc-Provable {{impl lp:F lp:B}} {{Pimpl lp:F lp:B lp:Q}} :-  
  link Qm Q, % a constraint relating the local variable Qm with Q  
  pi p \ tc-Provable {{fact lp:F}} p => tc-Provable B (Qm p).
```

In the body of the rule, we have added a new higher-order *Elpi* variable $\mathbf{Qm}$ which has the type `term` $\rightarrow$ `term`. In the recursive call to `tc-Provable` we pass as argument the variable $\mathbf{Qm}$ applied, using *Elpi* application, to $\mathbf{p}$, allowing the the unification problem $\mathbf{Qm} \mathbf{p} = \mathbf{p}$ to get a solution, namely, $\mathbf{Qm} = \mathbf{x} \setminus \mathbf{x}$.

The role of the `link` predicate is to relate the *Elpi* variable $\mathbf{Qm}$ back to the *Rocq* variable $\mathbf{Q}$. For instance, when $\mathbf{Qm}$ is assigned, then $\mathbf{Q}$ is assigned to the HOAS term `fun _ _ x => x`. With this encoding, the query `tc-Provable {{imp a (fact a)}} S` has a solution.

The purpose of this compilation is to detect, at compile time, all subterms that may cause unification difficulties up to $\beta\eta$ -conversion. Each such subterm t is replaced by a unification variable v . This transformation simplifies unification in the meta-language, while an additional premise is introduced to relate t and v .

Our compilation strategy is described as a general framework based on two languages: $\mathcal{O}$ for the object language, and $\mathcal{M}$ for the meta-language. We explain how $\mathcal{O}$ is encoded into $\mathcal{M}$, and we show that the higher-order unification procedure of the meta-language, denoted $\simeq_m$, can simulate a higher-order unification procedure $\simeq_o$ for the object language, including $\beta\eta$ -conversion.

Section section 4.1 states the problem formally and gives the intuition behind our solution; section 4.2 sets up a basic simulation of first-order logic programs, section 4.3 and section 4.4 extend it to higher-order logic programs in the pattern fragment while section 4.6 goes beyond the pattern fragment. Section section 4.7 discusses the implementation in *Elpi*.

4.1 Problem statement and solution

Although the problem was first observed while working on *Rocq*'s Calculus of Inductive Constructions, we introduce a simplified setting to study it more easily. In this setting, we consider a language $\mathcal{O}$ with a rich equational theory, and a language $\mathcal{M}$ with a simpler one.

Both $\mathcal{O}$ and $\mathcal{M}$ are encoded in the *Elpi* language. The syntax of terms is shown in figure 4.1. Terms of $\mathcal{O}$ are called as $\mathbf{tm}_o$, while the HOAS encoding of these terms in $\mathcal{M}$ are written as $\mathbf{tm}_m$. These correspond, respectively, to object-level terms and meta-level terms.

The main difference between these two representations is in the way variables are encoded. In both cases, a variable is linked to a memory address. However, in $\mathcal{O}$, variables are first-order objects and they do not carry any notion of scope. On the other hand, variables in $\mathcal{M}$ are equipped with an explicit scope, represented as a list of terms.

This scope is intended to capture the application of meta-level variables to terms. By contrast, the constructor `appm` encodes the application of $\mathcal{O}$ terms at the meta-language level.

This distinction can be seen in the example introduced in the previous section for the rule `tc-Provable`. The term `(Qm p)` is represented as `uvam [Qm, p]`, while the term `{{lp:Q lp:p}}` is encoded as `appm [Q, p]`.

When we write $\mathcal{M}$ terms outside code blocks we follow the usual λ -calculus notation, reserving f, g, a, b for constants, x, y, z for bound variables and X, Y, Z, F, G, H for unification variables. However, we need to distinguish between the “application” of a unification

```

kind tmo type.
type appo list tmo → tmo .
type lamo (tmo → tmo) → tmo .
type cono string → tmo .
type uvao addr → tmo .

kind tmm type.
type appm list tmm → tmm .
type lamm (tmm → tmm) → tmm .
type conm string → tmm .
type uvam addr → list tmm → tmm .

```

Figure 4.1: The $\mathcal{O}$ and $\mathcal{M}$ languages

```

pred (=o) tmo , tmo → .
cono C =o cono C.
appo A =o appo B :- forall2 (=o) A B.
lamo F =o lamo G :- pi x\ x =o x => F x =o G x.      % rλλ
uvao N =o uvao N.
lamo F =o T :-                                         % rη
  pi x\ beta T [x] (T' x), x =o x => F x =o T' x.
T =o lamo F :-                                         % rηr
  pi x\ beta T [x] (T' x), x =o x => T' x =o F x.
appo [lamo X|L] =o T :- beta (lamo X) L R, R =o T.    % rβl
T =o appo [lamo X|L] :- beta (lamo X) L R, T =o R.    % rβr

```

Figure 4.2: Equality implementation for $\mathcal{O}$

variable to its scope and the application of a term to a list of arguments. We write the scope of unification variables in subscript while we use juxtaposition for regular application. Here are few examples:

```

f a      appm [conm "f", conm "a"]
λx.λy.Fxy lamm x\ lamm y\ uvam F [x, y]
λx.Fx a  lamm x\ appm [uvam F [x], conm "a"]
λx.Fx x  lamm x\ appm [uvam F [x], x]

```

When it is clear from the context, we shall use the same syntax for $\mathcal{O}$ terms (although we never use subscripts for unification variables). We use $s, s_1, \dots$ for terms in $\mathcal{O}$ and $t, t_1, \dots$ for terms in $\mathcal{M}$.

4.1.1 Equational theories and unification

To reproduce the unification issue in *Elpi* when dealing with *Rocq* terms, we extend our setting with two main routines: one for unification, and one for testing equality between terms. The second routine is required to check that the substitution produced by the unification process indeed makes the two terms equal.

Term equality: $=_o$ and $=_m$ For both languages, we extend the equational theory from ground terms to the full language by adding reflexivity for unification variables. That is, a unification variable is equal to itself, but different from any other term. The corresponding implementation is given in figures 4.2 and 4.3.

The first four rules are shared by both equalities and define the standard congruence on terms. Because we use an HOAS encoding, these rules also account for α -equivalence. In addition, $=_o$ includes rules for η - and β -equivalence. This is because, in the object language, the term `appo [lamo x\ x, cono "3"]` and `cono "3"` are definitionally equal.

```

1346 pred ( $=_m$ )  $\text{tm}_m$  ,  $\text{tm}_m$   $\rightarrow$  .
1347 con $_m$   $C =_m$  con $_m$   $C$  .
1348 app $_m$   $A =_m$  app $_m$   $B :- \text{forall2 } (=_m) A B$  .
1349 lam $_m$   $F =_m$  lam $_m$   $G :- \text{pi } x \backslash x =_m x \Rightarrow F x =_m G x$  .
1350 uva $_m$   $N A =_m$  uva $_m$   $N B :- \text{forall2 } (=_m) A B$  .

```

Figure 4.3: Equality implementation for $\mathcal{M}$

Both procedures use implication to assume that the fresh nominal constants introduced by **pi** are equal to themselves.

The main point in showing these equality tests is to remark how weaker $=_m$ is, and to identify the four rules that need special treatment in the simulation of $\simeq_o$ in $\mathcal{M}$. For brevity, we omit the code of **beta**: it is sufficient to know that **beta** $F L R$ computes in $\mathcal{R}$ the weak-head normal form of **app** $_o$ $[F \mid L]$.

Substitution: ρs and σt We reuse the notations for substitution from section 2.2.2: we write $\sigma = \{ X \mapsto t \}$ for the substitution that assigns the term t to the variable X , we write σt for the application of the substitution to a term t , and $\sigma X = \{ \sigma t \mid t \in X \}$ when X is a set of terms. We write $\sigma \subseteq \sigma'$ when σ is more general than σ' . We will use ρ for representing substitutions in $\mathcal{O}$, and σ for the ones in $\mathcal{M}$. In this section, we consider the substitution for $\mathcal{O}$ and $\mathcal{M}$ identical. We defer to section 4.2.1 a more precise description pointing out their differences.

Term unification: $\simeq_o$ vs. $\simeq_m$ As the signature of **unify** and **matching** in section 2.2.2, $\mathcal{M}$'s unification signature is:

```

1361 pred ( $\simeq_m$ )  $\text{tm}_m$  ,  $\text{tm}_m$  , subst  $\rightarrow$  subst .

```

We write $(t_1 \simeq_m t_2) \sigma \rightsquigarrow \sigma'$ when σt_1 and σt_2 unify with substitution σ' . σ' is a substitution extension of σ (see definition 2.2.2). We write $t_1 \simeq_m t_2 \rightsquigarrow \sigma'$ when the initial substitution σ is empty. We write $\mathcal{L}$ as the set of terms that are in the pattern fragment (definition 2.2.4).

The specification of a “good” unification procedure restricted to a domain $\mathcal{D}$ is the following:

Proposition 4.1.1 (Good unification). *A good unification $\simeq$ for equality $=$ in the domain $\mathcal{D}$ satisfies the following properties:*

$$\{ t_1, t_2 \} \subseteq \mathcal{D} \rightarrow t_1 \simeq t_2 \rightsquigarrow \rho \rightarrow \rho t_1 = \rho t_2 \quad (4.1)$$

$$\{ t_1, t_2 \} \subseteq \mathcal{D} \rightarrow \rho t_1 = \rho t_2 \rightarrow \exists \rho', t_1 \simeq t_2 \rightsquigarrow \rho' \wedge \rho' \subseteq \rho \quad (4.2)$$

The meta and (resp. object) language of choice are expected to provide an implementation of $\simeq_m$ (resp. $\simeq_o$) that is a good unification for $=_m$ (resp. $\simeq_m$) in $\mathcal{L}$.

4.1.2 The problem: logic-program simulation

We simulate the execution, called *run*, of a sequence of unification problems, denoted $\mathbb{P}$, from the object language $\mathcal{O}$ into the meta-language $\mathcal{M}$. The sequence $\mathbb{P}$ is given as a list of *steps*. Each step $\mathbb{P}_p$ corresponds to the unification of two terms, $\mathbb{P}_{p_l}$ and $\mathbb{P}_{p_r}$. We write

1375 $\mathbb{P}_p = \{ \mathbb{P}_{p_l}, \mathbb{P}_{p_r} \}$. A term s is said to belong to $\mathbb{P}$ if it appears either on the left-hand side
 1376 or on the right-hand side of some unification step in $\mathbb{P}$.

The composition of these steps starting from the empty substitution ρ_0 produces the final substitution ρ_n , which is the result of the logic program execution.

$$\begin{aligned} \text{step}_o(\mathbb{P}, p, \rho) &\rightsquigarrow \rho' \stackrel{\text{def}}{=} (\mathbb{P}_{p_l} \simeq_o \mathbb{P}_{p_r}) \rho \rightsquigarrow \rho' \\ \text{run}_o(\mathbb{P}, n) &\rightsquigarrow \rho_n \stackrel{\text{def}}{=} \bigwedge_{p=1}^n \text{step}_o(\mathbb{P}, p, \rho_{p-1}) \rightsquigarrow \rho_p \end{aligned}$$

1377 In order to simulate the execution of run_o in $\mathcal{M}$, we start by converting each $\mathcal{O}$ -term
 1378 $s \in \mathbb{P}$ is translated to a $\mathcal{M}$ -term $t \in \mathbb{Q}$. We write this translation as $\langle s \rangle \rightsquigarrow (t, m, l)$. The
 1379 implementation of the compiler is detailed in sections 4.2, 4.4 and 4.6, here we just point
 1380 out that it additionally produces a variable map m and a list of links l . The final goal
 1381 of this simulation in $\mathcal{M}$ is to generate a substitution which is a valid substitution for $\mathcal{O}$.
 1382 Therefore, we need a variable map that connects unification variables in $\mathcal{M}$ to variables
 1383 in $\mathcal{O}$ and is used to “decompile” the substitution. Decompile is noted $\langle \sigma, m, l \rangle^{-1} \rightsquigarrow \rho$.
 1384 Links are an accessory piece of information whose description is deferred to section 4.1.3.

Then we simulate each run in $\mathcal{O}$ with a run in $\mathcal{M}$ as follows:

$$\begin{aligned} \text{step}_m(\mathbb{Q}, p, \sigma, \mathbb{L}) &\rightsquigarrow (\sigma'', \mathbb{L}') \stackrel{\text{def}}{=} \\ &\sigma \mathbb{Q}_{p_l} \simeq_m \sigma \mathbb{Q}_{p_r} \rightsquigarrow \sigma' \wedge \text{progress}(\mathbb{L}, \sigma') \rightsquigarrow (\mathbb{L}', \sigma'') \\ \text{run}_m(\mathbb{P}, n) &\rightsquigarrow \rho_n \stackrel{\text{def}}{=} \\ &\mathbb{Q} \times \mathbb{M} \times \mathbb{L}_0 = \{ (t, m, l) \mid s \in \mathbb{P}, \langle s \rangle \rightsquigarrow (t, m, l) \} \\ &\bigwedge_{p=1}^n \text{step}_m(\mathbb{Q}, p, \sigma_{p-1}, \mathbb{L}_{p-1}) \rightsquigarrow (\sigma_p, \mathbb{L}_p) \\ &\langle \sigma_n, \mathbb{M}, \mathbb{L}_n \rangle^{-1} \rightsquigarrow \rho_n \end{aligned}$$

1385 By analogy with $\mathbb{P}$, we write $\mathbb{Q}_{p_l}$ and $\mathbb{Q}_{p_r}$ for the two $\mathcal{M}$ terms being unified at step p ,
 1386 and we write $\mathbb{Q}_p$ for the set $\{ \mathbb{Q}_{p_l}, \mathbb{Q}_{p_r} \}$.

1387 The step_m procedure performs the unification of the p^{th} problem in $\mathbb{Q}$ under a substi-
 1388 tution σ and a list of links $\mathbb{L}$. It consists of two sub-steps: first, a call to the meta-language
 1389 unification procedure; second, a check of the progress condition on the set of links. This
 1390 check compensates for the weaker equational theory supported by $\simeq_m$.

1391 Since we simulate unification of $\mathcal{O}$ within $\mathcal{M}$, the purpose of run_m is to take a list of
 1392 unification problems $\mathbb{P}$ and produce a substitution ρ_n for $\mathcal{O}$. This substitution is equiva-
 1393 lente to the one returned by run_o when applied to $\mathbb{P}$.

1394 The run_m procedure first compiles all terms in $\mathbb{P}$, then executes each step_m to compute
 1395 a substitution σ_n , and finally decompiles σ_n to obtain ρ_n .

1396 We claim:

Proposition 4.1.2 (Run equivalence). *$\forall \mathbb{P}, \forall n$, if all terms in $\mathbb{P}$ belongs to $\mathcal{L}$*

$$\text{run}_o(\mathbb{P}, n) \rightsquigarrow \rho \wedge \text{run}_m(\mathbb{P}, n) \rightsquigarrow \rho' \rightarrow \forall s \in \mathbb{P}, \rho s =_o \rho' s$$

1397 That is, the two executions give equivalent results if all terms in $\mathbb{P}$ are in the pattern
 1398 fragment. Moreover:

Proposition 4.1.3 (Simulation fidelity). *In the context of run_o and run_m , if all term in $\mathbb{P}$ belongs to $\mathcal{L}$ we have that $\forall p \in 1 \dots n$,*

$$\text{step}_o(\mathbb{P}, p, \rho_{p-1}) \rightsquigarrow \rho_p \leftrightarrow \text{step}_m(\mathbb{Q}, p, \sigma_{p-1}, \mathbb{L}_{p-1}) \rightsquigarrow (\sigma_p, \mathbb{L}_p)$$

In particular, this property ensures that a *failure* in the $\mathcal{O}$ execution is reflected by a failure in $\mathcal{M}$ at the *same step*. We consider this aspect important in practice, as it guarantees a strong correspondence between execution traces. As a consequence, a user can debug a logic program written in $\mathcal{O}$ by inspecting its execution trace in $\mathcal{M}$. We also claim that run_m handles terms outside $\mathcal{L}$ in the following sense:

Proposition 4.1.4 (Fidelity recovery). *In the context of run_o and run_m , if $\rho_{p-1}\mathbb{P}_p \subseteq \mathcal{L}$ (even if $\mathbb{P}_p \notin \mathcal{L}$) then*

$$\text{step}_o(\mathbb{P}, p, \rho_{p-1}) \rightsquigarrow \rho_p \leftrightarrow \text{step}_m(\mathbb{Q}, p, \sigma_{p-1}, \mathbb{L}_{p-1}) \rightsquigarrow (\sigma_p, \mathbb{L}_p)$$

In other words, if the two terms involved in a step re-enter $\mathcal{L}$, then step_m and step_o are again related, even if $\mathbb{P} \notin \mathcal{L}$ and hence proposition 4.1.3 does not apply. Indeed, the main difference between proposition 4.1.3 and proposition 4.1.4 is that the assumption of the former is purely static, it can be checked upfront. When this assumption is not satisfied, one can still simulate a logic program and have guarantees of fidelity if, at run time, decidability of higher-order unification is restored.

This property has practical relevance since in many logic programming implementations, including *Elpi*, the order in which unification problems are executed does matter. The simplest example is the sequence $F \simeq \lambda x.a$ and $F a \simeq a$: the second problem is not in $\mathcal{L}$ and has two unifiers, namely $\sigma_1 = \{ F \mapsto \lambda x.x \}$ and $\sigma_2 = \{ F \mapsto \lambda x.a \}$. If we execute the two unification problem in order, the first unification return σ_2 , making the second problem re-enter $\mathcal{L}$.

4.1.3 The solution (in a nutshell)

The role of the compiler is to translate terms from the object language into terms of the meta-language.

In particular, a term s is compiled into a term t in which every “problematic” subterm e is replaced by a fresh unification variable h . Each such variable is equipped with an additional *link*, which encodes a suspended unification problem between h and e . In this way, the unification procedure $\simeq_m$ behaves well on t , since it no longer contradicts with $=_o$ in the way it would on the original problematic subterms.

We now give a more precise definition of what we call “problematic” and “well behaved”. We use the symbol $\diamond$, which in modal logic means “possibly”, since problematic terms are characterized by a certain uncertainty.

Definition 4.1.1 ($\diamond\beta$). $\diamond\beta$ is the set of terms of the form $F x_1 \dots x_n$ such that $x_1 \dots x_n$ are distinct names (of bound variables).

An example of a $\diamond\beta$ term is the application $F x$. This term is problematic since the application node of its syntax tree cannot be used to justify a unification failure, i.e. by properly instantiating F , the term head constructor may become a λ , or a constant, or a name, or remain an application.

Definition 4.1.2 ($\diamond\eta$). $\diamond\eta$ is the set of terms $\lambda x.s$ such that $\exists \rho, \rho(\lambda x.s)$ is an η -redex.

An example of a term s in $\diamond\eta$ is $\lambda x.\lambda y.F y x$ since the substitution $\rho = \{ F \mapsto \lambda x.\lambda y.f y x \}$ makes $\rho s =_o \lambda x.\lambda y.f x y$, which is the eta-long form of f . This term is problematic since its leading λ abstraction cannot justify a unification failure against a constant f .

Definition 4.1.3 ($\diamond\mathcal{L}$). $\diamond\mathcal{L}$ is the set of terms of the form $F t_1 \dots t_n$ such that $t_1 \dots t_n$ are not distinct names.

These terms are problematic for the very same reason terms in $\diamond\beta$ are, but they cannot be handled directly by the unification of the meta language, which is only required to handle terms in $\mathcal{L}$. Still, given $s \in \diamond\mathcal{L}$ there may exist a substitution ρ such that $\rho s \in \mathcal{L}$.

We write $\mathcal{P}(t)$ for the set of sub-terms of t , and we write $\mathcal{P}(X) = \bigcup_{t \in X} \mathcal{P}(t)$ when X is a set of terms.

Definition 4.1.4 (Well behaved set). Given a set of terms X ,

$$\mathcal{W}(X) \leftrightarrow \forall t \in \mathcal{P}(X), t \notin (\diamond\beta \cup \diamond\eta \cup \diamond\mathcal{L})$$

We write $\mathcal{W}(t)$ as a short for $\mathcal{W}(\{t\})$. We claim our compiler validates the following properties:

Proposition 4.1.5 ($\mathcal{W}$ -enforcing). *Given a term s in $\mathcal{O}$,*

$$\langle s \rangle \rightsquigarrow (t, m, l) \rightarrow \mathcal{W}(t)$$

Lemma 4.1.6. *Given two terms s_1 and s_2 , if $\exists \rho, \rho s_1 =_o \rho s_2$, then*

$$\langle s_1 \rangle \rightsquigarrow (t_1, m_1, l_1) \wedge \langle s_2 \rangle \rightsquigarrow (t_2, m_2, l_2) \rightarrow t_1 \simeq_m t_2 \rightsquigarrow \sigma$$

In other words, proposition 4.1.5 states that any term s in $\mathcal{O}$ is compiled into a term t in $\mathcal{M}$ that satisfies $\mathcal{W}$, even when s itself does not satisfy $\mathcal{W}$.

Proof sketch. By proposition 4.1.5 and the (still high-level) definition of term compilation, we know that t_1 and t_2 belong to $\mathcal{W}(t)$. This means that t_1 and t_2 are the HOAS representations of the source language terms s_1 and s_2 , where problematic subterms are replaced by fresh variables. By hypothesis, there exists a substitution that makes s_1 and s_2 equal. It follows that their compiled forms, t_1 and t_2 , must unify. $\square$

This theorem states that if two terms can be unified in the source language, then their compiled forms can also be unified in the target language. Note that a compiler that maps every term to a unification variable would satisfy lemma 4.1.6, but would clearly fail to meet proposition 4.1.3.

Proposition 4.1.7 ($\mathcal{W}$ -preservation). $\forall \mathbb{Q} \ \mathbb{L} \ p \ \sigma \ \sigma',$

$$\begin{aligned} \mathcal{W}(\sigma \mathbb{Q}) \wedge \sigma \mathbb{Q}_{p_l} \simeq_m \sigma \mathbb{Q}_{p_r} \rightsquigarrow \sigma' &\rightarrow \mathcal{W}(\sigma' \mathbb{Q}) \\ \mathcal{W}(\sigma \mathbb{Q}) \wedge \text{progress}(\mathbb{L}, \sigma) \rightsquigarrow (_, \sigma') &\rightarrow \mathcal{W}(\sigma' \mathbb{Q}) \end{aligned}$$

Proposition 4.1.7 is central for proving propositions 4.1.2 and 4.1.3. Informally, it states that the problematic subterms set aside by the compiler are not reintroduced, neither by $\simeq_m$ nor by the progress relation. As a consequence, each step_m , whose definition relies on these two notions, produces terms that still belong to $\mathcal{W}$.

In the next sections of this chapter, namely sections 4.3, 4.4 and 4.6, we explain how the compiler recognizes terms in $\diamond\beta$, $\diamond\eta$, and $\diamond\mathcal{L}$, respectively. We also focus on how progress handles links, and how it preserves $\mathcal{W}$ while ensuring propositions 4.1.2 to 4.1.4.

We prove proposition 4.1.3 in sections 4.4 to 4.6 and proposition 4.1.2 only in section 4.3, where links are not present yet. In order to prove proposition 4.1.2 in the presence of links one needs to refine proposition 4.1.3 by relating the two substitutions ρ_p and σ_p : they are equivalent only by using decompilation to take into account terms in $\mathbb{L}$. Then, by induction on the number of steps, proposition 4.1.3 implies proposition 4.1.2. We discuss proposition 4.1.4 in section 4.6.

4.2 Basic compilation and simulation

In this section, we present our encoding in *Elpi* for memory management, and explain how variables in $\mathcal{O}$ and $\mathcal{M}$ are assigned, allocated, and mapped from one language to the other. At the end of the section, we introduce a first simple compiler translating terms from $\mathcal{O}$ to $\mathcal{M}$.

4.2.1 Memory map ($\mathbb{M}$) and substitution (ρ and σ)

Unification variables are identified by a (unique) memory address. The memory and its associated operations are described below:

```
typeabbrev (memory A) (list (option A)).
pred set? addr, memory A → A.
pred unset? addr, memory A → .
pred assign addr, memory A, A → memory A.
pred new memory A → addr, memory A.
```

If a memory cell is `none`, then the corresponding unification variable is unset. The operation `assign` sets an unset cell to a given value, while `new` selects the first unused address and initializes it with `none`.

Since each unification variable in $\mathcal{M}$ appears together with a scope, its assignment must be abstracted over that scope. This allows the same assignment to be instantiated in different scopes. This abstraction is captured by the `inctx` container, in particular by its binding constructor `abs`.

```
kind inctx type → type.
type abs (tmm → inctx A) → inctx A.
type val A → inctx A.
typeabbrev assignment (inctx tmm ).
typeabbrev subst (memory assignment).
```

For example, in a context where y is a bound variable and X a unification variable, the unification $X_y \simeq_m \lambda z.y$ yields the assignment $\{X_y \mapsto \lambda z.y\}$, which corresponds in *Elpi* concrete syntax to `abs y\ val (lamm z\ y)`.

Intuitively, the `abs` node represents abstraction in the meta-language, while `lamm` encodes object-level abstraction of in the HOAS of $\mathcal{M}$, in the same way `uvam` represents application of the meta-language, and `appm` encodes object-level applications in the HOAS of $\mathcal{M}$.

In contrast, assignments for variables in $\mathcal{O}$ are represented as plain terms. In particular, `fsubst` is an abbreviation for `memory tmo`. In this setting, the unification $X.y \simeq_o \lambda z.y$

1494 produces the assignment $\{X \mapsto \lambda yz.y\}$, which corresponds to `lamo y \ lamo z \ y` in *Elpi*
 1495 concrete syntax.

1496 The compiler defines a mapping between variables of the two languages, so as to
 1497 preserve the correspondence between variables in $\mathcal{O}$ and variables in $\mathcal{M}$. This mapping is
 1498 represented in the following code snippet.

```

kind ovariable type.
type ovar addr → ovariable.
kind mvariable type.
type mvar addr → arity → mvariable.
kind mapping type.
type (⇒) ovariable → mvariable → mapping.
typeabbrev mmap (list mapping).

```

1499 The variable mapping store is called `mmap` and consists of a list of `mapping`. Each mapping
 1500 relates variables of the object language (`ovvariable`) with variables of the meta-language
 1501 (`mvariable`). We associate a number to each `mvariable` in order to record its arity.

1502 **Invariant 4.2.1** (Unification-variable arity). *Each variable A in $\mathcal{M}$ has a (unique) arity*
 1503 *N and each occurrence $(uva_m \ A \ L)$ is such that L has length N .*

```

pred m-alloc ovariable, mmap, subst → mvariable, mmap, subst.
m-alloc Ov M S Mv M S :- mem M (Ov ↦ Mv), !.
1504 m-alloc Ov M S Mv [Ov ↦ Mv|M] S1 :- Mv = mvar N _, new S N S1.

```

Figure 4.4: Malloc code

1506 When a single `ovvariable` occurs multiple times with different numbers of arguments, the
 1507 compiler generates multiple mappings for it, on a first approximation, and then ensures the
 1508 mapping is bijective by introducing η -link; this detail is discussed in section section 4.5.

1509 It is worth examining the code of the application of substitution to terms in $\mathcal{M}$. This
 1510 is called `deref` and implements definition 2.2.3. Notice how assignments are moved to the
 1511 current scope, i.e. the `abs`-bound variables are renamed with the names in the scope of
 1512 the unification variable occurrence.

```

pred deref subst, tmm → tmm .
deref _ (conm C) (conm C).
deref S (appm A) (appm B) :- map (deref S) A B.
deref S (lamm F) (lamm G) :-
1513   pi x \ deref S x x => deref S (F x) (G x).
deref S (uvam N L) R :- set? N S A,
   move A L T, deref S T R.
deref S (uvam N A) (uvam N B) :- unset? N S,
   map (deref S) A B.
pred move assignment,
   list tmm → tmm .
move (abs Bo) [H|L] R :-
   move (Bo H) L R.
move (val A) [] A.

```

1514 Note that `move` relies crucially on invariant invariant 4.2.1, which ensures that all
 1515 occurrences of a unification variable are applied to argument lists of the same length. As
 1516 a consequence, these occurrences share the same simple type at the meta-level, and the
 1517 number of `abs` nodes in the resulting assignment matches this length.

4.2.2 Links ($\mathbb{L}$)

As mentioned in section 4.1.3, the compiler replaces terms in $\Diamond\eta$, $\Diamond\beta$, and $\Diamond\mathcal{L}$ with fresh variables linked to the problematic terms. Terms in $\Diamond\beta$ do not need a link since $\mathcal{M}$ variables faithfully represent the problematic term thanks to their scope.

```
kind baselink type.
type ( $=_\eta$ )  $\text{tm}_m \rightarrow \text{tm}_m \rightarrow \text{baselink}$ .
type ( $=_\beta$ )  $\text{tm}_m \rightarrow \text{tm}_m \rightarrow \text{baselink}$ .
typeabbrev links (list (inctx baselink)).
```

The right-hand side of a link, the problematic term, can occur under binders. To accommodate this situation, the compiler wraps `baselink` using the `inctx` container.

Invariant 4.2.2 (Link left hand side). *The left-hand side of a suspended link is a variable.*

New links are suspended by construction. If the left-hand side is assigned during a step, then the link is considered for progress and possibly eliminated. This is discussed in section 4.4 and section 4.6.

In the examples we represent links as equations under a context. The equality sign is subscripted with the kind of `baselink`. For example $x \vdash A_x =_{\mathcal{L}} F_x a$ corresponds to:

```
abs x \ val (uvam A [x] =ℒ appm [uvam F [x], conm "a"])
```

4.2.3 Compilation

The simple compiler described in this section, and shown in figure 4.5, serves as a basis for the extensions presented in sections 4.3, 4.4 and 4.6. Its entry point (4.5d) first performs β -normalisation of the input term s in $\mathcal{O}$, and then compiles s into its corresponding HOAS representation in $\mathcal{M}$.

Together with the term s , the compiler is initially given a memory map, a list of links, and a substitution. These three components act as accumulators and are progressively updated during the compilation phase.

The auxiliary predicate `comp`, shown in figure 4.5a, performs the recursive traversal of the $\mathcal{O}$ term. Its current implementation is partial and will be extended in the following sections to handle problematic subterms. At this stage, the list of links is left unchanged. Rules r_c and $r_@$ in figure 4.5a implement the straightforward translation between the two term representations.

Rule r_v handles variables in the object language. A variable v_o in $\mathcal{O}$ is mapped to a variable in $\mathcal{M}$ using the memory map. If v_o is not yet present in the variable mapping, a fresh entry is created and the substitution is updated with a pointer for this new variable. This pointer is initially set to `none`. If, instead, v_o already occurs in the mapping and is associated with a meta-language variable v_m in $\mathcal{M}$, then v_m is simply returned. All these operations are delegated to the `m-alloc` procedure, which performs the actual allocation.

The λ -case is handled by rule r_λ , which invokes the auxiliary function `comp-mlam`. This procedure recursively calls `comp` in an extended context containing two local variables, `x` and `y`, representing respectively abstractions at the object and meta-level.

Finally, the auxiliary function `close-links` recursively wraps the abstractions in the links in an additional `abs` node binding. In this way, links generated deep inside the compiled term can be lifted outside their original binding context.

```

pred comp tmo , mmap, links, subst → tmm , mmap, links, subst.
comp (cono C) M1 L1 S1 (conm C) M1 L1 S1.           % rc
comp (lamo F) M1 L1 S1 (lamm G) M2 L2 S2 :-          % rλ
  comp-lam F M1 L1 S1 G M2 L2 S2.
comp (uvao A) M1 L1 S1 (uvam B []) M2 L1 S2 :-       % rv
  m-alloc (ovar A) M1 S1 (mvar B 0) M2 S2.
comp (appo A) M1 L1 S1 (appm B) M2 L2 S2 :-         % r@
  fold6 comp A M1 L1 S1 B M2 L2 S2.

```

(a) Recursive compilation on terms

```

pred comp-lam (tmo → tmo) , mmap, links, subst →
  (tmm → tmm) , mmap, links, subst.
comp-lam F M1 L1 S1 G M2 L3 S2 :-
  pi x y \ (pi M L S \ comp x M L S y M L S) =>
    comp (F x) M1 L1 S1 (G y) M2 (L2 y) S2,
  close-links L2 L3.

```

(b) Compilation of λ -abstractions

```

pred close-links (tmm → links) → links.
close-links (v \ [X v | L v]) [abs X | R] :- close-links L R.
close-links (_ \ []) [].

```

(c) Close links

```

pred compile tmo , mmap, links, subst → tmm , mmap, links, subst.
compile F M1 L1 S1 G M2 L2 S2 :-
  beta-normal F F', comp F' M1 L1 S1 G M2 L2 S2.

```

(d) Entry point

Figure 4.5: First-order compiler

Lemma 4.2.3 ($\mathcal{W}$ -partial-enforcement). $\forall s, \langle s \rangle \rightsquigarrow (t, m, l)$, if $\mathcal{W}(s)$ then $\mathcal{W}(t)$.

Proof. By the definition of `comp`, the compilation procedure is a simple mapper from terms in $\mathcal{O}$ to terms in $\mathcal{M}$. By hypothesis, $\mathcal{W}(s)$, therefore $\mathcal{W}(t)$. $\square$

4.2.4 Execution

A step in $\mathcal{M}$ consists in unifying two terms and reconsidering all links for progress. If either of these tasks fails, we consider the entire step to fail. It is at this granularity that we can relate steps in the two languages.

```

pred stepm tmm , tmm , links, subst → links, subst.
stepm T1 T2 L1 S1 L2 S3 :-
  (T1 ≈m T2) S1 S2,
  progress L1 S2 L2 S3.

```

Reconsidering links is a fixpoint process because the progress of a link can update the substitution, which may then enable another link to progress.

```

pred progress links, subst → links, subst.
progress L S L3 S3 :-
  progress1 L L1 S S1,
  occur-check-links L1,
  scope-check L1 L2 S1 S2
  if (L = L2, S = S2) (L3 = L2, S3 = S2)
  (progress L2 S2 L3 S3).

```

Progress In the base compilation scheme, `progress1` is the identity function on both the links and the substitution, so the fixpoint trivially terminates. Sections section 4.4 and section 4.6 add rules to `progress1` and explain why they do not hinder termination.

Occur check Since compilation moves problematic terms out of the sight of $\approx_m$, that procedure can only perform a partial occur check. For example, the unification problem $X \approx_m f Y$ cannot generate a cyclic substitution alone, but should be disallowed if $\mathbb{L}$ contains a link like $\vdash Y =_{\eta} \lambda z. X_z$: we don't know yet if Y will feature a lambda in head position, but we surely know it contains X . The procedure `occur-check-links` is in charge of performing this check that is needed in order to guarantee proposition 4.1.3.

Scope check The predicate `scope-check` replaces each link of the form $\Gamma \vdash X_{x_1 \dots x_n} =_{\eta} t$ with a link $x_1 \dots x_n \vdash X_{x_1 \dots x_n} =_{\eta} t'$ whenever $\{x_1 \dots x_n\} \subseteq \Gamma$. The term $t' = \lambda x. Y_{x_1 \dots x_n} x$ is obtained after executing $\lambda x. Y_{x_1 \dots x_n} x \approx_m t$ using a fresh Y . This unification ensures that t' cannot contain variables in $\Gamma / \{x_1 \dots x_n\}$, and if it fails then progress hence `stepm` fails. In turn this grants fidelity in the case where the lhs of an η -link is pruned of a name that occurs (rigidly) in t : links represent suspended unification problems *that may succeed*.

4.2.5 Substitution decompilation

Decompiling the substitution involves three steps.

First and foremost, problematic terms stored in $\mathbb{L}$ have to be moved back into the game: a suspended link must be turned into a valid assignment. This operation is possible thanks to invariant 4.2.2, and that no link causes an occur-check (section 4.2.4) and the fact that $\mathbb{L}$ is duplicate-free (see section 4.5).

The second step allocates in the memory of $\mathcal{O}$ new variables used to implement the pruning of higher-order variables. For example, $F.x y = F.x z$ requires allocating a variable G in order to express the assignment $F_{ab} \mapsto G_a$.

The final step is to decompile each assignment in the $\mathcal{M}$ substitution. The `val` node carries a term that is easy to decompile since $\mathbb{M}$ is a bijection. Since the `lamo` node carries no additional information (other than the function body), each `abs` node can be decompiled to a `lamo` one.

However, if the `lamo` node was carrying more information, as is the case for *Rocq* where it holds the type of the bound variable, one would need to store this piece of information in the memory map, were we store the arity (that is a very simple function type).

Proposition 4.2.4 (Compilation round trip). *If $\langle s \rangle \rightsquigarrow (t, m, l)$ and $l \in \mathbb{L}$ and $m \in \mathbb{M}$ and $\sigma = \{A \mapsto t\}$ and $X \mapsto A^n \in \mathbb{M}$ then $\langle \sigma, \mathbb{M}, \mathbb{L} \rangle^{-1} \rightsquigarrow \rho$ and $\rho X =_o \rho s$.*

4.2.6 Definition of $\simeq_o$ and its properties

We already have all the ingredients to show how to implement the simulation of $\simeq_o$ in the meta-language.

```

pred ( $\simeq_o$ ) tmo , tmo → fsubst.
(A  $\simeq_o$  B) F :-
  compile A [] [] A' M1 L1 S1,
  compile B M1 L1 S1 B' M2 L2 S2,
  stepm A' B' L2 S2 L3 S3,
  decompile M2 L3 S3 [] F.

```

So far the compiler is very basic. It does not really enforce that the terms passed to `stepm` are in $\mathcal{W}$, and indeed makes no use of the higher-order capabilities of the meta language (all generated variables have an empty scope). Still, we can prove that $\simeq_o$ is a good “first-order” unification algorithm if the input already happens to be in $\mathcal{W}$. Later, when the compiler will ensure proposition 4.1.5, the proof will be refined to cover for the new cases.

Theorem 4.2.5 (Properties of $\simeq_o$ in $\mathcal{W}$). *The implementation of $\simeq_o$ above is a good unification for $=_o$ (as per proposition 4.1.1) in the domain $\mathcal{W}$.*

Proof sketch. In this setting, $=_m$ is as strong as $=_o$ on ground terms. What we have to show is that whenever two different $\mathcal{O}$ terms can be made equal by a substitution ρ , we can produce this ρ by finding a σ via $\simeq_m$ on the corresponding $\mathcal{M}$ terms and by decompiling it. If we look at the syntax of $\mathcal{O}$, the only interesting case is `uvao X  $\simeq_o$  s`. In this case, after compilation, we have `uvam Y []  $\simeq_m$  t` that succeeds with $\sigma = \{Y \mapsto t\}$ and σ is decompiled to $\rho = \{X \mapsto s\}$ by proposition 4.2.4. $\square$

Theorem 4.2.6 (Fidelity in $\mathcal{W}$). *Proposition 4.1.2 and proposition 4.1.3 hold if $\mathcal{W}(\mathbb{P})$.*

Proof sketch. Since `progress1` is a no-op, step_o and step_m coincide. By theorem 4.2.5, the unification relation $\simeq_m$ is equivalent to $\simeq_o$ on terms in $\mathcal{W}$. $\square$

4.2.7 Notational conventions

In the following sections we use the following notation to represent the behavior of the compiler. $\mathbb{P}$ represent the input unification problems in $\mathcal{O}$ while $\mathbb{Q}$ their corresponding compiled version with memory mapping $\mathbb{M}$ and links $\mathbb{L}$. For example:

$$\begin{aligned}\mathbb{P} &= \{ \quad p_1 \quad \simeq_o \quad p_2 \quad \quad p_3 \quad \simeq_o \quad p_4 \quad \} \\ \mathbb{Q} &= \{ \quad t_1 \quad \simeq_m \quad t_2 \quad \quad t_3 \quad \simeq_o \quad t_4 \quad \} \\ \mathbb{M} &= \{ \quad X_1 \mapsto A_1^n \quad X_2 \mapsto A_2^m \quad \} \\ \mathbb{L} &= \{ \quad \Gamma \vdash a =_\eta b \quad \}\end{aligned}$$

We index each sub-problem, sub-mapping, and sub-link with its position starting from 1 and counting from left to right. For example, $\mathbb{Q}_2$ corresponds to the $\mathcal{M}$ problem $t_3 \simeq_m t_4$.

As we deal with each category of problematic terms we weaken the assumptions of theorem 4.2.6 using the following definitions:

Definition 4.2.1 ($\mathcal{W}_\beta$). $\mathcal{W}_\beta(X) \leftrightarrow \forall t \in \mathcal{P}(X), t \notin (\diamond\eta \cup \diamond\mathcal{L})$

Definition 4.2.2 ($\mathcal{W}_{\beta\eta}$). $\mathcal{W}_{\beta\eta}(X) \leftrightarrow \forall t \in \mathcal{P}(X), t \notin \diamond\mathcal{L}$

4.3 Handling of $\diamond\beta$

The basic compiler given in the previous section is unable to make the following higher-order unification problem succeed.

$$\begin{aligned}\mathbb{P} &= \{ \quad \lambda x.(f.(X\ x)\ a) \quad \simeq_o \quad \lambda x.(f.\ x\ a) \quad \} \\ \mathbb{Q} &= \{ \quad \lambda x.(f.(A\ x)\ a) \quad \simeq_m \quad \lambda x.(f.\ x\ a) \quad \} \\ \mathbb{M} &= \{ \quad X \mapsto A^0 \quad \}\end{aligned}$$

The unification problem $\mathbb{Q}_1$ fails while trying to unify $A\ x$ and x , which is equivalent to `appm [uvam A [], x]` versus `x`. In order to exploit the higher-order unification algorithm of the meta language, we compile the $\mathcal{O}$ term $X\ x$ into the $\mathcal{M}$ term A_x .

We add the following rule before rule $r_\oplus$ in figure 4.5a, where `pattern-fragment` is a predicate checking if a list of terms is a list of distinct names.

```
comp (appo [uvao A|Ag]) M1 L S1 (uvam B Bg) M2 L S2 :-
  pattern-fragment Ag, !,
  fold6 comp Ag M1 L S1 Bg M1 L S1,
  len Ag Arity,
  m-alloc (ovar A) M1 S1 (mvar B Arity) M2 S2.
```

Note that compiling `Ag` cannot create new mappings nor links, since `Ag` is made of bound variables, and the hypothetical rule in figure 4.5b loaded by `comp-mlam` grants this property.

Decompilation No change to `commit-link` since no link is added.

1638 **Progress** No change to **progress** since no link is added.

1639 **Lemma 4.3.1** (Properties of $\simeq_o$ in $\mathcal{W}_\beta$). *Given the updated compilation scheme, the $\simeq_o$*
 1640 *of section 4.2.6 is a good unification (as per proposition 4.1.1) for $=_o$ in the domain $\mathcal{W}_\beta$.*

1641 *Proof sketch.* If we look at the $\mathcal{O}$ terms, there is one more interesting case, namely
 1642 $\text{app}_o \text{ [uva}_o \text{ x|w]} \simeq_o s$ when w is a list of distinct names compiled to $\bar{w}$. In this case the
 1643 $\mathcal{M}$ problem is $Y_{\bar{w}} \simeq_m t$ that succeeds with $\sigma = \{Y_{\bar{y}} \mapsto t[\bar{w}/\bar{y}]\}$, which in turn is decompiled
 1644 to $\rho = \{Y \mapsto \lambda \bar{y}.s[\bar{w}/\bar{y}]\}$. Thanks to rule r_{β_l} in figure 4.2 we have $(\lambda \bar{y}.s[\bar{w}/\bar{y}]) \bar{w} =_o s$. $\square$

1645 **Lemma 4.3.2** ($\mathcal{W}$ -partial-enforcement). $\forall s, \langle s \rangle \rightsquigarrow (t, m, l)$, if $\mathcal{W}_\beta(s)$ then $\mathcal{W}(t)$.

1646 *Proof sketch.* We need to extend the proof lemma 4.2.3, since we need to consider the
 1647 case of terms in $\diamond\beta$. Terms in $\diamond\beta$ are captured by the new rule for **comp** at the beginning
 1648 of this section. The compiled term is a variable and its arguments are distinct names,
 1649 therefore the returned term is also in $\mathcal{W}_\beta$. $\square$

1650 In other words the compiler eliminates all subterms that are in $\diamond\beta$.

1651 **Theorem 4.3.3** (Fidelity in $\mathcal{W}_\beta$). *Proposition 4.1.2 and proposition 4.1.3 hold if $\mathcal{W}_\beta(\mathbb{P})$.*
 1652

1653 *Proof sketch.* Thanks to lemma 4.3.2, $\simeq_m$ is as powerful as $\simeq_o$ in $\mathcal{W}_\beta$, as well as in $\mathcal{W}$ by
 1654 theorem 4.2.6. $\square$

1655 4.4 Handling of $\diamond\eta$

1656 A term $\lambda x.t x$ is said to be the η -redex of t if x does not occur free in t , and conversely
 1657 we say that t is the η -contraction of $\lambda x.t x$. The equational theory of $\mathcal{O}$ identifies these
 1658 terms, but the current compilation scheme does not, as shown by the following example:
 1659 while $\lambda x.X x \simeq_o f$ does admit the solution $\rho = \{X \mapsto f\}$, the corresponding problem in $\mathbb{Q}$
 1660 does not.

$$\begin{aligned} \mathbb{P} &= \{ \lambda x.(X x) \simeq_o f \} \\ \mathbb{Q} &= \{ \lambda x.A_x \simeq_m f \} \\ \mathbb{M} &= \{ X \mapsto A^1 \} \end{aligned}$$

1661 The reason is that **lam_m** **x** **uva_m** **A** **[x]** and **con_m** **"f"** start with different term con-
 1662 structors.

1663 In order to guarantee proposition 4.1.2, we detect lambda abstractions that can dis-
 1664 appear by η -contraction (section 4.4.1), and we modify the compiler so that it generates
 1665 fresh unification variables in their place and delegates the problematic term t to $\mathbb{L}$ (sec-
 1666 tion 4.4.2) by replacing t with a unification variable. The compilation of the problem $\mathbb{P}$
 1667 above is refined to:

$$\begin{aligned} \mathbb{P} &= \{ \lambda x.(X x) \simeq_o f \} \\ \mathbb{Q} &= \{ A \simeq_m f \} \\ \mathbb{M} &= \{ X \mapsto B^1 \} \\ \mathbb{L} &= \{ \vdash A =_\eta \lambda x.B_x \} \end{aligned}$$

As per invariant 4.2.2 the η -link left-hand side is a variable while the right-hand side is a term in $\Diamond\eta$ that has the following property:

Invariant 4.4.1 (η -link rhs). *The rhs of any η -link has the shape $\lambda x.t$ and t is in $\mathcal{W}$.*

Each η -link is kept in the link store $\mathbb{L}$ during execution and is activated under specific conditions, in particular, when invariant 4.2.2 or invariant 4.4.1 are no more true, or the rhs is for sure a term that cannot be η -contracted. Activation is implemented in section 4.4.3 by extending the `progress1` predicate defined in section 4.2.4.

4.4.1 Detection of $\Diamond\eta$

When compiling a term t , we need to determine if any subterm $s \in \mathcal{P}(t)$ that is of the form $\lambda x.r$ can be an η -redex, i.e. if there exists a substitution ρ such that $\rho(\lambda x.r) =_o s$. The detection of lambda abstractions that can “disappear” is not as trivial as it may seems. Here a few examples:

$$\begin{array}{ll} \lambda x.f(Ax) & \in \Diamond\eta \quad \rho = \{ A \mapsto \lambda x.x \} \\ \lambda x.f(Ax).x & \in \Diamond\eta \quad \rho = \{ A \mapsto \lambda x.a \} \\ \lambda x.f.x(Ax) & \notin \Diamond\eta \\ \lambda x.\lambda y.f(Ax).(Byx) & \in \Diamond\eta \quad \rho = \{ A \mapsto \lambda x.x ; B \mapsto \lambda y.\lambda x.y \} \end{array}$$

The first two examples are straightforward, and illustrate how a unification variable can either expose or hide a bound name in its scope, thereby turning the resulting term into an η -redex or not.

The third example shows that when a variable occurs outside the scope of a unification variable, it cannot be erased and may therefore prevent the term from being an η -redex.

The last example highlights the recursive nature of the check we need to implement. The term begins with a spine of two λ -abstractions, so the whole term belongs to $\Diamond\eta$ if and only if the inner term $\lambda y.f(Ax).(Byx)$ also belongs to $\Diamond\eta$. If this is the case, it could η -contract to $f(Ax)$, making $\lambda x.f(Ax)$ a potential η -redex.

We now describe how terms in $\Diamond\eta$ are detected.

Definition 4.4.1 (*may-contract-to*). A β -normal term s *may-contract-to* a name x if there exists a substitution ρ such that $\rho s =_o x$.

Lemma 4.4.2. *A β -normal term $s = \lambda x_1 \dots x_n.t$ may-contract-to x only if one of the following three conditions holds:*

1. $n = 0$ and $t = x$;
2. t is the application of x to a list of terms l and each l_i may-contract-to x_i (e.g. $\lambda x_1 \dots x_n.x x_1 \dots x_n =_o x$) ;
3. t is a unification variable with scope W , and for any $v \in \{x, x_1 \dots x_n\}$, there exists a $w \in W$, such that w may-contract-to v (if $n = 0$ this is equivalent to $x \in W$).

Proof sketch. Since our terms are in β -normal form the only rule that can play a role is r_{η} in figure 4.2, and note that that rule uses `beta` to add an argument to an application or to the scope of a variable. If the term s is not exactly x (case 1) it can only be an

```

comp (lamo F) M1 L1 S1 (uvam A Scope) M2 L3 S3 :-
  maybe-eta (lamo F) [], !,
  new S1 A S2,                                % allocates a fresh variable A in S1
  comp-lam F M1 L1 S2 G M2 L2 S3,
  get-scope (lamm G) Scope,                    % returns the free names in lamm
  L3 = [val (uvam A Scope =η lamm G) | L2].

```

Figure 4.6: Compilation of $\Diamond\eta$ terms

1703 η -redex of x , or a unification variable that can be assigned to x , or a combination of both.
 1704 If s begins with a lambda, then the lambda can only disappear by η contraction. In that
 1705 case, the term t under the spine of binders $x_1 \dots x_n$ can either be x applied to terms that
 1706 *may-contract-to* these variables (case 2), or a unification variable that can be assigned to
 1707 the application $x x_1 \dots x_n$ (case 3). $\square$

1708 **Definition 4.4.2** (*occurs-rigidly*). A name x *occurs-rigidly* in a β -normal term t , if for
 1709 any substitution ρ , x is a subterm of ρt

1710 In other words, x *occurs-rigidly* in t if it occurs in t outside of the scope of a unification
 1711 variable X ; otherwise, an instantiation of X can make x disappear from t . Note that
 1712 η -contracting t cannot make x disappear, since x is not a locally bound variable inside t .

1713 We can now describe the implementation of $\Diamond\eta$ detection:

1714 **Definition 4.4.3** (*maybe-eta*). Given a β -normal term $s = \lambda x_1 \dots x_n. t$, *maybe-eta* s holds
 1715 if any of the following holds:

- 1716 1. t is a constant c or a name y applied to the arguments $l_1 \dots l_m$ such that $m \geq n$
 1717 and for every i such that $m - n < i \leq m$ the term l_i *may-contract-to* x_i , and no x_i
 1718 *occurs-rigidly* in $l_1 \dots l_{m-n} y$;
- 1719 2. t is a unification variable with scope W and for each x_i there exists a $w_j \in W$ such
 1720 that w_j *may-contract-to* x_i .

1721 **Lemma 4.4.3** ($\Diamond\eta$ detection). *If t is a β -normal term and $t \in \Diamond\eta$ then maybe-eta t holds.*

1722 *Proof sketch.* Follows from definition 4.4.2 and lemma 4.4.2 $\square$

1723 Remark that the converse of lemma 4.4.3 does not hold: *maybe-eta* performs an over-
 1724 approximation of terms that are in $\Diamond\eta$. There exists, in fact, a term t satisfying the
 1725 criteria (1) of definition 4.4.3 that is not in $\Diamond\eta$, i.e. there exists no substitution ρ such
 1726 that ρt is an η -redex. A simple counter example is $\lambda x. f(Ax).(Ax)$ since x does not
 1727 *occur-rigidly* in the first argument of f , and the second argument of f *may-contract-to*
 1728 x . In other words Ax may either use or discard x , but our analysis does not take into
 1729 account that *the same term* cannot have two contrasting behaviors.

1730 As we will see in the rest of this section, this is not a problem since it does not break
 1731 proposition 4.1.2 nor proposition 4.1.3.

1732 4.4.2 Compilation and decompilation

1733 **Compilation** In order to take into account the compilation of $\Diamond\eta$ terms, we insert the
 1734 rule in figure 4.6 just before rule r_λ in figure 4.5a.

Whenever a term of the form $\text{lam}_o \ F$ is detected to belong to $\Diamond\eta$, it is compiled into $\text{lam}_m \ F1$ and replaced by a fresh variable A . This variable sees all names that are free in $\text{lam}_m \ F1$, and it is linked to $\text{lam}_m \ F1$ through a η -link. The invariant invariant 4.2.2 is satisfied for this link. Moreover:

Corollary 4.4.4. *The rhs of any η -link has exactly one lambda abstraction, hence the rule above respects invariant 4.4.1.*

Proof sketch. By contradiction, suppose that the rule above is applied and that the rhs of the link is $\lambda x.\lambda y.t$, where x and y occur in t . If *maybe-eta* $\lambda y.t$ holds then the recursive call to *comp* (made by *comp-mlam*) must have put a fresh variable in its place, so this case is impossible. Otherwise, if *maybe-eta* $\lambda y.t$ does not hold, then also *maybe-eta* $\lambda x.\lambda y.t$ does not hold either, contradicting the assumption that the rule was applied. $\square$

Lemma 4.4.5 ($\mathcal{W}$ -partial-enforcement). $\forall s, \langle s \rangle \rightsquigarrow (t, m, l)$, if $\mathcal{W}_{\beta\eta}(s)$ then $\mathcal{W}(t)$.

Proof sketch. We need to extend the proof of lemma 4.4.5, since we need to consider the case of terms $\Diamond\eta$. By lemma 4.4.3, we know that any $\Diamond\eta$ term is captured by *maybe-eta*, therefore the rule in figure 4.6 is used. The output is a variable, where in $\mathcal{L}$, therefore, the new terms is $\mathcal{W}$. $\square$

Decompilation The decompilation of a η -link is performed by unifying the lhs with the rhs. Note that this unification never fails, since the lhs is a flexible term not appearing in any other η -link (by definition 4.4.5).

4.4.3 Progress

η -links are meant to delay the unification of “problematic” terms until we know for sure if their head lambdas can (or cannot) be η -contracted.

Definition 4.4.4 (η -progress-lhs). A link $\Gamma \vdash X =_{\eta} t$ is removed from $\mathbb{L}$ when X becomes in a substitution σ . Let $y \in \Gamma$. There are two cases:

1. if $\sigma X = a$ or $\sigma X = y$ or $\sigma X = f a_1 \dots a_n$ we unify the η -redex of X with t , that is we run $(\lambda x.X x \simeq_m t) \sigma$
2. if $\sigma X = \lambda x.s$ we run $(X \simeq_m t) \sigma$.

Definition 4.4.5 (η -progress-deduplicate). A link $\Gamma \vdash X_{\vec{s}} =_{\eta} T$ is removed from $\mathbb{L}$ when another link $\Delta \vdash X_{\vec{r}} =_{\eta} T'$ is in $\mathbb{L}$. By invariant 4.2.1 the length of $\vec{s}$ and $\vec{r}$ is the same; hence we can move the term T' from Δ to Γ by renaming its bound variables, i.e. $T'' = T'[\vec{r}/\vec{s}]$. We then run $T \simeq_m T''$ (under the context Γ).

Lemma 4.4.6. *Let $\lambda x.t$ be the rhs of a η -link, then $\mathcal{W}(t)$ holds.*

Proof sketch. By lemma 4.4.5. $\square$

Lemma 4.4.7. *Given a η -link $\Gamma \vdash X =_{\eta} \lambda x.t$, the unification done by η -progress-lhs is between terms in $\mathcal{W}$.*

1770 *Proof sketch.* Let σ be a substitution such that $\mathcal{W}(\sigma X)$ (since $\simeq_m$ preserves $\mathcal{W}$). By
 1771 invariant 4.4.1, we have $\mathcal{W}(t)$. If $\sigma X = \lambda x.r$, then r is unified with t and both terms are
 1772 in $\mathcal{W}$. Otherwise, $\lambda x.Xx$ is unified with $\lambda x.t$, and Xx and t are both in $\mathcal{W}$. $\square$

1773 **Lemma 4.4.8.** *The unification done by η -progress-deduplicate is between terms in $\mathcal{W}$.*

1774 *Proof.* Given two η -links $\Gamma_1 \vdash X =_\eta \lambda x.t$ and $\Gamma_2 \vdash Y =_\eta \lambda x.t'$ the unification is performed
 1775 between t and t' . By lemma 4.4.5 both terms are in $\mathcal{W}$. $\square$

1776 **Lemma 4.4.9.** *The progress of η -link guarantees proposition 4.1.7*

1777 *Proof sketch.* By lemmas 4.4.7 and 4.4.8, every unification performed by the activation of
 1778 a η -link is between terms in $\mathcal{W}$. $\square$

1779 **Lemma 4.4.10.** *progress terminates.*

1780 *Proof sketch.* Rules definition 4.4.4 and definition 4.4.5 remove one link from $\mathbb{L}$, hence they
 1781 cannot be applied indefinitely. Moreover each rule only relies on terminating operations
 1782 such as $\simeq_m$, η -contraction, η -redex, relocation (a recursive copy of a finite term). $\square$

1783 **Theorem 4.4.11** (Fidelity in $\mathcal{W}_{\beta\eta}$). *Given a list of unification problems $\mathbb{P}$ such that
 1784 $\mathcal{W}_{\beta\eta}(\mathbb{P})$, if the memory map is bijective then the introduction of η -link guarantees propo-
 1785 sition 4.1.3.*

1786 *Proof sketch.* We want to prove that step_o succeeds iff step_m succeeds, where step_m is
 1787 made by a unification step u and a link progression p . Without loss of generality we
 1788 consider a unification problem in $\mathcal{O}$ of the form $s_1 \simeq_o s_2$ where $s_1 \in \Diamond\eta$ and is compiled to
 1789 a variable X with the accessory link $\Gamma \vdash X =_\eta \lambda x.t_1$. The unification u always succeeds,
 1790 since X is a fresh variable, we only have to consider the link progression p . Two cases
 1791 should be analyzed:

1792 $s_2 \in \Diamond\eta$, in this case, the unification problem becomes $X \simeq_m Y$ with the extra link
 1793 $\vdash Y =_\eta \lambda x.t_2$. After the unification of X and Y , η -progress-deduplicate is fired and,
 1794 once the lambdas are crossed, t_1 and t_2 are unified. This unification succeeds iff
 1795 $s_1 \simeq_o s_2$ succeeds by theorem 4.3.3.

1796 $s_2 \notin \Diamond\eta$, in this case s_2 is compiled to t_2 and the unification step u assigns t_2 to
 1797 X and triggers η -progress-lhs. If s_1 and s_2 are equal thanks to r_{η_l} in figure 4.2
 1798 then progression p unifies $\lambda x.t_1$ with the η -redex of t_2 namely $\lambda x.t_2 x$, otherwise they
 1799 are equal thanks to rule $r_{\lambda\lambda}$ in figure 4.2 and the link progress p unifies $\lambda x.t_1$ with
 1800 t_2 . Alternatively, if s_1 are different s_2 it cannot be because of the λ constructor in
 1801 the head of t_1 , since rule $r_{\lambda\lambda}$ in figure 4.2 or rule r_{η_l} in figure 4.2 move under it,
 1802 eventually finding two different subterms s'_1 and s'_2 . By theorem 4.3.3, $\simeq_m$ fails on
 1803 terms t'_1 and t'_2 even if η -progress-lhs η -expands t_2 .

1804 $\square$

1805 **Corollary 4.4.12** (Properties of $\simeq_o$ in $\mathcal{W}_{\beta\eta}$). *Given the updated compilation scheme,
 1806 procedure $\simeq_o$ of section 4.2.6 is a good unification (as per proposition 4.1.1) for $=_o$ in the
 1807 domain $\mathcal{W}_{\beta\eta}$.*

Example of η -progress-lhs The example at the beginning of section 4.4, once $\sigma = \{ A \mapsto f \}$, triggers η -progress-lhs since the link becomes $\vdash f =_\eta \lambda x. B_x$ and the lhs is a constant. This rule runs $\lambda x. f x \simeq_m \lambda x. B_x$, resulting in $\sigma = \{ A \mapsto f ; B_x \mapsto f \}$. Since X is mapped to B and B_x is decompiled into $\lambda x. f x$, by the definition of η -link decompilation σ is decompiled to $\rho = \{ X \mapsto \lambda x. f x \}$.

4.5 Making $\mathbb{M}$ a bijection

We want to allow for partial application of functions. As a consequence the same unification variable X can be used multiple times with different arities. When it is the case the memory map $\mathbb{M}$ generated by the compiler is not a bijection. For example:

$$\begin{aligned} \mathbb{P} &= \{ \lambda x. (X x) \simeq_o f \quad X \simeq_o \lambda x. a \} \\ \mathbb{Q} &= \{ A \simeq_m f \quad C \simeq_m \lambda x. a \} \\ \mathbb{M} &= \{ X \mapsto C^0 \quad X \mapsto B^1 \} \\ \mathbb{L} &= \{ x \vdash A =_\eta \lambda y. B_y \} \end{aligned}$$

in the unification problems $\mathbb{P}$ above, X is used with arity 1 in $\mathbb{P}_1$ and with arity 0 in $\mathbb{P}_2$. In order to preserve invariant 4.2.1 the compiler did generate two entries in $\mathbb{M}$ for X , namely B and C . However, there is no connection between these two variables and any incompatible assignments to them would not be detected, in turn breaking proposition 4.1.3. To address this issue, we post-process $\mathbb{M}$ to ensure the following property:

Proposition 4.5.1 ($\mathbb{M}$ is a bijection). *After compilation, for each $\mathcal{O}$ -variable X in $\mathbb{P}$ and for each $\mathcal{M}$ -variable A in $\mathbb{Q}$ there is exactly one entry $X \mapsto A^n$ in $\mathbb{M}$.*

Note that running $\simeq_m$ may require allocating new variables in $\mathcal{M}$ as explained in section 4.2.5. The property above ignores these variables since they are fresh and have no corresponding variable in $\mathcal{O}$.

The core procedure of this post-processing step is *align-arity* that is iterated by *map-deduplication*:

Definition 4.5.1 (*align-arity*). Given two mappings $m_1 : X \mapsto A^m$ and $m_2 : X \mapsto C^n$ where $m < n$ and $d = n - m$, *align-arity* $m_1 \ m_2$ generates the following d links, one for each i such that $0 \leq i < d$,

$$x_0 \dots x_{m+i} \vdash B_{x_0 \dots x_{m+i}}^i =_\eta \lambda x_{m+i+1}. B_{x_0 \dots x_{m+i+1}}^{i+1}$$

where B^i is a fresh variable of arity $m + i$, and $B^0 = A$ as well as $B^d = C$.

The intuition is that we η -expand the occurrence of the variable with lower arity to match the higher arity. Since each η -link can add exactly one lambda, we need as many links as the difference between the two arities.

Definition 4.5.2 (*map-deduplication*). For all mappings $m_1, m_2 \in \mathbb{M}$ such that $m_1 : X \mapsto A^m$ and $m_2 : X \mapsto C^n$ and $m < n$ we remove m_1 from $\mathbb{M}$ and add to $\mathbb{L}$ the result of *align-arity* $m_1 \ m_2$.

Theorem 4.5.2 (Fidelity with *map-deduplication*). *Given a list of unification problems $\mathbb{P}$, such that $\mathcal{W}_{\beta\eta}(\mathbb{P})$, if $\mathbb{P}$ contains the same $\mathcal{O}$ -variable used at different arities, then *map-deduplication* guarantees proposition 4.1.3*

Proof sketch. Let X be a $\mathcal{O}$ -variable appearing in two different subterms s_1 and s_2 . In s_1 variable X is applied to the list of terms $x_1 \dots x_m$ while, in s_2 , it is applied to $y_1 \dots y_n$. Without loss of generality we take $m < n$. After *map-deduplication* we have two distinct $\mathcal{M}$ -variables for X , say A and C , related by a chain $\vec{c}$ of η -links of length $n - m$. We prove that if A and C are assigned respectively to the $\mathcal{W}$ terms $\lambda x_1 \dots x_p.t_1$ and $\lambda x_1 \dots x_q.t_2$, then a failure occurs iff these terms do not unify. By *η -progress-lhs*, the assignment of A activates p links of $\vec{c}$. In particular, when $p \geq n$, then the entire chain collapses and unification is performed between $\lambda x_n \dots x_p.t_1$ and $\lambda x_1 \dots x_q.t_2$. By theorem 4.3.3 and since we work with $\mathcal{W}$ terms, this unification succeeds iff it does in $\mathcal{O}$. Otherwise if $p < n$ then we have to consider two cases. 1) If t_1 is a rigid term then t_2 must be the η -redex of t_1 , implying $q = 0$. If it is not the case, then *η -progress-lhs* would eventually cause a failure. 2) If t_1 is a unification variable then *scope-check* guarantees that t_2 does not use $x_1 \dots x_p$ as free variables, causing otherwise a unification failure. After this check, $m - n - p$ links of $\vec{c}$ remain suspended, as we expect: decompilation is charged to wake all links in $\mathbb{L}$ and eventually relate t_1 with t_2 . $\square$

Example of *map-deduplication* If we take back the example given at the beginning of this section, *map-deduplication* produces the following result:

$$\begin{aligned} \mathbb{P} &= \{ \lambda x.(X x) \simeq_o f \quad X \simeq_o \lambda x.a \} \\ \mathbb{Q} &= \{ A \simeq_m f \quad C \simeq_m \lambda x.a \} \\ \mathbb{M} &= \{ X \mapsto B^1 \} \\ \mathbb{L} &= \{ \vdash C =_\eta \lambda x.B_x \quad x \vdash A =_\eta \lambda y.B_y \} \end{aligned}$$

Note that $X \mapsto C^0$ disappears from $\mathbb{M}$ in favour of the auxiliary η -link $\vdash C =_\eta \lambda x.B_x$. The resolution of $\mathbb{Q}_1$ assigns a to A . This wakes up $\mathbb{L}_2$ by *η -progress-lhs*, assigning $f x$ to B_x . The resolution of $\mathbb{Q}_2$ instantiates C to $\lambda x.a$, which, in turn triggers $\mathbb{L}_1$ by *η -progress-lhs*. In turn, this performs $\lambda x.a \simeq_m \lambda x.(f x)$ that fails as expected.

4.6 Handling of $\Diamond\mathcal{L}$

As observed in [43] it is worth handling terms in $\Diamond\mathcal{L}$ since, in practice, these terms often re-enter $\mathcal{L}$ at runtime.

In the following example problem $\mathbb{P}_2$, namely $X a \simeq_o a$, admits two different solutions: $\rho_1 = \{ X \mapsto \lambda x.x \}$ and $\rho_2 = \{ X \mapsto \lambda x.a \}$. The unification algorithm alone cannot chose the right one, since no solution is more general than the other, but for the first problem the choice is obvious, and in turn this choice makes $\mathbb{P}_2 \subseteq \mathcal{L}$:

$$\begin{aligned} \mathbb{P} &= \{ X \simeq_o \lambda x.a \quad (X a) \simeq_o a \} \\ \mathbb{Q} &= \{ A \simeq_m \lambda x.a \quad (A a) \simeq_m a \} \\ \mathbb{M} &= \{ X \mapsto A^0 \} \end{aligned}$$

In order to support this scenario we have to improve a little our compiler and progress routines. In particular $\mathbb{Q}_1$ generates $\sigma = \{ A \mapsto \lambda x.a \}$ making $\sigma\mathbb{Q}_2$ equal to $(\lambda x.a) a \simeq_m a$ that $\simeq_m$ cannot solve since it lacks rules r_{β_l} and r_{β_r} in figure 4.2.

To address this problem the compiler must recognize $\diamond\mathcal{L}$ terms and replace them with fresh variables and generate a new kind of links that we call $\mathcal{L}$ -link.

In addition to invariant 4.2.2, the term on the rhs of a $\mathcal{L}$ -link has the following property:

Invariant 4.6.1 ($\mathcal{L}$ -link rhs). *The rhs of any $\mathcal{L}$ -link has the shape $X_{s_1 \dots s_n} t_1 \dots t_m$ where X is a unification variable in $\mathcal{L}$ (with scope $s_1 \dots s_n$) and $t_1 \dots t_m$ is a list of terms such that $m > 0$ and t_1 is either a variable occurring in $s_1 \dots s_n$ or a term other than a variable.*

Note that the shape of such as rhs is $\text{app}_m \text{ [uva}_m \text{ X S | L]}$, where S is $[\text{s}_1, \dots, \text{s}_n]$ and L is $[\text{t}_1, \dots, \text{t}_m]$.

4.6.1 Compilation and decompilation

We insert this rule just before rule $r_{@}$ in figure 4.5a

```
comp (appo [uvao A|Ag]) M1 L1 S1 (uvam B Sc) M3 L3 S4 :- !,
  pattern-fragment-prefix Ag Pf Extra, new S1 B S2,
  len Pf Ar, m-alloc (ovar A) M1 S2 (mvar C Ar) M2 S3,
  fold6 comp Pf M2 L1 S3 Pf1 M2 L1 S3,
  fold6 comp Extra M2 L1 S3 Extra1 M3 L2 S4,
  Beta = appm [uvam C Pf1 | Extra1],
  get-scope Beta Sc,
  L3 = [val (uvam B Sc =\beta Beta) | L2].
```

The list Ag is split into two parts: Pf is in $\mathcal{L}$, and can be empty; Extra cannot be empty and is such that $\text{append Pf Extra Ag}$. The rhs of the $\mathcal{L}$ -link is the application of a fresh variable C having in scope all names in Pf1 (the compilation of Pf). The variable B , returned as the compiled term, is a fresh variable having in scope all the free variables occurring in Pf1 and Extra1 (the compilation of Extra). This construction enforces invariant 4.6.1.

Lemma 4.6.2 ($\mathcal{W}$ -enforcement). $\forall s, \langle s \rangle \rightsquigarrow (t, m, l)$, then $\mathcal{W}(t)$.

Proof sketch. By lemmas 4.3.2 and 4.4.5 and the rule above. $\square$

Decompilation All $\mathcal{L}$ -link should be solved before decompilation. If any $\mathcal{L}$ -link remains in $\mathbb{L}$, decompilation fails.

4.6.2 Progress

We add the followings rules to **progress1**. Let l be a $\mathcal{L}$ -link of the form $\Gamma \vdash T =_{\mathcal{L}} X_{s_1 \dots s_n} t_1 \dots t_m$

Definition 4.6.1 ($\mathcal{L}$ -progress-refine). Let σ be a substitution such that σt_1 is a name s not occurring in $s_1 \dots s_n$. If $m = 1$, then l is removed and the lhs is unified with $X_{s_1 \dots s_n} s$. If $m > 1$, then l is replaced by the link $\Gamma \vdash T =_{\mathcal{L}} Y_{s_1 \dots s_n} s t_2 \dots t_m$ (where Y is a fresh variable of arity $n + 1$) and link $\Gamma \vdash X_{s_1 \dots s_n} =_{\eta} \lambda x. Y_{s_1 \dots s_n} x$ is added to $\mathbb{L}$.

1901 **Definition 4.6.2** ($\mathcal{L}$ -progress-rhs). Link l is removed from $\mathbb{L}$ if $X_{s_1 \dots s_n}$ is instantiated to
 1902 a term t and $t t_1 \dots t_m$ β -reduces to a $t' \in \mathcal{L}$.

1903 **Definition 4.6.3** ($\mathcal{L}$ -progress-fail). progress fails when either
 1904 there exists another link $l' \in \mathbb{L}$ with the same lhs as l ; or
 1905 the lhs of l become rigid.

1906 We relax this condition and accommodate for heuristics in section 4.6.3.

1907 Finally we introduce the following η -link progress rule.

1908 **Definition 4.6.4** (η -progress-rhs). A link $\Gamma \vdash X =_\eta T$ is removed from $\mathbb{L}$ when either
 1909 *maybe-eta* T does not hold (anymore), in this case X is unified with T ; or
 1910 T η -contracts to a term T' not starting with the lam_m constructor. In this case X
 1911 is unified with T'

1912 The idea behind this rule is to instantiate the lhs variable of a η -link whenever its
 1913 rhs is for sure a $\mathcal{W}$ term.

1914 **Lemma 4.6.3.** *progress terminates*

1915 *Proof sketch.* Let $l = \Gamma \vdash T =_{\mathcal{L}} X_{s_1 \dots s_n} t_1 \dots t_m$ a $\mathcal{L}$ -link in the store $\mathbb{L}$. The progression
 1916 made by $\mathcal{L}$ -progress-rhs terminates: l is removed from $\mathbb{L}$ after having performed termi-
 1917 nating instructions. If l is not activated by $\mathcal{L}$ -progress-rhs, then X is a variables. The
 1918 activation of $\mathcal{L}$ -progress-refine replaces l with the new $\mathcal{L}$ -link $\Gamma \vdash T =_{\mathcal{L}} Y_{s_1 \dots s_n} s t_2 \dots t_m$.
 1919 This progression can be triggered at most m times, since at each time the number of terms
 1920 applied to X decreases. In the end l_m will be removed from $\mathbb{L}$ after the unification between
 1921 lhs with rhs. We also note that each $\mathcal{L}$ -progress-refine generates a η -link, yet, according
 1922 to lemma 4.4.10, this does not affect termination. $\mathcal{L}$ -progress-fail terminates since it stop
 1923 progression with failure. Finally, η -progress-rhs performs terminating operations and, if
 1924 its permises succeed, the considered η -link is removed from $\mathbb{L}$, ensuring termination. $\square$

1925 **Theorem 4.6.4** (Fidelity in $\mathcal{O}$). *The introduction of $\mathcal{L}$ -link guarantees proposition 4.1.4*
 1926 *if $\mathbb{P} \notin \mathcal{L}$.*

1927 *Proof sketch.* Let $\mathbb{P}_i$ be the first problem in $\mathbb{P}$ such that it is not in $\mathcal{L}$ and let σ be the
 1928 substitution obtained solving $\mathbb{Q}_1 \dots \mathbb{Q}_{i-1}$. After the execution of step_m for the $i - 1$ time,
 1929 each variable X corresponding to a $\diamond \eta$ subterm in the original $\mathcal{O}$ unification problem,
 1930 is instantiated iff it is known for sure that X is no more in $\diamond \eta$. If $\sigma \mathbb{Q}_i$ is in $\mathcal{L}$, then by
 1931 definitions 4.6.1 and 4.6.2 the associated $\mathcal{L}$ -link is solved and removed. In this case, all
 1932 calls to $\simeq_m$ are between terms in $\mathcal{L}$ and by theorem 4.3.3 fidelity is guaranteed. If $\sigma \mathbb{Q}$
 1933 is still in $\diamond \mathcal{L}$, then by definition 4.6.3 unification fails as the corresponding unification in $\mathcal{O}$
 1934 would (it is called outside its domain). $\square$

4.6.3 Relaxing definition 4.6.3

Working with terms in $\mathcal{L}$ is sometime too restrictive [1] and we could find in literature a few strategies to go beyond $\mathcal{L}$ without implementing Huet’s algorithm [29]. Some implementations of λ Prolog [44] such as Teyjus [43] delay the resolution of $\diamond\mathcal{L}$ unification problems until the substitution makes them reenter $\mathcal{L}$. Other systems apply heuristics like preferring projection over mimic, and commit to that solution. This is the case for the unification algorithm *Rocq* uses in its type-class solver [50],

In this section we show how we can implement these strategies by simply adding (or removing) rules to the `progress` predicate. In the example below $\mathbb{P}_1$ is in $\diamond\mathcal{L}$.

$$\begin{aligned} \mathbb{P} &= \{ (X\ a) \simeq_o a \quad X \simeq_o \lambda x.Y \} \\ \mathbb{Q} &= \{ A \simeq_m a \quad B \simeq_m \lambda x.C \} \\ \mathbb{M} &= \{ Y \mapsto C^0 \quad X \mapsto B^0 \} \\ \mathbb{L} &= \{ x \vdash A =_{\mathcal{L}} (B\ a) \} \end{aligned}$$

If we want the object-language unification to delay the first unification problem (waiting for X to be instantiated), we can relax definition 4.6.3. Instead of failing when the lhs of $\mathbb{L}_1$ becomes rigid (equal to a), we keep it in $\mathbb{L}$ until the head of its rhs also become rigid. In this case, since both the lhs and rhs have rigid heads, they can be unified. While this relaxed rule does not break proposition 4.1.3 per se, the `occur-check-links` procedure becomes incomplete since invariant 4.2.2 is broken. Also note that delaying unification outside $\mathcal{L}$ can leave $\mathcal{L}$ -link for the decompilation phase. Therefore `commit-links` should be modified accordingly.

If instead we want $\simeq_o$ to follow the second strategy and pick an arbitrary solution we can modify `progress` by applying the desired heuristic instead of failing. For instance, in $X\ a\ b = Y\ b$, the last argument of the two terms is the same and unification can succeed by assigning Xa to Y . This heuristic is used by [50].

4.7 Actual implementation in *Elpi*

In this chapter we did study a compiler and a simulation loop on a minimal language. The actual implementation uses the *Rocq-Elpi* meta-language to both compile the sequence of problems (the rules) and execute them, that is *Elpi* plays the role of $\mathcal{M}$ as well.

The main difference is that we cannot implement `runm` since it is the runtime of the programming language. In particular the runtime iterates $\simeq_m$, but `stepm` also needs to check links for progress. Luckily *Elpi* extends λ Prolog with constraints (suspended goals) and Constraint Handling Rules (*CHR*) to operate on them section 2.2.5. Similarly to [39] a constraint is a goal suspended on a list of variables that is resumed *as soon as one of these variable is assigned*, before any other existing goal is considered. In turn link activation grants proposition 4.1.3. We point out that [39] delays $\diamond\mathcal{L}$ unification problems while we also delay problems that are in $\diamond\eta$.

In the following snippet, we show how η -link is either suspended or allowed to make progress.

```
link-eta L R :- not (var L), !, eta-progress-lhs L R.
link-eta L R :- not (maybe-eta R), !, eta-progress-rhs L R.
link-eta L R :- declare_constraint (link-eta L R) [L,R].
```

The first two clauses describe when progress can occur. If L is no longer a variable, the invariant invariant 4.2.2 is violated; therefore, we trigger unification between L and R . Similarly, if R is no longer a $\diamond\eta$ term, the invariant invariant 4.4.1 does not hold anymore, and we again force unification of L and R .

If neither condition holds, no progress is possible, and the constraint is suspended again. Similar rules are implemented for $\mathcal{L}$ -link.

```
rule (N1 ▷ G1 ?- link-eta (uvar X LX1) T1) % match
  / (N2 ▷ G2 ?- link-eta (uvar X LX2) T2) % remove
  | (relocate LX1 LX2 T2 T2') % condition
  <=> (N1 ▷ G1 ?- T1 = T2'). % new goal
```

In the snippet above, we illustrate the deduplication of η -link. The syntax $N \triangleright G \text{ ?- } P$ denotes a λ Prolog sequent, that is a goal P under a set G of hypothetical rules (introduced by $\Rightarrow$) and where the program context binds (via the π operator) eigenvariables in N . Sequents are presented to *CHR* with their higher-order unification variables replaced by “frozen constants”, that is A_{xyz} becomes $\text{uvar } f_A \text{ [x,y,z]}$ for a fresh constant f_A . Frozen constants are “defrost” when they are part of a new goal (see [25, section 4.3]).

The first directive matches a constraint whose lhs is a variable X ; the second matches and removes a constraint on the same variable; the third relocates $T2$ and the latter crafts the new goal that unifies $T1$ with $T2'$ under $G1$ and the set of heigenvariables $N1$.

4.8 Related work and conclusion

Object-language terms can be unified using different strategies.

The first approach that comes to mind consist in implementing $\simeq_o$ as a regular routine, i.e. write rules as follows

```
tc-Provable {{impl lp:F lp:B}} {{Pimpl lp:F lp:B lp:Q}} :-
  pi p \ tc-Provable {{fact lp:F}} p => tc-Provable B X,
  unify X {{lp:Q lp:p}}.
```

where `unify` is a regular predicate. This method fails to take advantage of the logic programming engine provided by the meta language since it removes all the data from the rule’s heads and makes indexing degrade. Additionally, implementing a unification procedure in the meta language is likely to be significantly slower compared to the built-in one on the common domain of first order terms.

Another possibility is to avoid having application and abstraction nodes in the syntax tree, and use the meta language ones, as in:

```
tc-Provable ({{impl}} F B) ({{Pimpl}} F B Q) :-
  pi p \ tc-Provable ({{fact}} F) p => tc-Provable B (Q p).
```

However, this encoding has two big limitations. First it is not always feasible to adopt it for *Rocq* due to the fact that the type system of the meta language is too limited to accommodate the one of the object language, e.g. *Rocq* can typecheck variadic functions [8]. Second, the encoding of *Rocq* terms provided by *Elpi* is primarily used for meta programming, i.e. to extend the *Rocq* system. Consequently, it must be able to manipulate terms

that are not known in advance without relying on introspection primitives such as Prolog’s `functor/3` and `arg/3`. To avoid that constants cannot be symbols of the meta language, but must rather live in an open world, akin to the `string` data type used in the `conm` constructor.

In the literature we could find a related encoding of the Calculus of Constructions (CC) [15]. The goal of that work is to exhibit a logic program performing proof checking for CC and hence relate the proof system of intuitionistic higher-order logic (that animates λ Prolog programs) with the one of CC. The encoding is hence tailored toward a different goal, for example it utilizes three relations to represent the equational theory of CC, and that choice alone makes things harder for us. Section 6 contains a discussion about the use of the unification procedure of the meta language in presence of non ground goals, but the authors are not interested in exploiting a decidable fragment of higher-order unification but are rather leaning towards an interactive system where the user is given control over the search procedure.

Another work that is, somewhat surprisingly, only superficially related to ours is the type-class engine built in Isabelle’s meta language. In [61] classes identify (simple) types, they are not higher order predicates as in *Rocq*, hence the solver does not require a higher-order unification procedure.

The approach proposed in this chapter addresses the issues discussed earlier. It exploits the unification mechanism of the meta-language, while handling problematic subterms separately when needed.

With this design, the encoding can benefit from indexing data structures and mode analysis for clause filtering. In particular, subterms are replaced by variables *only* when strictly necessary, so that most of the original structure is preserved and remains indexable.

Benchmarks on the *stdpp* library [30] show promising results. Our type-class solver introduces some overhead on small goals, but becomes consistently faster than the *Rocq* one for goals of size 32 nodes or more. The use of links also turns out to be useful in practice, as about 15% of the instances in *stdpp* require them.

Another advantage is the flexibility of the approach, which allows different strategies and heuristics to deal with terms outside the pattern fragment.

Finally, the encoding remains sufficiently shallow to: (1) extend certain forms of static analysis from the meta-language, such as determinacy, to the object language; (2) benefit from future improvements in the logic programming engine of *Elpi*, for example tabled search.

We considered mechanizing our results in Abella [20], in particular at an early stage of this work, but we were not able to complete it.

The main limitation is that the logic of Abella does not support quantification over predicates, nor the use of the cut operator. It is in principle possible to eliminate cut and higher-order combinators such as `map` or `forall2`, but this leads to a significant increase in verbosity, since these constructs are used pervasively.

More importantly, this transformation introduces a gap between the verified version and the actual implementation, which we found to hinder the development and exploration of the system.

Nevertheless, we established a test suite of about one hundred cases, which is available at <https://github.com/FissoreD/ho-unif-for-free>.

Determinacy checking for *Elpi*

Over the years *Elpi* was put in the hands of quite a number of *Rocq* users and the most widely cited barrier to entry has been the paradigm shift to a logic programming one. *Rocq* users are comfortable with typed and functional programming but often lack experience with backtracking and its control. Uncontrolled backtracking can produce slower programs and complicate debugging.

To lower this barrier we design a system of determinacy signatures that allows programmers to declare when a predicate behaves like a function: at most one result is produced and the runtime will not consider alternative results. This notion of determinism is called *semidet* by Henderson et al. [28] and named *operationally deterministic* by Nakamura [45]. Alternative notions exist, for example Warren [12] considers predicates *observably deterministic* when they produce the same result multiple times or diverge. We chose operational determinacy since it matches the behavior of functions in mainstream functional languages.

Our determinacy checker covers *Elpi*'s higher-order features, including higher-order predicates (for example map and fold, see section 5.1.1) and dynamic predicates that extend themselves at runtime (see section 5.1.2).

We apply these signatures to a large portion of publicly available *Elpi* code in the *Rocq* ecosystem and report our experience. Our findings indicate that most *Elpi* programs are intended to be backtracking-free and require only minimal signatures for the static analyzer we implemented to verify determinacy.

Our contributions are: 1) we design a language of signatures to assert the determinacy of predicates that covers higher-order logic programming constructs such as first-class predicates and dynamic predicates; 2) we implement a static analyzer for that language; 3) we apply the static analyzer to most of the *Elpi* code available in the *Rocq* ecosystem.

5.1 *Elpi* and determinacy signatures by examples

As in most logic programming languages, *Elpi* programs are organized in rules. When rules are not *mutually exclusive*, i.e. multiple rules apply on the same query, the runtime continues the computation using the rule with highest priority. Moreover it stores the other rules as a *choice point* and can backtrack to that point to continue the computation. What makes backtracking tricky to *Elpi* users is that it is an opt-out feature of logic programming: it is left to the programmer to tell the runtime to commit to one rule and

discard the choice point, rather than the other way around. If the user forgets to commit to a rule, his program still works but becomes much harder to debug and typically slower to run in case of failures.

The simplest example is the list-membership *relation* that can succeed more than once if the element we are looking for occurs multiple times in the list:

```
mem X [Y|_ ] :- X = Y.      % the head is the X we are looking for, or
mem X [_|YS] :- mem X YS. % recurse on the tail
```

The first rule for `mem` is the base case: it succeeds if the head of the list is equal to the `X` we are looking for. The second one discards the list's head and continues looking for `X` in the tail. A query such as `mem 2 [2,3,2]` can be solved in two different ways: either using rule 1 immediately, or using rule 2 twice to discard the first two items, and then conclude with rule 1.

Elpi users often have a background in functional programming and simply do expect `mem` to be a *function* testing membership, and not a relation. Moreover one needs to look, carefully, at how `mem` is implemented in order to see the peril. This misconception becomes a problem when they use `mem` in a larger piece of code and observe its execution.

```
main L :- code_before L, mem 1 L, code_after L.
```

For example, in the code above, one may see `code_after` being executed multiple times: each choice point left by `mem` enables the runtime to backtrack to that point in time and continue from there. Even worse, if `code_after` is expensive and fails, one pays its cost multiple times.

It must be said that backtracking can be very useful in some situations and that a programmer with experience in logic programming can take advantage of that. We want the *Elpi* casual user to be able to write code without mastering all that, and simply declare that they expect *their code* to compute functions.

```
func main list int -> . % the signature for main
```

The signature starts by a keyword declaring the determinacy of the predicate: `func` for deterministic predicates, that we also call functions, and `pred` for any predicate, including both functions and relations. Inputs are separated from outputs by the arrow symbol, and in the case of `main` there is no output.

```
main L :- code_before L, mem 1 L, code_after L.
%                ~~~~~ error: non-deterministic atom
```

Since `main` was declared as a function, our static analysis invites the user to discard the choice points left by `mem`, for example by inserting a `!` (a *cut* directive) after it. Assuming `code_after` is a functions, this code is accepted, since the cut discards all choice points generated by `mem` and `code_before L`.

```
main L :- code_before L, mem 1 L, !, code_after L. % OK
```

Note that the programmer may also write a functional membership test as follows, and use it in functional code with no additional safeguard. In this case the cut discards the second rule as soon as `X = Y` succeeds.

```

func in int, list int -> .
in X [Y|_ ] :- X = Y, !. % commit to this rule, discard the next one
in X [_|YS] :- in X YS.

main L :- code_before L, in 1 L, code_after L. % OK

```

Statically tracking the use of backtracking is a well-known technique in logic programming, usually referred to as determinacy analysis [12, 49]. However, the literature does not provide a detailed treatment of features that are specific to *Elpi* and not available in plain *Prolog*, such as higher-order programming, dynamic programming, and meta-programming.

These features were introduced in section 2.2.4 and play an important role in determinacy. In the following three subsections, we revisit those examples and analyze them from the perspective of determinacy.

5.1.1 Higher-order programs

A notable higher-order program is the one turning relations into functions.

```

func once (pred) -> . % a function taking a fully applied predicate
once P :- P, !.      % the ! commits to the first success

```

We say that the signature of the `once` predicate marks it as *unconditionally* a function: `once` leaves no choice points, no matter what it receives as the higher-order argument `P`.

An example of a *conditional* function is the predicate to map over a list:

```

func map (func A -> B), list A -> list B.
map _ [] [].
map F [X|XS] [Y|YS] :- F X Y, map F XS YS.

```

We say that the signature of `map` has a *precondition*, namely that `F` is a function, and a *postcondition*, namely that `map` is a function, i.e., the call `map F L L'` leaves no choice points. If the precondition is not met the postcondition is not guaranteed to hold and we say that `map` is *miscalled*. In this case we weaken its signature to^{*}:

```

pred map (func A -> B), list A -> list B.

```

Our static analysis tracks miscalls to functions and treats them as relations when analyzing the surrounding code. This avoids code duplication, since only one definition of `map` is needed in the library. We illustrate this with four scenarios involving correct calls and miscalls to `map`.

<pre> % a function func incr int -> int. incr 1 2. </pre>	<pre> % a relation pred get_pdiv int -> int. get_pdiv 6 3. get_pdiv 6 2. </pre>
<pre> % four predicates calling map or miscalling map </pre>	
<pre> func p list int -> list int. p L R :- map incr L R. % OK </pre>	<pre> pred q list int -> list int. q L R :- map get_pdiv L R. % OK </pre>

^{*}We explain in detail this weakening operation in section 5.4

```

func r list int -> list int.      func r1 list int -> list int.
r L R :- map get_pdiv L R.        r1 L R :- map get_pdiv L R, !. % OK
% ~~~~~ error: non-deterministic atom

```

In this example, `incr` is a function, while `get_pdiv` is a relation. In particular, `get_pdiv` associates more than one result to the input 6, making it non-deterministic.

The predicate `p` is deterministic because it satisfies the precondition of `map`: the argument `incr` is a function, so the call does not introduce choice points.

The predicate `q` instead contains a miscall to `map`, since `get_pdiv` is a relation. As a result, the execution may leave choice points. However, this is accepted because `q` is declared as a relation (with `pred`), and therefore non-determinism is allowed.

The situation is different for `r`. Its body is the same as `q`, but `r` is declared as a function. For this reason, the static analysis reports an error: a deterministic predicate cannot contain a non-deterministic computation.

Finally, `r1` has the same signature as `r` and also miscalls `map`, but it explicitly removes choice points using a `cut`. This makes the overall behavior deterministic, and the definition is accepted by the analyzer. Similarly, an implementation such as `once (map get_pdiv L R)` would also be valid, since the `once` wrapper ensures that any potential choice points are discarded.

5.1.2 Dynamic programs

In section 2.2.4, we pointed out the interest of dynamic programming in a language like *Elpi*. In this paragraph, we point out why the predicate `copy` can be considered as a deterministic predicate.

```

func copy tm -> tm.
copy (app A B) (app C D) :- copy A C, copy B D.
copy (lam F) (lam G)      :- pi x\ copy x x => copy (F x) (G x).

```

`copy` is a function because no two rules can apply to the same input: `lam` and `app` are different constants, and each constant `x` generated dynamically is guaranteed to be fresh by the `pi` operator, hence each `copy x x` rule is mutually exclusive with the other rules.

5.1.3 Meta programs

The meta-program we have show in section 2.2.4 is a good example of how our determinacy checker interacts well in program inspecion and composition.

We take back the implementation for the predicates `comp` and `fuse`, but this time, we pay attention to their determinacy in the signatures.

```

func comp (func A -> B), (func B -> C), A -> C.
comp F G X Z :- F X Y, G Y Z.

func fuse (func list A -> list B), (func list B -> list C) ->
(func list A -> list C).
fuse (map F) (map G) (map (comp F G)).

```

The signature of `comp` informs the checker that `comp` behaves deterministically if its two higher-order arguments are functions. This piece of information is important when the checker analyses the rule for `fuse`.

The signature of `fuse` has two preconditions, both inputs should be functions, and two postconditions: `fuse` is a function and its output is also a function. The code of `fuse` inspects two programs: if both are mapping a list, respectively with `F` and `G`, then it generates a new program `map (comp F G)`. The static analysis is able to verify that this output is a function following this reasoning: since `map F` and `map G` are functions then also `F` and `G` must be; from that we have that `comp F G` is a function and so is `map (comp F G)`.

5.2 *Elpi*'s abstract syntax and semantics

Any static analysis to rule out non-determinism must depend on the operational semantics of the language. In particular the order in which rules are applied and the order in which rule premises are executed directly impact the scope of the cut directive.

The *Elpi* syntax and semantics considered in this section are those presented in sections 2.2.1 and 2.2.2. To simplify the presentation, we treat `pi` and `=>` as a single operator. In this setting, the construct `pi x. r => a` represents the introduction of a fresh symbol `x`, which is visible in both the rule `r` and the atom `a`. The local rule `r` is added to the scope of the atom `a`.

Using the combined form `pi x. r => a` does not reduce the expressiveness of *Elpi*. The behavior of the `pi` operator can be simulated by introducing a dummy rule `r` in `a`, while the `=>` operator can be recovered by using a vacuous variable `x`.

We omit from the syntax the construction of data with binders, since it does not play a relevant role in the static analysis of determinacy.

We use the following notation: `s` ranges over symbol names (including predicates, data constructors, and variables), `v` over variables, `k` over predicate and data constructors, `a` and `b` over atoms, and `r` over rules.

In summary, the algorithm proceeds as follows:

- it checks that each rule is mutually exclusive with the others in the program;
- it verifies that the sequence of calls in the body consists of correctly used functional predicates;
- it ensures that the preconditions of rules entail their postconditions.

5.3 Mutual Exclusion

We focus on how to address the first source of non-determinism. The interesting aspects for this section are the `Cut` operator and the `matching/unify` procedure depending on the modes of a predicate.

In the next definition we distinguish between global rules, i.e., rules that are part of the program $\mathcal{P}$, and local rules that are loaded via the `pi` operator. For example, the `copy` predicate from section 5.1.2 has two global rules and one local rule.

Definition 5.3.1 (Mutual exclusion ($\text{mut_excl } \mathbb{P} \mathbb{R}$)). Given a program $\mathcal{P}$ and a rule r , such that $r \notin \mathcal{P}$, we say that r is mutually exclusive with the rules in $\mathcal{P}$ if it has the highest priority and one of the following additional conditions holds:

- (a) there is a **Cut** in the body of r , or
- (b) r is a global rule and all rules r' in $\mathcal{P}$ have an input argument not unifying with the corresponding input in r
- (c) r is a local rule introduced via **pi** x r b , x is an input of r and for all rules r' in the program, the corresponding input is not a variable

```

pred p int -> int.      func q int -> int.      func r int -> int.
p 1 3.                  q 1 3.                  r X R :- p X R, !.
p 1 2.                  q 2 4.                  r X R :- q X R.

```

For example, the rules for predicate **p** above are not mutually exclusive, so choice points may remain after a query resolution, hence introducing non-determinism. For instance, **p** 1 **X** has two solutions, namely **X** = 3 and **X** = 2.

Conversely, **q** satisfies mutual exclusion: no query can trigger both rules. Note that, since inputs are matched, a query like **q** **X** **Y** has no solution in *Elpi*. Both rules for **r** match any input. If the first rule succeeds, the **Cut** removes the alternatives created by the call to **p** and also discards the other rule **r** **X** **R** :- **q** **X** **R**. If the first rule fails, the **Cut** is not reached hence the second rule is not discarded. Thus, at most one rule can succeed for **q**.

The predicates **s** and **s1** below illustrate the role of **pi**.

```

func s int -> int.      func s1 int -> int.
s X R :- pi y\ q y 3 => q X R.  s1 X R :- pi y\ q 1 5 => q X R.
                                % error: mutexcl violated ~~~~~

```

The local rule **q** **y** 3 is mutually exclusive with the rules for **q** above since **y** is fresh: each time the rule is loaded **y** is different from 1 and 2 and any other fresh constants generated before. On the contrary the local rule **q** 1 5 is not mutually exclusive: the query **s1** 1 **R** has two results, namely **R** = 5 and **R** = 3.

Note that condition (c) also rules out the following example, since the **pi**-quantified symbol **y** is not used in input position.

```

func s2 int -> int.
s2 1 R :- pi y\ q 3 5 => s2 2 R.
s2 2 R :- pi y\ q 3 6 => q 3 R.
% ~~~~~ error: mutexcl violated

```

Both rules for **s2** load a local rule for **q**. All these local rules are mutually exclusive with the global rules in the program, but the query **s2** 2 **R** has two results namely **R** = 5 and **R** = 6.

5.3.1 Checking mutual exclusion: discrimination trees

This subsection briefly describes our implementation of mutual exclusion verification in *Elpi*. It is not required for understanding the rest of the chapter and can be skipped on a first reading.

To support mutual exclusion checking in *Elpi*, we rely on the discrimination tree data structure [38], a standard mechanism for indexing rules in logic programming systems.

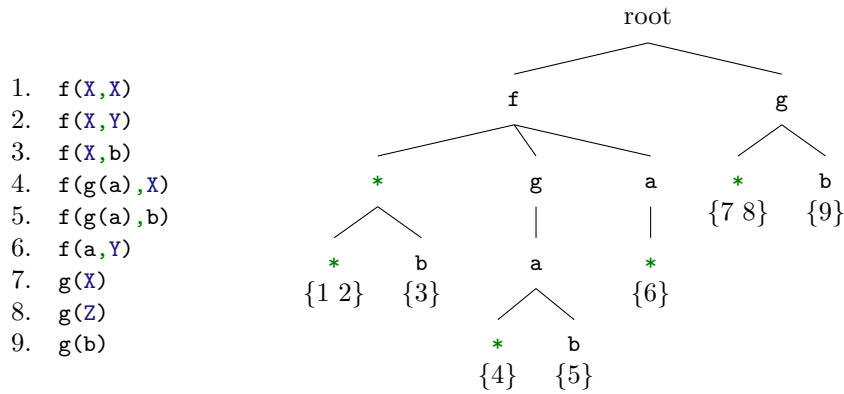


Figure 5.1: Example of a discrimination tree, from [38]

Discrimination trees are widely used in *Prolog*-like systems to improve the efficiency of clause retrieval. Our implementation follows, in part, the presentation given in Pfenning's notes[†].

A discrimination tree is a trie-based structure. Inner nodes are labeled, while leaves store buckets of objects. Tries represent dictionaries by sharing common prefixes of keys. A *path* is defined as the sequence of labels encountered from the root to a leaf.

Two basic operations are supported: insertion and retrieval. To insert an element e with key k , the algorithm traverses the trie symbol by symbol, starting from the root. For each symbol, it follows the corresponding edge, creating a new node if necessary. Once all symbols have been consumed, the element e is stored in the resulting leaf. Retrieval works similarly: the key is read symbol by symbol, and the tree is traversed accordingly. If the path exists, the elements stored in the corresponding leaf are returned.

In *Prolog* systems, the indexed objects are the rules of the database, and the head of each rule serves as its key. In figure 5.1, the database (left) and its corresponding discrimination tree (right) are shown. Leaves are annotated with the rules stored in each bucket. The purpose of the tree is to enable fast clause retrieval.

Variables are represented in an approximate way. In particular, they are encoded as generic symbols that match any term. For instance, in figure 5.1, rules 1 and 2 share the same path and are stored in the same bucket. Retrieval must also take modes into account. For example, if the predicate g has one input and one output argument, the query $g\ X\ Y$ returns only rules 7 and 8. If both arguments are outputs, then rules 7, 8, and 9 are returned.

For efficiency, terms are encoded as sequences of integers, where each integer represents multiple pieces of information at the bit level. Modes are included in this encoding, allowing the system to determine whether a rigid term should be matched or unified during retrieval.

Unification in the discrimination tree is only an approximation of the full *Elpi* unification algorithm, which is more expensive. By construction, the set of rules returned by the tree is a superset of the rules that actually match the query. A second phase applies the full *Elpi* unification to filter out non-applicable rules.

[†]<https://www.cs.cmu.edu/~fp/courses/99-atp/lectures/lecture28.html>

We also optimize the encoding of query paths. In practice, query terms are often larger than the terms stored in the database, so encoding them entirely is not useful. Instead, we use a hit-map to decide which subterms should be encoded and at which depth. Furthermore, to avoid complex unification, terms headed by a λ -symbol are treated as variables. For example, $\lambda x. x$ may unify with id or $(\lambda x. 3)$, but all such terms are encoded as variables in the tree.

Discrimination trees are effective for indexing rules in the *Elpi* database, making retrieval faster in many cases. In this chapter, however, their main role is to support mutual exclusion checking.

During compilation, an auxiliary discrimination tree is built. Each rule of a deterministic predicate is inserted into this structure. When a new rule is added, the tree is queried to retrieve the set of potentially overlapping rules r . By invariant, r is an ordered (possibly empty) list in which all rules, except possibly the last one, contain a **Cut**. The mutual exclusion check then verifies that the new rule is inserted either at the end of the list; otherwise it should contain the **Cut** operator.

5.4 Static analysis of determinacy signatures

The abstract syntax of signatures is given below. Predicate signatures $\mathbb{T}_y$ start with input arguments from the S_i non-terminal, continue with output arguments from the S_o non-terminal and end with a determinacy marker $\mathbb{D}$: **Fun** for functions and **Rel** for relations (respectively **func** and **pred** in the concrete syntax).

$$\begin{array}{llll}
 S_i & ::= S_o \mid S_i \xrightarrow{i} S_i \mid \star \xrightarrow{i} S_i & \text{sign. start} & \mathbb{D} ::= \text{Fun} \mid \text{Rel} \quad \text{determinacy} \\
 S_o & ::= \mathbb{D} \mid S_i \xrightarrow{o} S_o \mid \star \xrightarrow{o} S_o & \text{sign. stop} & \star ::= \star \quad \text{all data} \\
 \mathbb{T}_y & ::= S_i & \text{signature} &
 \end{array}$$

We collapse all data types into a single type $\star$ and we disallow to store predicates into data, i.e., one cannot form a list of predicates. Note that the grammar of $\mathbb{T}_y$ forces all input arguments to come first. We reserve d for instances of type $\star$, ρ and τ for signatures and $\mathcal{D}$ for instances of $\mathbb{D}$. For example the signature **func map (func A $\rightarrow$ B), list A $\rightarrow$ list B** corresponds to $(\star \xrightarrow{i} \star \xrightarrow{o} \text{Fun}) \xrightarrow{i} \star \xrightarrow{i} \star \xrightarrow{o} \text{Fun}$.

We denote booleans $\mathbb{B} = \{\top, \perp\}$; contexts Γ being partial maps from symbols to signatures. We use $\emptyset$ for the empty list and $x :: xs$ for prepending the element x to a list xs . We reserve Γ for contexts.

5.4.1 Operations on signatures

We compare signatures using the standard subtyping relation of functional languages. In particular, we compare inputs contravariantly and outputs covariantly. It is similar to [27], except for the absence of the **Any** terminal. **Fun** is smaller than **Rel** since functions generate a “smaller number of results” than relations.

$$\begin{array}{lllll}
 \star \subseteq \star & \text{Fun} \subseteq \text{Fun} & \text{Rel} \subseteq \text{Rel} & \text{Fun} \subseteq \text{Rel} & \text{base cases} \\
 \rho \xrightarrow{i} \tau \subseteq \rho' \xrightarrow{i} \tau' & \text{iff} & \rho' \subseteq \rho \wedge \tau \subseteq \tau' & & \text{covariance} \\
 \rho \xrightarrow{o} \tau \subseteq \rho' \xrightarrow{o} \tau' & \text{iff} & \rho \subseteq \rho' \wedge \tau \subseteq \tau' & & \text{contravariance}
 \end{array}$$

The static analysis needs often to keep the stronger or weaker signature among two. We hence derive from $\subseteq$ a **min** and **max** operators defined below.

$\begin{aligned} \mathbf{min} \star \star &= \star \\ \mathbf{min} \text{ Rel Rel} &= \text{Rel} \\ \mathbf{min} \text{ Rel Fun} &= \text{Fun} \\ \mathbf{min} \text{ Fun Rel} &= \text{Fun} \\ \mathbf{min} \text{ Fun Fun} &= \text{Fun} \\ \mathbf{min} (\rho \xrightarrow{i} \tau) (\rho' \xrightarrow{i} \tau') &= \mathbf{max} \rho \rho' \xrightarrow{i} \mathbf{min} \tau \tau' \\ \mathbf{min} (\rho \xrightarrow{o} \tau) (\rho' \xrightarrow{o} \tau') &= \mathbf{min} \rho \rho' \xrightarrow{o} \mathbf{min} \tau \tau' \end{aligned}$	$\begin{aligned} \mathbf{min} \star \star &= \star \\ \mathbf{min} \text{ Rel Rel} &= \text{Rel} \\ \mathbf{min} \text{ Rel Fun} &= \text{Rel} \\ \mathbf{min} \text{ Fun Rel} &= \text{Rel} \\ \mathbf{min} \text{ Fun Fun} &= \text{Fun} \\ \mathbf{max} (\rho \xrightarrow{i} \tau) (\rho' \xrightarrow{i} \tau') &= \mathbf{min} \rho \rho' \xrightarrow{i} \mathbf{max} \tau \tau' \\ \mathbf{max} (\rho \xrightarrow{o} \tau) (\rho' \xrightarrow{o} \tau') &= \mathbf{max} \rho \rho' \xrightarrow{o} \mathbf{max} \tau \tau' \end{aligned}$
---	---

2290 When a predicate is miscalled we need to weaken its signature.

We define strong and weak as follows:

$\begin{aligned} \mathbf{strong} \star &= \star \\ \mathbf{strong} \text{ Rel} &= \text{Fun} \quad \mathbf{strong} \text{ Fun} = \text{Fun} \\ \mathbf{strong} (\rho \xrightarrow{i} \tau) &= \mathbf{weak} \rho \xrightarrow{i} \mathbf{strong} \tau \\ \mathbf{strong} (\rho \xrightarrow{o} \tau) &= \mathbf{strong} \rho \xrightarrow{o} \mathbf{strong} \tau \end{aligned}$	$\begin{aligned} \mathbf{weak} \star &= \star \\ \mathbf{weak} \text{ Fun} &= \text{Rel} \quad \mathbf{weak} \text{ Rel} = \text{Rel} \\ \mathbf{weak} (\rho \xrightarrow{i} \tau) &= \mathbf{strong} \rho \xrightarrow{i} \mathbf{weak} \tau \\ \mathbf{weak} (\rho \xrightarrow{o} \tau) &= \mathbf{weak} \rho \xrightarrow{o} \mathbf{weak} \tau \end{aligned}$
--	--

2291 The set of signatures that only differ by a leaf in $\mathcal{D}$ form a lattice w.r.t. the order relation
 2292 $\subseteq$, the join operator **max** and meet operator **min**. The infimum of the lattice is computed
 2293 by strong and the supremum is computed by weak.

2294 In section 5.1.1 we have weakened the signature of **map** from $\tau = (\star \xrightarrow{i} \star \xrightarrow{o} \text{Fun}) \xrightarrow{i}$
 2295 $\star \xrightarrow{i} \star \xrightarrow{o} \text{Fun}$ to $(\star \xrightarrow{i} \star \xrightarrow{o} \text{Fun}) \xrightarrow{i} \star \xrightarrow{i} \star \xrightarrow{o} \text{Rel}$.

2296 One could define the supremum of τ as $\tau' = (\star \xrightarrow{i} \star \xrightarrow{o} \text{Rel}) \xrightarrow{i} \star \xrightarrow{i} \star \xrightarrow{o} \text{Rel}$, which
 2297 intuitively amounts to considering all **Fun** as **Rel**, i.e., dropping all **Fun** in case of error.
 2298 However, this would contradict the lattice property: $\mathbf{max} \tau (\mathbf{weak} \tau) = \mathbf{weak} \tau$, since we
 2299 would have $\mathbf{max} \tau \tau' = \tau'$, but $\tau \not\subseteq \tau'$. Moreover, since we compare input signatures using
 2300 $\subseteq$, this alternative definition of weak (resp. strong) would not make the checker accept
 2301 more programs.

2302 5.4.2 The idea

2303 The static analyzer accumulates knowledge on variables in a context Γ initially only con-
 2304 taining the signature of predicates. The checking algorithm considers one rule at a time,
 2305 following their priorities. It begins by updating Γ using the preconditions on the *in-*
 2306 *puts* of the head, then it scans all body atoms and finally checks the postconditions: the
 2307 determinacy of the body and the determinacy of the *outputs* of the head.

2308 For example, consider the predicates **comp** and **compose** below: **comp** is a function that
 2309 takes as input two functions **F** and **G**, and calls them in sequence, whereas **compose** builds a
 2310 new function from the two received in input. At first, the checker assumes that the inputs
 2311 of **comp** satisfy the preconditions given in its signature. In particular, **F** and **G** are marked
 2312 as functions in Γ . Then the algorithm scans the atoms in the body. The determinacy of
 2313 the whole rule is computed as the **max** of the determinacies of its premises. Since both
 2314 **F X Y** and **G Y Z** have determinacy **Fun**, their **max** is also **Fun**. Finally, the analyzer verifies
 2315 that the determinacy of the body is consistent with the one declared in the signature.

For `compose`, two conditions must be checked: the term `comp F G` must be a function, and `compose` itself must also be a function. The algorithm assumes again that `F` and `G` are functions. The term `comp F G` is a partial application, and it returns a function when its arguments are functions. Since this holds for `F` and `G`, the preconditions of `comp` are satisfied, and the result is a function. Finally, the body of `compose` is empty, so it is deterministic. Therefore, `compose` behaves deterministically.

We now move to an example that involves a miscall to a predicate, that is, a rule with a non-deterministic body atom.

```
func r (func int -> int), (pred int -> int), int -> int.
r F G X Y :- map G [X] [Z], !, compose F F H, H Z Y.
```

After analyzing the head of the rule, the context asserts that `F` (resp. `G`) is a function (resp. a relation). In the body of `r`, the first atom is non-deterministic, since `G` violates the preconditions of `map`. The signature of `map G [X] [Z]` is hence weakened to `Rel`. Even if this atom could generate multiple alternatives, the `Cut` commits to the first one restoring the determinism of the body of `r` up to that point. The call to `compose` satisfies its preconditions, so its postconditions apply: the entire atom `compose F F H` is considered deterministic and `H` is marked as a function in Γ . The final atom `H Z Y` is also deterministic because of the knowledge about `H` accumulated so far in Γ .

5.4.3 The algorithm

We present the static analyzer using derivation rules, they are listed in figures 5.2 and 5.3. Each procedure has an associated entails-like symbol, e.g., context $\frac{\text{flag}}{\text{name}} \text{intake} = \text{result}$. We use the equal sign to “mode” the rule, i.e., separate what is given to the rule from what is generated by it. The static analyzer is composed of seven procedures for a total of 24 rules.

The entry point is `check.-` (figure 5.2a). This procedure takes a program and a context of signatures, it verifies if a rule validates the signature ascribed to its predicate. The procedure stores in Γ all the knowledge available on variables and verifies that the precondition made by the signature entails the postcondition.

In particular, the **assume head input** procedure (figure 5.3a) loads into Γ_1 the information assumed from the head inputs. This procedure is applied under the assumption that the predicate is well-called. Hence, all variables that occur (recursively) in input positions are assumed to satisfy the determinacy specified in the signature.

On the other hand, the **check head output** procedure (figure 5.3b) verifies that Γ_{n+1} entails the required postcondition. More precisely, the inferred type of the terms appearing in output positions must be smaller or equal (with respect to $\sqsubseteq$) than the type declared in the signature. In addition, the determinacy of the body must also be smaller or equal than the expected one. For instance, if the predicate is declared deterministic, then the determinacy flag inferred for the body must be `Fun`.

Each of the n body atoms b_i is processed by **check atom** (figure 5.2b). This procedure updates the context, step by step, and computes the determinacy of the whole body, producing $\mathcal{D}_{n+1}$.

The **check atom** procedure includes specific rules to handle `Cut` (a_l in figure 5.2b) and `pi` (a_v in figure 5.2b), as well as rules for predicate calls (a_c) and miscalls (a_1).

$$\frac{\Gamma \Big|_{\text{hd}}^i c = \mathcal{D}_1, \Gamma_1 \quad \forall i \in 1 \dots n \text{ do } \mathcal{P}, \Gamma_i, \mathcal{D}_i \Big|_{\bar{\alpha}} b_i = \mathcal{D}_{i+1}, \Gamma_{i+1} \quad \Gamma_{n+1} \Big|_{\text{hd}}^{\circ} c : \mathcal{D}_{n+1}}{\mathcal{P}, \Gamma \Big|_{\bar{r}} c :- b_1 \dots b_n} \text{ (r)}$$

(a) Check rule

$$\frac{x \text{ fresh in } \Gamma \quad \mathcal{P}, \Gamma + \{x \mapsto \star\} \Big|_{\bar{r}} r \quad \text{mut_excl } \mathcal{P} \Gamma x r \quad (r :: \mathcal{P}), \Gamma + \{x \mapsto \star\}, \mathcal{D} \Big|_{\bar{\alpha}} a = \mathcal{D}', \Gamma'}{\mathcal{P}, \Gamma, \mathcal{D} \Big|_{\bar{\alpha}} \text{pi } x. r \Rightarrow a = \mathcal{D}', \Gamma'} \text{ (a}_v\text{)}$$

$$\frac{}{\mathcal{P}, \Gamma, _ \Big|_{\bar{\alpha}} \text{Cut} = \text{Fun}, \Gamma} \text{ (a}_i\text{)} \quad \frac{\Gamma \Big|_{\text{tm}}^i c = \text{Rel}, \perp}{\mathcal{P}, \Gamma, _ \Big|_{\bar{\alpha}} c = \text{Rel}, \Gamma} \text{ (a}_\perp\text{)}$$

$$\frac{\Gamma \Big|_{\text{tm}}^i c = \mathcal{D}', \top \quad \Gamma \Big|_{\text{ca}}^{\circ} c : \mathcal{D}' = \Gamma'}{\mathcal{P}, \Gamma, \mathcal{D} \Big|_{\bar{\alpha}} c = \max \mathcal{D} \mathcal{D}', \Gamma'} \text{ (a}_c\text{)}$$

(b) Rules for **check atom**: its type is $\mathbb{P}, \mathbb{F}, \mathbb{D} \Big|_{\bar{\alpha}} \mathbb{A} = \mathbb{D}, \mathbb{F}$

$$\frac{\Gamma s = \tau}{\Gamma \Big|_{\text{tm}}^i s = \tau, \top} \text{ (tm}_k^i\text{)} \quad \frac{\Gamma \Big|_{\text{tm}}^i f = \star \xrightarrow{i} \tau, b}{\Gamma \Big|_{\text{tm}}^i f d = \tau, b} \text{ (tm}_\star^i\text{)} \quad \frac{\Gamma \Big|_{\text{tm}}^i f = _ \xrightarrow{o} \tau, b}{\Gamma \Big|_{\text{tm}}^i f t = \tau, b} \text{ (tm}_o^i\text{)}$$

$$\frac{\Gamma \Big|_{\text{tm}}^i f = \rho \xrightarrow{i} \tau, \top \quad \Gamma \Big|_{\text{tm}}^i t = \rho', \top \quad \rho' \subseteq \rho}{\Gamma \Big|_{\text{tm}}^i f t = \tau, \top} \text{ (tm}_\top^i\text{)}$$

$$\frac{\Gamma \Big|_{\text{tm}}^i f = \rho \xrightarrow{i} \tau, b_1 \quad \Gamma \Big|_{\text{tm}}^i t = \rho', b_2 \quad b_1 = \perp \vee b_2 = \perp \vee \rho' \not\subseteq \rho}{\Gamma \Big|_{\text{tm}}^i f t = \text{weak } \tau, \perp} \text{ (tm}_\perp^i\text{)}$$

(c) Rules for **check input**: its type is $\mathbb{F} \Big|_{\text{tm}}^i \mathbb{Tm} = \mathbb{T}_y, \mathbb{B}$

$$\frac{\tau' = \text{if } X \in \Gamma \text{ then } \Gamma X \text{ else } \tau}{\Gamma \Big|_{\text{tm}}^{\circ} X : \tau = \Gamma + \{X \mapsto \min \tau \tau'\}} \text{ (tm}_x^{\circ}\text{)} \quad \frac{\Gamma k = \tau' \quad \tau' \subseteq \tau}{\Gamma \Big|_{\text{tm}}^{\circ} k : \tau = \Gamma} \text{ (tm}_k^{\circ}\text{)}$$

$$\frac{\Gamma \Big|_{\text{tm}}^{\circ} f : \star \xrightarrow{i} \tau = \Gamma'}{\Gamma \Big|_{\text{tm}}^{\circ} f d : \tau = \Gamma'} \text{ (tm}_\star^{\circ}\text{)} \quad \frac{\Gamma \Big|_{\text{tm}}^{\circ} f : _ \xrightarrow{o} \tau = \Gamma'}{\Gamma \Big|_{\text{tm}}^{\circ} f t : \tau = \Gamma'} \text{ (tm}_o^{\circ}\text{)}$$

$$\frac{\Gamma \Big|_{\text{tm}}^{\circ} f : \rho \xrightarrow{i} \tau = \Gamma' \quad \Gamma' \Big|_{\text{tm}}^{\circ} t : \rho = \Gamma''}{\Gamma \Big|_{\text{tm}}^{\circ} f t : \tau = \Gamma''} \text{ (tm}_i^{\circ}\text{)}$$

(d) Rules for **assume output**: its type is $\mathbb{F} \Big|_{\text{tm}}^{\circ} \mathbb{Tm} : \mathbb{T}_y = \mathbb{F}$

Figure 5.2: Determinacy checker algorithm part 1

$$\begin{array}{c}
\frac{\Gamma \ k = \tau}{\Gamma \Big|_{\mathbf{hd}}^{\mathbf{i}} k = \tau, \ \Gamma} \ (\mathbf{hd}_k^{\mathbf{i}}) \quad \frac{\Gamma \Big|_{\mathbf{hd}}^{\mathbf{i}} f = _ \overset{\circ}{\rightarrow} \tau, \ \Gamma' \quad \vee \quad \Gamma \Big|_{\mathbf{hd}}^{\mathbf{i}} f = \star \overset{\mathbf{i}}{\rightarrow} \tau, \ \Gamma'}{\Gamma \Big|_{\mathbf{hd}}^{\mathbf{i}} f \ t = \tau, \ \Gamma'} \ (\mathbf{hd}_o^{\mathbf{i}}) \\
\\
\frac{\Gamma \Big|_{\mathbf{hd}}^{\mathbf{i}} f = \rho \overset{\mathbf{i}}{\rightarrow} \tau, \ \Gamma' \quad \Gamma' \Big|_{\mathbf{tm}}^{\mathbf{o}} t : \rho = \Gamma''}{\Gamma \Big|_{\mathbf{hd}}^{\mathbf{i}} f \ t = \tau, \ \Gamma''} \ (\mathbf{hd}_i^{\mathbf{i}})
\end{array}$$

(a) Rules for **assume head input**: its type is $\mathbb{T} \Big|_{\mathbf{hd}}^{\mathbf{i}} \mathbb{Tm} = \mathbb{T}_y, \ \mathbb{T}$

$$\begin{array}{c}
\frac{\Gamma \ k = \tau' \quad \tau \subseteq \tau'}{\Gamma \Big|_{\mathbf{hd}}^{\mathbf{o}} k : \tau} \ (\mathbf{hd}_k^{\mathbf{o}}) \quad \frac{\Gamma \Big|_{\mathbf{hd}}^{\mathbf{o}} f : _ \overset{\mathbf{i}}{\rightarrow} \tau \quad \vee \quad \Gamma \Big|_{\mathbf{hd}}^{\mathbf{o}} f : \star \overset{\circ}{\rightarrow} \tau}{\Gamma \Big|_{\mathbf{hd}}^{\mathbf{o}} f \ t : \tau} \ (\mathbf{hd}_i^{\mathbf{o}}) \\
\\
\frac{\Gamma \Big|_{\mathbf{hd}}^{\mathbf{o}} f : \rho \overset{\circ}{\rightarrow} \tau \quad \Gamma \Big|_{\mathbf{tm}}^{\mathbf{i}} t = \rho', \ \top \quad \rho' \subseteq \rho}{\Gamma \Big|_{\mathbf{hd}}^{\mathbf{o}} f \ t : \tau} \ (\mathbf{hd}_o^{\mathbf{o}})
\end{array}$$

(b) Rules for **check head output**: its type is $\mathbb{T} \Big|_{\mathbf{hd}}^{\mathbf{o}} \mathbb{Tm} : \mathbb{T}_y$

$$\begin{array}{c}
\frac{\Gamma \ s = \tau}{\Gamma \Big|_{\mathbf{ca}}^{\mathbf{o}} s : \tau = \Gamma} \ (\mathbf{a}_k^{\mathbf{o}}) \quad \frac{\Gamma \Big|_{\mathbf{ca}}^{\mathbf{o}} f : \rho \overset{\circ}{\rightarrow} \tau = \Gamma' \quad \Gamma' \Big|_{\mathbf{tm}}^{\mathbf{o}} t : \rho = \Gamma''}{\Gamma \Big|_{\mathbf{ca}}^{\mathbf{o}} f \ t : \tau = \Gamma''} \ (\mathbf{a}_o^{\mathbf{o}}) \\
\\
\frac{\Gamma \Big|_{\mathbf{ca}}^{\mathbf{o}} f : _ \overset{\mathbf{i}}{\rightarrow} \tau = \Gamma}{\Gamma \Big|_{\mathbf{ca}}^{\mathbf{o}} f \ t : \tau = \Gamma} \ (\mathbf{a}_i^{\mathbf{o}})
\end{array}$$

(c) Rules for **assume term output**: its type is $\mathbb{T} \Big|_{\mathbf{ca}}^{\mathbf{o}} \mathbb{Tm} : \mathbb{T}_y = \mathbb{T}$

Figure 5.3: Determinacy checker algorithm part 2

The **Cut** operator resets the current determinacy to **Fun**. This follows from its semantics: once a **Cut** is executed, all alternative choice points are discarded, so no nondeterminism can arise.

The **pi** operator introduces a fresh symbol together with a local rule. This rule is checked to ensure that determinacy is preserved. In particular, it must satisfy **checkc** in the current context and must not violate mutual exclusion. Then, the underlying atom is analysed in the program extended with this local rule.

Rule a_c checks that the preconditions of the predicate c are satisfied. This is ensured by **check input**, which verifies that the inputs meet the expected conditions. If this holds, the postcondition is assumed, producing an updated context Γ' . The determinacy is updated as the maximum between the current one and that declared for the predicate.

If the preconditions are not satisfied, the context remains unchanged and the determinacy is set to **Rel**. This value reflects that, in case of a miscalled predicate, no deterministic behaviour can be guaranteed.

The most relevant procedure is the one used to check whether the preconditions of a predicate call are satisfied, namely **check input** (figure 5.2c). This procedure ignores data arguments and output arguments, and simply retrieves from the context the signature of constants and variables (tm_k^i in figure 5.2c).

Rule tm_τ^i corresponds to the standard typing rule for application. When the input arguments match the expected signature, it computes the resulting type and reports no miscal (i.e. τ).

Otherwise, the head or the argument contain a miscal, or when the argument does not match the expected signature, rule $tm_\perp^i$ is applied. In this case, a miscal is reported and the resulting signature is obtained by weakening the signature inferred from the head. Intuitively, the predicate call is treated as a relation, and the types of the output arguments are weakened accordingly.

The procedure to update the context when an atom is found to be a good call is **assume term output** (figure 5.3c). It is an iterator over output arguments: other than ignoring input arguments ($\mathbf{a}_i^o$) and fetching the signature of symbols in the context ($\mathbf{a}_k^o$), it calls **assume output** on output arguments.

The **assume output** procedure (figure 5.2d) updates the context when it encounters variables (rule tm_X^o). If a variable already has a signature in the context, the stronger one is kept.

Data and output arguments are ignored, while input arguments are traversed recursively. In this way, the signatures of variables occurring in input positions are also updated in the context.

The processing of the head of a rule is similar to that of body atoms, but symmetric with respect to how arguments are treated. The **assume head input** procedure (figure 5.3a), called at the beginning of figure 5.2a, ignores output arguments and applies **assume output** to the inputs.

The **check head output** procedure (figure 5.3b) is invoked at the end. It ignores input arguments and checks that the output ones satisfy the declared signature in the current context. This context results from assuming the rule preconditions together with the postconditions of all well-called body atoms.

5.5 Determinism guarantees

In order to express *operational determinacy* we need to use the operational semantics for *Elpi*. The semantics is given in section 2.2.2.

The properties of `runE` relevant to the current presentation are:

1. `Cut` behaves as per section 2.2.3
2. `pi x. r ⇒ a` generates a fresh constant for x available in r and a
3. `pi x. r ⇒ a` locally loads r in a giving it the highest priority
4. input arguments are matched against the query as per definition 2.2.6

Definition 5.5.1 (Operationally deterministic execution ($\text{deterministic}_{\mathbf{a}} \mathbb{P} \mathbb{A}$)). We say that a query atom b is operationally deterministic in a program $\mathcal{P}$ iff

$$\forall \sigma, \text{runE} ((\sigma, (\mathcal{P}, b) :: \emptyset) :: \emptyset) \rightsquigarrow (_, a) \rightarrow a = \emptyset$$

In other words a deterministic query leaves no alternatives when it succeeds. We now link the static check to the deterministic execution of a query.

Definition 5.5.2 (Checked program ($\text{checked}_{\mathbf{p}} \mathbb{P} \mathbb{P}$)). A program $\mathcal{P}$ is checked against a context of signatures Γ iff it validates these rules:

$$\frac{}{\text{checked}_{\mathbf{p}} \Gamma \emptyset} \quad \frac{\text{checked}_{\mathbf{p}} \Gamma \mathcal{P} \quad \text{mut_excl } \mathcal{P} \ r \quad \mathcal{P}, \Gamma \big|_{\mathbf{r}} r}{\text{checked}_{\mathbf{p}} \Gamma (r :: \mathcal{P})}$$

The intuition is that we build the program from the end, always adding new rules with highest priority.

Definition 5.5.3 (Checked query atom ($\text{checked}_{\mathbf{a}} \mathbb{P} \mathbb{A}$)). An atom b is checked against a context of signatures Γ iff

$$\emptyset, \Gamma, \text{Fun} \big|_{\mathbf{a}} b = \text{Fun}, _$$

Theorem 5.5.1 (Static analysis). *Given program $\mathcal{P}$, a context of signatures Γ and a query atom b , $\text{checked}_{\mathbf{p}} \Gamma \mathcal{P} \wedge \text{checked}_{\mathbf{a}} \Gamma b \Rightarrow \text{deterministic}_{\mathbf{a}} \mathcal{P} b$.*

The proof of theorem 5.5.1 has been mechanized in the *Rocq* proof assistant, for the first-order part and is available at <https://github.com/FissoreD/elpi-formalization>. We discuss of the first-order formalization in detail at chapter 6. The mechanization for the higher-order part is ongoing.

5.6 Experience report

The static analyzer we describe was integrated in the *Elpi* language version 3.0. It consists of about 200 lines of *OCaml* code for the implementation of `mut_excl` and about 800 lines of code for the implementation of `check_`, along with all the related auxiliary functions. We have ported some *Rocq* libraries written using *Elpi*: *Rocq-Elpi*, *Hierarchy Builder*, *Algebra Tactics*, *Trocq*, and *BRiCk*.

Software	LOC	% Fun	Cut added	Cut needed (bugs)
<i>Elpi</i>	1448	97%	1	1 (100%)
<i>Rocq-Elpi</i>	13164	79%	91	78 (85%)
<i>Hierarchy Builder</i>	5326	94%	33	30 (90%)
<i>Algebra Tactics</i>	1799	–	1	1 (100%)
<i>Trocq</i>	2918	–	4	4 (100%)
<i>BRiCk</i> (public)	2500	–	11	11 (100%)
<i>BRiCk</i> (private)	7500	–	6	6 (100%)
Total	34655	85%	147	131 (89%)

Table 5.1: Experience report

We can divide the porting of these libraries into two categories: libraries in which we have traversed all the predicates and carefully changed `pred` markers to `func` (*Elpi*, *Rocq-Elpi*, and *Hierarchy Builder*), and libraries that have been ported to just pass the static analyzer. In particular, the latter class of libraries measures the minimal cost for adapting code to the changes we made to the *Elpi* and *Rocq-Elpi* standard libraries. The typical example of code requiring a fix is code that loads a local rule containing no cut, typically `copy`.

We summarize these results in table 5.1. In the *Elpi* standard library, we annotated 272 predicates out of 281 as functions, with 37 occurrences of higher-order arguments, as in `map`, `filter`, `fold`, etc. We identified 1 error, i.e., a missing cut. In *Rocq-Elpi*, we have 928 functions out of 1181, with 62 higher-order predicates. We added 91 cuts to please the static analyzer, 78 of which were clearly bugs in the code. The remaining 13 cuts are due to `mut_excl`, which is not powerful enough to discriminate among rules. The algorithm could be improved to better distinguish mutual exclusion by extending definition 5.3.1, as in [34], but the problem is, in general, undecidable. An example of this issue is given below:

```
func fact int -> int.
fact 0 1.
fact N R :- N > 0, N' is N - 1, fact N' R', R is N * R'.
```

The two rules for `fact` are in mutual exclusion: the input cannot be 0 and greater than 0 at the same time. But a complete algorithm identifying all of these scenarios does not exist, and we require an explicit `Cut` in the first rule.

In *Hierarchy Builder*, we have 460 functions out of 488, with 33 higher-order arguments, and we have corrected 30 bugs in the code.

Overall, we can conclude that 85% of the predicates were functions.

In *Algebra Tactics*, we added 1 cut over 1799 lines of codes; in *Trocq*, we added 4 cuts over 2918 lines of code. None of these libraries were using `copy` as a relation, i.e. to generate all possible copies of a term. The *BRiCk* application has some public code we ported ourselves and some closed source code ported by the BlueRock company, that we thank for sharing these numbers. In the public code we added 11 cuts over 2500 lines of code, while in the private one they added 11 cuts over 7500 lines.

Authors	System	Mode analysis	Dynamic	H.O.	Miscalls
[12]	SB- <i>Prolog</i>	required	✗	✗	✗
[34]	Ciao	required	✗	✗	✗
[33]	Red Alert	required	✗	✗	✗
[49]	Mercury	required	✗	✓‡	✗†
[27]	Curry	no need	✗	✓	✓
Fissore and Tassi	<i>Elpi</i>	no need	✓	✓	✓

Table 5.2: Comparison with related work

5.7 Related work and concluding remarks

We summarize several related works in table 5.2. One key difference between our work and [12, 34, 49, 33, 27] is that they focus on *closed* predicates, that do not evolve during execution, while *λ-Prolog*, hence *Elpi*, allows rules to be dynamically loaded. For closed predicates, programs can be transformed into *super-homogeneous* form by merging all rules into one with disjunctions. This simplifies determinacy inference and may allow to compile the code to a faster binary. However, this transformation is not feasible in our dynamic setting (section 5.1.2).

Regarding modes, *Elpi*’s matching procedure over input arguments allows us to dispense with explicit mode analysis – an essential component in [12, 34, 49, 33].

We differ from Red Alert [33] in that we pick an operational semantics that explicitly represents choice points as a list of alternatives, and that hence is very close to the actual behavior of *Elpi*’s runtime. They pick a denotational semantics over the super-homogeneous form that in turn lets them formulate the static analyzer as an abstract interpreter. It is unclear if a denotational semantics can be written without transforming the program in super-homogeneous form.

Some of the higher-order features of the *Elpi* language are available in Mercury [49], although in a more constrained way (§). For example `once` from section 5.1.1 is considerably more useful if the input relation is allowed to be non-ground, e.g., `once (map get_pdiv [6] L)`. We allow higher-order predicates to be miscalled, while Mercury does not. Instead, it requires higher-order functions to be annotated with explicit mode and determinacy signatures for each possible behavior. As a result, Mercury ascribes 11 signatures to the `foldr` predicate and any call that does not match exactly one of these signatures results in a compilation failure (†). *Elpi* targets less technically inclined users, more familiar with functional programming languages than logic ones, hence we strive for simplicity, defining only one determinacy class and immediately classifying wrong calls as non-deterministic. As a result the signature for `foldr` is just `func foldr (func L, A -> A), list L, A -> A`, familiar to our audience.

The “determinism types” of Curry [27] are much more similar to our signatures, and Curry supports miscalled functions as well. In the work of Hanus and Prott the `Any` type stands for non-determinism and like a black hole it absorbs the determinism of the surrounding sub-terms. This is a convenient way to capture all miscalls and treat them uniformly. In our presentation, we have no equivalent; instead, we use `weak` to change the signature of a term when a miscall is detected (such as, $tm_{\perp}^i$ in 5.2c or `atom` calls $a_{\perp}$

in 5.2b). The advantage is that weak only changes the leaves ($\mathbb{D}$) of the signature while leaving its structure intact. Although we do not analyze data in this chapter, our set of rules can be complemented to perform full type checking (in addition to determinacy checking) in one pass.

In Curry the `allValues` builtin can be used to turn a relation into a function. *Prolog*, and also *Elpi*, provide a similar `findall` builtin, but in addition to that they also let one commit to the first result by either using `once`, or by placing a `Cut` after the non-deterministic call. It is this “after the fact” commitment that make the analysis a bit more complex. In fact, in the body of a deterministic predicate we can have calls to relations (or miscalled functions) and still consider the rules valid with respect to determinacy (see a_i in 5.2b). The notion of *functional context* in [12] is closely related to this difficulty.

Our experience report confirms most of the *Elpi* code out there is made of functional predicates and that the static analyzer accepts the vast majority of it requiring no changes. The main improvement we will consider is about *mode subtyping* for higher-order arguments. In the definition of $\subseteq$ we currently accept only arrows that have the same input/output mode. However, it seems possible to relax this condition. For example we could allow passing to `map` a function of type $\star \overset{\circ}{\rightarrow} \star \overset{\circ}{\rightarrow} \mathbf{Fun}$, since the modes of higher-order arguments seem to play no role in the determinacy of code that receives and calls them.

CHAPTER 6

A Mechanized First-Order Static Determinacy Checker

In the previous chapter we have explained the implementation of the determinacy checker algorithm that we have implemented in *Elpi*, starting from version 3. The checker helps users control backtracking: the user asserts which predicates should be considered deterministic and then the checker verifies that these predicates behave like functions, i.e., they commit to their first solution, leaving no *choice points*.

This chapter reports our first steps towards a mechanization, in the *Rocq* prover, of the determinacy analysis from [17]. We focus on the control operator *cut*, which is useful to restrict backtracking but makes the semantics depart from a pure logical reading.

We formalize two operational semantics for *Prolog* with *cut*. The first is a stack-based semantics that closely models a sublanguage of *Elpi*: we exclude from the semantics the interpretation for the *pi* and *=>* operators. This implementation is similar to the semantics mechanized by Pusch in *Isabelle* [46] and to the model of Debray and Mishra [11, Sec. 4.3]. This stack-based semantics is a good starting point to study further optimizations used by standard *Prolog* abstract machines [60, 2], but it makes reasoning about the scope of the cut operator difficult. To address this limitation we introduce a tree-based semantics in which the branches pruned by *Cut* are explicit and we prove the two semantics equivalent. Using the tree-based semantics, we then prove some properties about the impact of evaluating the cut in disjunctive queries. For example, unlike in classical logic, $q_1 \vee q_2$ is not equivalent to $q_2 \vee q_1$, since q_2 may be cut-away by q_1 .

Finally, we use the formal semantics to prove the correctness of a static determinacy analysis chapter 5. Intuitively, nondeterminism arises when a computation can be completed by applying two different rules. We exclude this situation in two ways. Firstly, at most one rule should be used on a given query. This condition is satisfied if (i) the rules have non-overlapping heads: if two heads do not overlap in the inputs they accept, then no query can unify with both, or if (ii) we account for the presence of cut in rule premises: once a rule whose premises contain a cut succeeds, all remaining rules are discarded. Secondly, each rule of a deterministic predicate should (i) either call functions, this ensure that no choice point is left after the execution of the premises of the chosen rule, (ii) or call relations provided that they are followed by the cut operator, so that any allocated choice points preceding the cut are discarded.

```

Inductive Tm := Symb of S | Pred of P | Var of V | App of Tm & Tm.
Inductive Atom := cut | call of Tm.
Record R := mkR { head : Tm; premises : list Atom }.
Record P := { rules : seq R; sig : sigT }.
Definition Σ := {fmap V -> Tm}.

```

Figure 6.1: Definition of term, rule, program and substitution

We show that if every rule defining a deterministic predicate passes this analysis, then any call to that predicate leaves no choice points.

6.1 Preliminaries: syntax and informal semantics of cut

In this chapter we use figure 2.4 as our running example, and we start by describing how we represent a program like that one.

The description of a program is given by the record $\mathbb{P}$ (in figure 6.1) and is shared by both semantics. A program consists of a list of rules R together with a signature environment sig . We delay the discussion of signatures to section 6.5 where we describe the static analysis that concerns them. The order in which rules appear in the list represent their priorities: rules coming first have higher priority.

A rule consists of a head term Tm and a list of premises. Each premise is an $Atom$, i.e. either the cut operator or a predicate call. Terms Tm include predicate symbols, data symbols, variables, and term application. The syntax tree allows variables to be applied and predicates to be mixed with data. These higher-order features play no role in the current chapter that focuses on first-order logic programming, but allows future extensions to full λ -Prolog to be based on the same mechanization.

The set of variables V , predicates P and data symbols S are countable. We represent substitutions Σ as finitely supported maps from variables to terms, using the finmap library [37]. The empty substitution is written ε , while the composition of two substitutions σ_1 and σ_2 with disjoint supports as $\sigma_1 + \sigma_2$.

The backchaining operation applies all rules to a query term; it is central to both semantics. Its implementation is very similar to the function $\mathcal{B}$ from section 2.2.2.

```

Definition backchain : P → FV → Tm → Σ → FV * list (Σ * list Atom)

```

Since a rule may be applied multiple times, its variables must be freshened before each application. Accordingly, the backchain function takes a program, the set of variable names used so far ($\mathcal{F}_V$), a query term, and the current substitution. It returns an updated set of variable names together with the list of alternatives, i.e. the rules that apply. More precisely, for each applicable rule it returns the rule premises together with an updated substitution obtained by unifying the rule head with the query term.

In our mechanization, we abstract over the unifier and its properties. The function `unify` takes two terms t_1 and t_2 , together with a substitution σ , as input, and optionally returns a substitution σ' that extends σ . The expected behavior of `unify` is the standard one (see for example [13, section 2.5]), limited to the first order. Therefore, if t_1 and t_2 unify under σ , then $\sigma' t_1 = \sigma' t_2$, where σt denotes substitution application and $=$ denotes syntactic equality.

```

Inductive tree :=
| KO | OK                                (* failure and success *)
| Todo of Atom                          (* unexplored branch *)
| Or   of option tree &  $\Sigma$  & tree      (*  $A \vee B_\sigma$  *)
| And  of tree & list Atom & tree.      (*  $A \wedge_{B_0} B$  *)

```

Figure 6.2: Definition of And-Or tree

Notational conventions In the following section we note $\mathbb{B}$ for the boolean type, and $\top$ and $\perp$ for `true` and `false`. The constructors of the option type are written $\cdot t$ for `Some t` and $\square$ for `None`. Following the Mathematical Components library, on which we depend, we use `x.1` and `x.2` to denote the first and second projection applied to the pair `x`. List concatenation is the infix `++`, while list comprehension (i.e. mapping a function on a list) is written `[seq f x | x <- l]`. The function `odflt d o` extracts the value from the option `o` and returns the default `d` if `o` is $\square$.

6.2 And-Or Tree semantics

An And-Or tree is a stratified, variable-width tree, defined in *Rocq* as in figure 6.2. It consists of two kinds of layers that alternate: a disjunctive layer is followed by a conjunctive layer and vice versa. At the root (a query), the tree records a list of alternatives (an Or-layer); for each alternative, it records a list of subqueries to be solved (an And-layer). Intuitively, solving a query requires solving one alternative in the Or-layer, and all subqueries in the corresponding And-layer.

The tree is built incrementally. The `Todo` node represents an unexplored branch (an `Atom`). When the atom is a `call`, we use the backchaining procedure from section 6.1 to compute the two layers to attach to the tree: alternatives form the Or-layer, and the premises of each applicable rule form the corresponding And-layer. This expansion step is performed by the `step` procedure described in section 6.2.2.

In addition to the `Todo` node we have two distinguished nodes, `KO` and `OK`, which represent respectively the smallest failed and successful tree.

The `Or` node, written $A \vee B_\sigma$, represents the addition of an alternative B_σ to an existing disjunction A . The left subtree A is optional and is absent once that branch has been fully explored. The right alternative is paired with a substitution σ , which becomes the current substitution when backtracking to B .

The `And` node, written $A \wedge_{B_0} B$, represents the addition of a subquery B to the first choice point in A . We call B_0 the *reset point* for B : it represents the continuation for all alternatives of A except the first. That is, when backtracking occurs within A , the subtree B is replaced by the atoms in B_0 . An example is shown in section 6.2.3.

A graphical representation of a tree is shown in figure 6.4b. For compactness, we depict `And` and `Or` nodes as n -ary, with children associating to the right. The `KO` node acts as the terminating element of an `Or` list, while `OK` is the terminating element of an `And` list.

```

Fixpoint next  $\sigma$   $t$  :  $\Sigma$  * tree :=
  match  $t$  with
  | Todo _ | KO | OK  $\Rightarrow$  ( $\sigma$ ,  $t$ )
  | Or  $\square$   $\sigma'$   $B \Rightarrow$  next  $\sigma'$   $B$ 
  | Or  $\neg A$  _  $\Rightarrow$  next  $\sigma$   $A$ 
  | And  $A$  _  $B \Rightarrow$ 
    let ( $\sigma'$ ,  $pA$ ) := next  $\sigma$   $A$  in
    if  $pA == \text{OK}$  then next  $\sigma'$   $B$ 
    else ( $\sigma'$ ,  $pA$ )
  end.

Definition next_subst  $\sigma$   $t$  := (next  $\sigma$   $t$ ).1.
Definition next_tree  $t$  := (next  $\varepsilon$   $t$ ).2.

Definition success  $t$  := next_tree  $t == \text{OK}$ .
Definition failed  $t$  := next_tree  $t == \text{KO}$ .
Definition incomplete  $t$  :=
  if next_tree  $t$  is Todo _ then  $\top$  else  $\perp$ .

```

Figure 6.3: next and derived functions

6.2.1 The semantics

The tree is constructed incrementally by two recursive functions, **step** and **prune**. Both traverse the tree depth-first, left-to-right. The node to be explored in a tree is retrieved thanks to the **next_tree** (in figure 6.3). When this node is **OK** we say the entire tree is a *success*, when it is **KO** we say the tree *failed*, otherwise the tree is *incomplete*. In figure 6.4, we provide some graphic representations of trees and mark with a black arrow the result of **next_tree**.

```

Inductive tag := Expanded | CutBrothers | Failed | Success.
Definition step :  $\mathbb{P} \rightarrow \mathcal{F}_v \rightarrow \Sigma \rightarrow \text{tree} \rightarrow (\mathcal{F}_v * \text{tag} * \text{tree})$ 
Definition prune :  $\mathbb{B} \rightarrow \text{tree} \rightarrow \text{option tree}$ 
Inductive runT :  $\mathbb{P} \rightarrow \mathcal{F}_v \rightarrow \Sigma \rightarrow \text{tree} \rightarrow$ 
  option ( $\Sigma$  * option tree)  $\rightarrow \mathbb{B} \rightarrow \mathcal{F}_v \rightarrow \text{Prop}$ 

```

Since all functions must terminate in *Rocq*, we define the transitive closure of **step** and **prune** as the inductive relation **runT**. More precisely **runT** iterates calls to **step** on incomplete trees to expand **Todo** nodes. When a tree fails it calls **prune** to find the next alternative and continues from there, if one exists.

6.2.2 The step procedure

The **step** procedure takes as input a program, a set of variables, a substitution, and a tree, and returns an updated set of variables, a **tag**, and an updated tree.

The set of variables is assumed to contain all variables occurring in the tree. It is passed to backchain so that rules can be refreshed correctly.

The **tag** indicates which kind of step was performed. **Failure** and **Success** are returned for trees that satisfy, respectively, the **failed** and **success** tests. **CutBrothers** indicates the interpretation of a superficial cut: **step** was called on an **And**-layer and its **next_tree** is the cut atom. Finally, **Expanded** accounts for the interpretation of predicate calls and non-superficial cuts.

The **step** procedure evaluates atoms. In particular, it leaves the tree unchanged when the tree is *success* or *failed*.

Lemma 6.2.1 (succ_step_iff). $\forall p v \sigma t$, $\text{success } t \leftrightarrow \text{step } p v \sigma t = (v, \text{Success}, t)$.

Lemma 6.2.2 (fail_step_iff). $\forall p v \sigma t$, $\text{failed } t \leftrightarrow \text{step } p v \sigma t = (v, \text{Failed}, t)$.

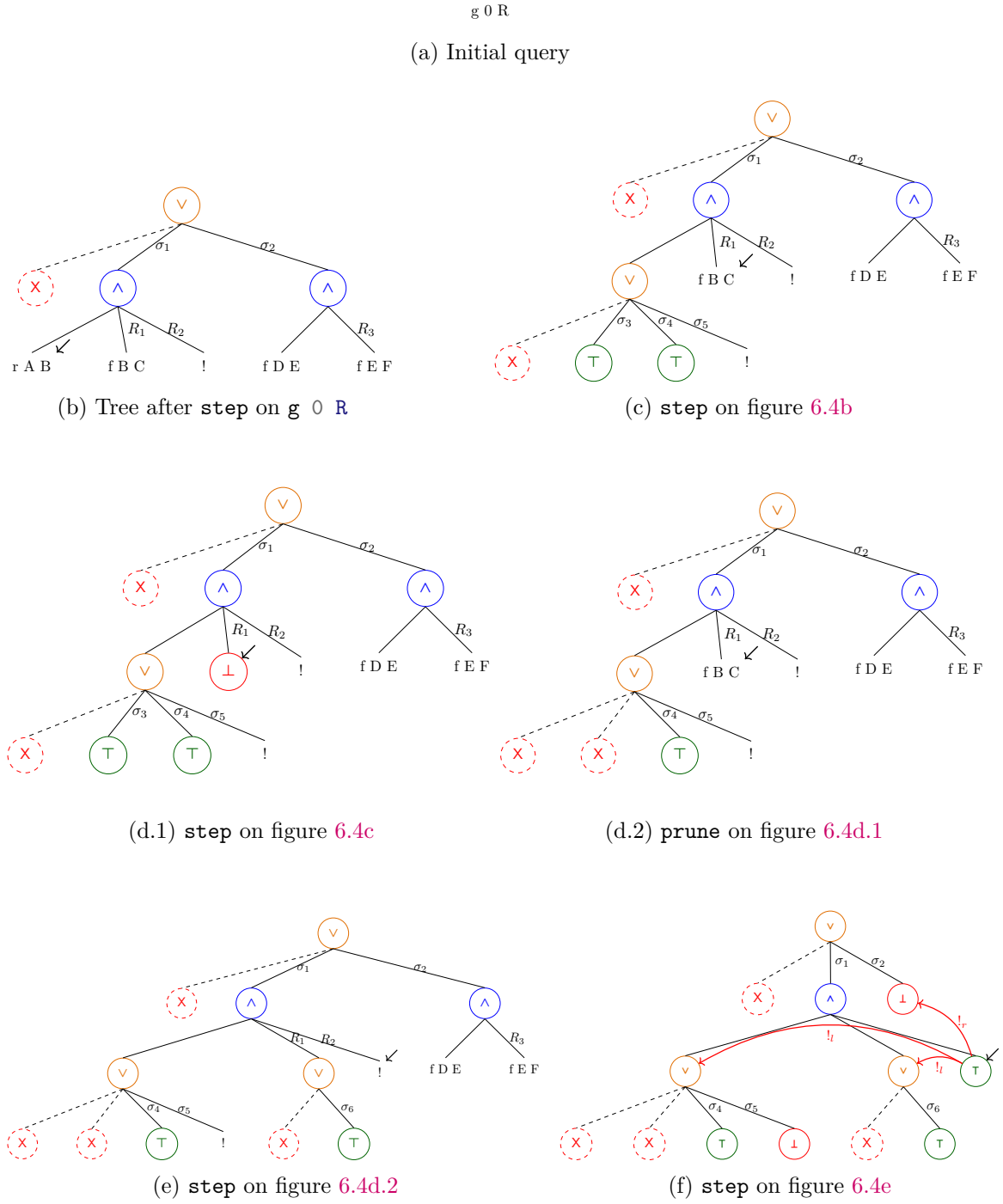


Figure 6.4: Execution in the tree semantics. The substitutions $\sigma_1 \dots \sigma_6$ are from section 2.2.3. R_1, R_2 , and R_3 are reset points and correspond respectively to the lists of atoms $[f \ Y \ Z, !]$, $[!]$, and $[f \ Y \ Z]$.

The **step** procedure expands **Todo** nodes by calling **backchain**, which returns a list l of alternatives. If l is empty, then the current call atom is replaced by **K0**. Otherwise, it is replaced by a right-skewed tree of **Or** nodes, where each leaf corresponds to a list of atoms $e \in l$. If e is empty, the leaf is **OK**; otherwise, it is a right-skewed tree of **And** nodes.

Figure 6.4 depicts the tree corresponding to the running example (section 2.2.3) as it undergoes calls to **step** and **prune**. We run query $q \ 0 \ R$ against the program P in figure 2.4. Let c_0 , c_1 , and c_2 be the children of the root. By construction, c_0 is the $\square$ tree, while c_1 and c_2 correspond respectively to the premises of rules r_1 and r_2 in P . Note that c_1 (resp. c_2) is associated with substitution σ_1 (resp. σ_2), whose values are the same as in section 2.2.3. Reset points are defines as follows: $R_1 := [\mathbf{f} \ Y \ Z, !]$, $R_2 := [!]$, and $R_3 := \mathbf{f} \ Y \ Z$. Intuitively, they record the lists of atoms used to generate the corresponding subtree. Other examples of **step** performing a tree expansion via **backchain** appear in figures 6.4c to 6.4e.

The interpretation of a cut The step corresponding to a cut performs two actions: (i) the cut node is replaced by **OK**; and (ii) the subtrees in the scope of **Cut** are pruned. More precisely, (ii) prunes the left siblings of the **Cut** node (in the same **And**-layer, denoted $!_l$) and prunes the right uncles of **Cut** (in the one-level-up **Or**-layer, denoted $!_r$).

An example of the impact of cut is shown in figure 6.4f. The red arrows indicate the scope of the cut and are labeled with $!_r$ and $!_l$ to indicate which pruning operation is applied to each pointed subtree. Note that the evaluation of a cut keeps the path leading to the cut node unchanged, in the sense that substitution σ_4 is preserved, while the path under substitution σ_5 (which also succeeds) is replaced by **K0**. This amounts to discarding the third rule of predicate r .

6.2.3 The prune procedure

When **backchain** returns an empty list, **step** produces a tree that *failed* and the computation must backtrack. The **prune** procedure is responsible for producing the new tree from which the computation can continue.

In figure 6.4d.1 we encounter a failure because no rule applies to the atom $\mathbf{f} \ B \ C$ under substitution σ_3 , which maps B to 0. The **prune** procedure prunes the path leading to this failure and resets the current sub-query to its original shape. More precisely, it kills the **OK** node under σ_3 (since that branch has been fully explored and produced no solution), and then replaces the **K0** node with the information stored in the reset point R_1 . This restores the call $\mathbf{f} \ B \ C$, which can then be reconsidered under substitution σ_4 .

Furthermore, **prune** serves another important purpose. Its behavior depends on the boolean flag it receives as input: **prune** $\perp t$ recursively removes all failures (if any) from t , whereas **prune** $\top t$ removes the first success in t . For example, the tree in figure 6.4f contains a successful path that maps R to 2. Calling **prune** on this tree returns $\square$, since there are no alternatives in the tree after the success.

Some interesting properties of **prune** are shown below; they illustrate how **prune** complements **step** with respect to failed, successful, and incomplete trees.

Lemma 6.2.3 (*incomplete_prune_id*). $\forall b \ t, \text{incomplete } t \rightarrow \text{prune } b \ t = t$.

Lemma 6.2.4 (*prune_Some*). $\forall b \ t \ t', \text{prune } b \ t = t' \rightarrow \text{failed } t' = \perp$.

$$\begin{array}{c}
\frac{\text{success } t \quad \text{next_subst } \sigma \ t = \sigma' \quad \text{prune } \top \ t = t'}{\text{runT } \mathcal{P} \ v \ \sigma \ t \cdot(\sigma', t') \perp v} \text{runT}_{\top} \\
\\
\frac{\text{incomplete } t \quad b' = (\text{tg} = \text{CutBrothers}) \vee b \quad \text{step } \mathcal{P} \ v \ \sigma \ t = (v', \text{tg}, t') \quad \text{runT } \mathcal{P} \ v' \ \sigma \ t' \ r \ b \ v''}{\text{runT } \mathcal{P} \ v \ \sigma \ t \ r \ b' \ v''} \text{runT}_{\mathcal{B}} \\
\\
\frac{\text{failed } t \quad \text{prune } \perp \ t = \cdot t' \quad \text{runT } \mathcal{P} \ v \ \sigma \ t' \ r \ n \ v'}{\text{runT } \mathcal{P} \ v \ \sigma \ t \ r \ n \ v'} \text{runT}_{\odot} \\
\\
\frac{\text{prune } \perp \ t = \square}{\text{runT } \mathcal{P} \ v \ \sigma \ t \ \square \perp v} \text{runT}_{\perp}
\end{array}$$

Figure 6.5: Rule system for `runT`

2675 **Lemma 6.2.5** (`prune_None`). $\forall t, \text{prune } \perp \ t = \square \rightarrow \text{failed } t$.

2676 6.2.4 The `runT` relation and its properties

2677 The inductive relation `runT` relates four inputs to three outputs. The inputs are a program,
 2678 a set of variables, an initial substitution σ , and a tree t . The outputs are an optional result
 2679 r , a boolean b , and an updated set of variable names.

2680 The main result r is $\square$ when the execution fails; otherwise when $r = \cdot(\sigma', t')$ the
 2681 execution succeeded and produced the solution σ' ; t' represents the remaining, not-yet-
 2682 explored alternatives. The boolean b records whether a superficial cut was evaluated
 2683 during execution.

2684 The `runT` predicate has four cases, depicted as inference rules in figure 6.5. (`runT` _{$\top$})
 2685 If `next_tree` t is a success, then the substitution σ' is extracted from t (starting from σ)
 2686 and the input tree is cleaned of its successful path. (`runT` _{$\mathcal{B}$}) If `next_tree` t is an atom,
 2687 then `step` is invoked to evaluate t , and `runT` is called recursively on the updated tree.
 2688 (`runT` _{$\odot$}) If `next_tree` t is a failure, then `prune` is called to clear the failed path; if the
 2689 resulting cleaned tree exists, `runT` is called recursively on it. Finally, (`runT` _{$\perp$}) accounts for
 2690 trees with no solutions.

2691 The set of rules defining `runT` is deterministic.

2692 **Lemma 6.2.6** (`runT_det`). $\forall p \ v_0 \ \sigma \ t \ r \ r' \ b \ b' \ v \ v', \text{runT } p \ v_0 \ \sigma \ t \ r \ b \ v \rightarrow$
 $\text{runT } p \ v_0 \ \sigma \ t \ r' \ b' \ v' \rightarrow r' = r \wedge v' = v \wedge b' = b$.

2693 The proof is by induction on the first `runT` derivation and is immediate, because
 2694 the rules are selected based on `next_tree` t . In particular, the hypotheses `success` t ,
 2695 `incomplete` t , and `failed` t are mutually exclusive. Moreover, by lemma 6.2.5, the hy-
 2696 pothesis of `runT` _{$\perp$} implies that t `failed`; hence `runT` _{$\perp$} is exclusive with `runT` _{$\top$} and `runT` _{$\mathcal{B}$} ,
 2697 and it is also exclusive with `runT` _{$\odot$} since they impose different requirements on the output
 2698 of `prune`.

2699 The boolean output of `runT` allows us to state interesting properties about the `Or` node
 2700 in the presence of cut. Here we report only a selection of those we proved.

Lemma 6.2.7 (runSST_or). $\forall p v v' \sigma \sigma' A A' \sigma_1 B, \text{runT } p v \sigma A \cdot (\sigma', \cdot A') \top v' \rightarrow$
 $\text{runT } p v \sigma (\cdot A \vee B_{\sigma_1}) \cdot (\sigma', \cdot (\cdot A' \vee KO_{\sigma_1})) \perp v'.$

Lemma 6.2.8 (runNT_or). $\forall p v v' \sigma A \sigma_1 B, \text{runT } p v \sigma A \square \top v' \rightarrow$
 $\text{runT } p v \sigma (\cdot A \vee B_{\sigma_1}) \square \perp v'.$

Lemma 6.2.9 (runNF_or). $\forall p v_0 v_1 v_2 \sigma A \sigma_1 \sigma_2 B B' b,$
 $\text{runT } p v_0 \sigma A \square \perp v_1 \rightarrow \text{runT } p v_1 \sigma_1 B \cdot (\sigma_2, B') b v_2 \rightarrow$
 $\text{let } sR := \text{omap } (\text{fun } x \Rightarrow \square \vee x_{\sigma_1}) B' \text{ in}$
 $\text{runT } p v_0 \sigma (\cdot A \vee B_{\sigma_1}) \cdot (\sigma_2, sR) \perp v_2.$

Lemmas 6.2.7 to 6.2.9 correspond to the introduction rules for disjunction. If A executes a cut, one *cannot* obtain $A \vee B$ from B : the execution of A , whether it ultimately succeeds or fails, makes the success of B irrelevant (the cut discards it). In the first lemma, B is forced to be KO ; in the second one, the failure of A absorbs B . The last lemma is connected to the depth-first exploration of the tree: proving a disjunction from its right-hand side can only occur after the left-hand side has been disproved. Depicted as rules:

$$\frac{A \quad (A \text{ cuts})}{A \vee B} (\vee_{il}) \quad \frac{\neg A \quad (A \text{ cuts})}{\neg(A \vee B)} (\vee_{ir1}) \quad \frac{\neg A \quad B \quad (A \text{ does not cut})}{A \vee B} (\vee_{ir2})$$

Lemmas 6.2.10 and 6.2.11 are elimination rules for disjunction.

Lemma 6.2.10 (run_orSST). $\forall p v v' \sigma \sigma' \sigma_1 A A' B B',$
 $\text{runT } p v \sigma (\cdot A \vee B_{\sigma_1}) \cdot (\sigma', \cdot (\cdot A' \vee B'_{\sigma_1})) \perp v' \rightarrow$
 $\exists b, \text{runT } p v \sigma A \cdot (\sigma', \cdot A') b v' \wedge B' = \text{if } b \text{ then } KO \text{ else } B.$

Lemma 6.2.11 (run_orSNT). $\forall p v v' \sigma \sigma' \sigma_1 A B B',$
 $\text{runT } p v \sigma (\cdot A \vee B_{\sigma_1}) \cdot (\sigma', \cdot (\square \vee B'_{\sigma_1})) \perp v' \rightarrow$
 $(\exists b, \text{runT } p v \sigma A \cdot (\sigma', \square) b v' \wedge \text{if } b \text{ then } B' = KO \text{ else } \text{prune } \perp B = \cdot B') \vee$
 $(\exists v_2 b, \text{runT } p v \sigma A \square \perp v_2 \wedge \text{runT } p v_2 \sigma_1 B \cdot (\sigma', \cdot B') b v').$

From $A \vee B$, when A is not inert (i.e. not already explored), we cannot deduce A or B independently. Instead, we can only derive a weakened conclusion of the form $\top \vee B'$, where B' provides no information in the case where the exploration of A performs a cut. This yields an elimination rule that is stronger than the usual one (depicted on the left). If A does not cut, the elimination rule has the usual two premises, together with the additional constraint that whenever B holds, A must have failed. This again reflects the depth-first traversal of the proof tree.

$$\frac{A \vee B \quad \frac{[A]}{C} \quad (A \text{ cuts})}{C} (\vee_{e1}) \quad \frac{A \vee B \quad \frac{[A]}{C} \quad \frac{[\neg A, B]}{C} \quad (A \text{ does not cut})}{C} (\vee_{e2})$$

It is worth recalling that the “ A cuts” side-condition in the rules above is precisely the boolean result returned by `runT`. Without this information, it is impossible to formulate these rules: it is not merely a matter of whether A contains a syntactic occurrence of cut, but whether that cut is actually executed during the (dis)proof of A (in rules $(\vee_{ir1})$ and $(\vee_{ir2})$). Indeed, an atom preceding the cut may fail, so the cut is never reached.

As anticipated in the introduction, these lemmas contradict the commutativity of disjunction.

$$\begin{aligned}
\mathcal{B}_S(\mathcal{P}, v, t, \sigma, a, g) &= \left[\sigma', ([(\mathcal{P}, q, a) \mid q \in b] \uparrow g) \mid (\sigma', b) \in \mathcal{B}(\mathcal{P}, v, t, \sigma) \right]. \\
\frac{}{\text{runS } \mathcal{P} \ v \ ((\sigma, \emptyset) :: a) \cdot (\sigma, a)} \text{runE}_\top & \quad \frac{\text{runS } \mathcal{P} \ v \ ((\sigma, g) :: ca) \ r}{\text{runS } \mathcal{P} \ v \ ((\sigma, (\text{cut}, ca) :: g) :: _) \ r} \text{runE}_! \\
\frac{\mathcal{B}_S(\mathcal{P}, v, t, \sigma, a, g) = (v_1, a') \quad \text{runS } \mathcal{P} \ v_1 \ (a' \uparrow a) \ r}{\text{runS } \mathcal{P} \ v \ ((\sigma, (\text{call } t, _) :: g) :: a) \ r} \text{runE}_\mathcal{B} & \quad \frac{}{\text{runS } \mathcal{P} \ _ \ \emptyset \ \square} \text{runE}_\perp
\end{aligned}$$

Figure 6.6: Rule system for runS

6.3 Stack-based semantics

The stack-based semantics we present is very close to what is implemented by the *Elpi* interpreter (see section 2.2.2). Since we encode a first-order interpreter, we have no `pi` and `=>` operators, therefore the corresponding rules (`runEv` and `runE=>`) are fired. This semantics is similar to the one proposed by Pusch in [46], which is in turn closely related to the semantics given by Debray and Mishra in [11, Section 4.3]. Pusch mechanized this semantics in Isabelle/HOL, together with several optimizations that are also present in the Warren Abstract Machine [60, 2].

The execution state is a stack of alternatives. Each alternative is a list of goals. Intuitively it amounts to a disjunctive normal form: each stack item is a self contained computation, coupling a substitution with all the goals to be solved.

Goals pair an atom with a list of the *cut-to* alternatives, making the two datatypes mutually recursive, but these alternatives have a very different role. They have to be understood as pointers to lower levels of the stack itself. This datum is used to implement the cut, i.e. to prune the stack of alternatives.

```

Inductive alts := nilA | consA of (Σ * goals) & alts
with
goals := nilG | consG of (Atom * alts) & goals

```

The stack-based semantics is described by the `runS` inductive predicates.

```

Inductive runS (p: P) : Fv → alts → option (Σ * alts) → Prop

```

The `runS` predicates relates a set of variables and a list of alternatives to optional result. $\square$ stands for failure, i.e., no solution, whereas $\cdot(\sigma, a)$ represent success with σ begin the solution and a as the list of non-yet-explored alternatives.

The rule system for `runS` is given in figure 6.6 and is made by 4 cases. We distinguish two main cases. If we run out of alternatives we fail: (`runE⊥`) stops the execution and return $\square$. If the list of alternatives is $(\sigma_0, h) :: a$, then we look at the list of goals h (under the substitution σ_0). If h is empty, (`runE⊤`) stops the execution with a success σ_0 leaving a as the unexplored alternatives. If h is not empty we look at its head goal (t, ca) where t is the atom and ca is its cut-to alternatives. If t is a cut rule (`runE!`) continues to evaluate the tail of h under with alternatives ca . Finally, when t is a call, rule (`runEB`) backchains: for each rule it pairs each premise with the current stack of alternatives, and prepends the obtained goals to the tail of h . Note that if backchain returns the empty list, (`runEB`) second premise amounts to exploring the next item in the current stack.

6.3.1 Inadequacy of the stack-based semantics for formal reasoning

The lemmas that we presented in section 6.2.4 are hard to express in the stack-based semantics, due to the lack of structure in the stack to delimit the scope of the cut operator. For example, consider a stack $a_0, a_1, \dots, a_n$. If a_0 succeeds but executes a cut, it is hard to relate the remaining alternatives $a_1, \dots, a_n$ to the alternatives $b_1, \dots, b_m$ that survive the cut, since the only simple invariant is that $b_1, \dots, b_m$ is a suffix of $a_1, \dots, a_n$. If $a_1, \dots, a_n$ were generated before the call whose execution generates the cut in a_0 , then they all stay; but if some were generated afterwards, they are discarded. Expressing this precisely would require attaching, at the very least, time-stamps (or depths) to goals, which amounts to a cumbersome encoding of a tree.

6.4 Equivalence of the two semantics

In order to relate the two semantics we have to relate the computations states, i.e., the And-Or tree and the stack of alternatives. We define a translation from trees to stacks, called `t2l`, but we do not explicitly provide the converse translation, an economy allowed by the proof technique we employ, inspired by [4].

Before describing the translation of trees to stacks (in section 6.4.2) we need to define the notion of *valid tree*. The `tree` datatype indeed contains trees that cannot be generated by `runT` via `step` or `prune`, for example all the trees that do not correspond to a depth-first left-to-right tree construction strategy.

6.4.1 Valid trees

The class of valid trees is captured by the `valid_tree` function shown in Figure 6.7. While all base cases of `tree` are valid, we need to analyze the `Or` and `And` constructors more carefully.

For the `Or` constructor, we distinguish two cases depending on whether the left-hand side is present. If it is not, then the right-hand side must be a valid tree. Otherwise, the left-hand side must be a valid tree, and the right-hand side must either be the `KO` tree or be unexplored. The former case occurs when the right-hand side is cut away by the evaluation of a superficial cut in the left-hand side. In the latter case we use the `base_or` predicate to check that `B` is the right-skewed tree formed by a disjunction of conjunctions, generated after a successful backchain for a `call`.

For the `And` constructor, the left hand side is required to be a valid tree. The shape of the right hand side depends on whether the left hand side is a success. If it is not successful, then the right hand side must not be explored, i.e. `B` must be the right-skewed tree containing conjunctions of premise atoms. This is enforced using the `big_and` procedure, which generates a tree from the reset point B_0 (the same procedure used to transform each alternative output by `backchain` into the And-layer of the resulting tree). When the left hand side is successful, then the right hand side may have been explored, and consequently must be a valid tree.

```

2788 Fixpoint valid_tree A :=
2789   match A with
2790   | Todo _ | OK | KO  $\Rightarrow$   $\top$ 
2791   | Or  $\square$  _ B  $\Rightarrow$  valid_tree B
2792   | Or  $\cdot$ A _ B  $\Rightarrow$  valid_tree A &&
2793     ((B == KO) || B.base_or B)
2794   | And A B0 B  $\Rightarrow$  valid_tree A &&
2795     if success A then valid_tree B
2796     else B == big_and B0
2797   end.
2798
2799 Fixpoint t2l t  $\sigma$  (cp: alts) : alts :=
2800   match t with
2801   | OK  $\Rightarrow$   $[(\sigma, \emptyset)]$ 
2802   | KO  $\Rightarrow$   $\emptyset$ 
2803   | TA a  $\Rightarrow$   $[(\sigma, [(a, \emptyset)])]$ 
2804   ...

```

Figure 6.7: Valid tree and Tree to stack

6.4.2 From trees to stacks

The `t2l` recursive function translates a tree to a stack. For space constraints figure 6.7 only gives the base cases, while we describe the recursive cases in the following text.

The function `t2l` takes as input the tree t_0 , a substitution σ_0 , and a list of choice points cp . These choice points represent alternatives that appear at the same level of the current tree, such as, for example, the right siblings of an `Or` node. The substitution is used to determine under which substitution the subtree being computed lives.

The `OK` node represents one successful alternative; therefore it is translated into a singleton list whose only element has an empty list of goals. The `KO` node represents failure, hence it is mapped to the empty list of alternatives. Finally, atoms are mapped to a singleton list containing one alternative whose only goal is that atom, with an empty cut-to list. At this stage, the cut-to list cannot point to cp : atoms are not right-uncle subtrees, whereas cp represents alternatives that will actually be discarded if the atom turns out to be a cut. The correct cut-to list is therefore determined later, once the trees corresponding to sibling subtrees have been inspected.

The translation of $A \vee B_\sigma$ creates two list of alternatives, one for the A and the other for B . The alternatives of B are valid choice points for A and should be passed to the translation of A . The two lists of alternatives can then be concatenated. Moreover, every atom occurring one level below cp (i.e. in A or B) should add cp as its cut-to alternatives.

The translation of $A \wedge_{B_0} B$ starts with the translation of A . If the translation of A is an empty list we return the empty list of alternatives without even touching B nor B_0 , since an empty translation for A indicates failure and this in turn implies the failure of the entire conjunction. When the translation of A is a list $(\sigma_0, g) :: a$, the B node represents the continuation of the first alternative g , whereas B_0 is the continuation for each alternative in a .

Example of t2l execution The translations of the trees in figure 6.4 are shown in figure 2.5. The letters in the figures relate each tree to the corresponding stack. For example, 6.4b is translated into 2.5b. We also note that 6.4d.1 and 6.4d.2 are both translated into 2.5d, showing that `t2l` is not injective: different trees can map to the same stack. Consequently, there are transitions in `runT` that are not visible in `runS`. As an example of a call to `t2l`, we take the tree and the stack in figure 6.8, which are respectively taken from figures 2.5c and 6.4c. The red arrows point from the head of an

alternative in the tree to the head of the corresponding cell in the stack. We focus on the deepest **Or** node in the tree. This subtree is a disjunction with four children. The first is ignored, since it is $\square$. The success node below σ_3 has $f \ B \ C$ and the cut operator as continuation, corresponding to the first cell in the stack. We note that the cut operator points outside the stack, since the scope of this cut eliminates its right uncle. The success node below σ_4 has R_1 as continuation, where R_1 is the list of goals $f \ B \ C, !$; this is translated into the second cell of the stack. Finally, the cut operator under σ_5 also has R_1 as continuation. It corresponds to the third alternative. We can see that the first cut points to the translation of the subtree under σ_2 , since it is one **Or**-layer above its right uncles.

6.4.3 Equivalence proof

The tree-based semantics exposes more information than needed, hence we hide it behind existentials.

Definition 6.4.1 ($\text{runT}' \ p \ v \ \sigma \ t \ r$). $\exists \ b \ v', \text{runT} \ p \ v \ \sigma \ t \ r \ b \ v'$.

From figures 2.5 and 6.4, we observe that, upon backtracking, runT may perform more steps than runS before returning a solution or a failure. These silent transitions must be skipped when proving the equivalence between the two semantics. Therefore, we prove the following lemma.

Lemma 6.4.1 (pruneF_t2l). $\forall \ t \ t' \ b, \text{valid_tree} \ t \rightarrow \text{failed} \ t \rightarrow \text{prune} \ b \ t = t' \rightarrow \forall \ l \ \sigma, \text{t2l} \ t \ \sigma \ l = \text{t2l} \ t' \ \sigma \ l$.

Proof. By induction on the tree t . □

The first direction of the equivalence states that the execution of a valid tree is matched by the execution of the stack obtained by translating the tree. Moreover, the unexplored alternatives returned as a tree correspond to those returned as a stack.

Theorem 6.4.2 (tree_to_elpi). $\forall \ p \ t \ \sigma \ r, \text{let } v := \text{vars_tree} \ t \cup \text{vars_sigma} \ \sigma \text{ in}$

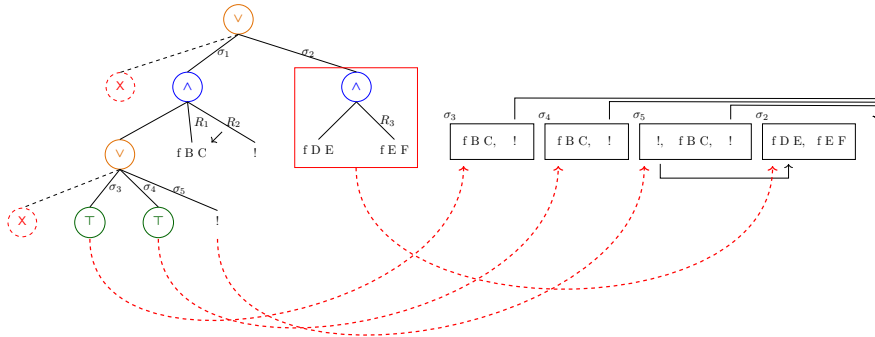


Figure 6.8: Example of t2l execution

```

valid_tree t →
  runT' p v σ t r →
    let a := t2l t σ ∅ in
    let r' := if r is ·(σ', t) then ·(σ', t2l (odflt KO t) σ ∅)
              else □ in
    runS p v a r'.

```

Proof. We proceed by induction on the execution of `runT'`. There are four cases to consider. The first case arises from `runTτ`; it deals with successful trees. A successful tree must start with an empty list, and therefore we conclude using `runEτ`. The case for `runTB` requires treating two subcases: `step` evaluates either a cut or a call. Accordingly, we apply `runE!` or `runEB`, followed by the induction hypothesis. The case for `runT∅` is silent: it represents the non-injective behavior of `t2l`; however, thanks to lemma 6.4.1 and the induction hypothesis, the conclusion is proved. The last case arises from the derivation `runT⊥`. It is easily proved using `runE⊥` and the fact that if `prune` returns `□` when trying to erase failure in `t`, then `t2l` applied to `t` returns the empty list. $\square$

The converse direction is stated following [4]: instead of using a function to translate a stack into a tree it states that any tree that translates to the given stack has an equivalent execution, i.e. gives the same solution and returns a unexplored tree that translates to the unexplored stack.

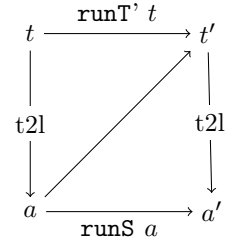
Theorem 6.4.3 (`elpi_to_tree`). $\forall p v a r, \text{runS } p v a r \rightarrow$

```

  ∀ σ t, valid_tree t → t2l t σ ∅ = a →
    if r is ·(σ', a') then
      ∃ t', runT' p v σ t ·(σ', t') ∧ t2l (odflt KO t') σ ∅ = a'
    else runT' p v σ t □.

```

Proof. The proof is based on the idea explained in [4, Section 3.5.1] and depicted in the figure on the right: we have a tree `t` whose translation is `a`, we proceed by induction on the execution of `runS`. This gives us not only the output `a'` but that hypothesis that it exists a tree `t'` such that its translation to a stack is `a'`. $\square$



Stating the converse direction in a symmetric way would require a function `l2t` transforming a stack `a` into a tree `t`. Writing this function is far from trivial, since the structure of the tree is lost by `save_as` and `runT` that intuitively keep the state as a big disjunction of conjunctions by distributing conjunctions over disjunctions. Moreover, one would need to define a notion of valid stack that is equally intricate.

The constructive nature of the logic of *Rocq* tells us that the proof above provides an effective function translating stacks into trees. However, that function also uses an execution trace of the stack-based semantics, from which it recovers the missing tree structure.

6.5 Case study: determinacy analysis

An interesting application of the tree semantics is determinacy checking, introduced in version 3 of the *Elpi* language [17]. The checker takes as input the signatures of the predicates declared by the user and verifies that the rules of a deterministic predicate generate at most one solution; we call this particular kind of predicate a function. Thanks to this static verification, the user can better control unexpected backtracking.

A predicate behaves deterministically if two conditions are satisfied: *mutual exclusion* guarantees that at most one rule can be successfully applied to any given query, whereas *rule checking* ensures that each rule does not create choice points, for example by calling non-deterministic predicates.

Determinacy checking is strongly related to the modes of predicates: we differentiate between input and output arguments. A deterministic call relates to its inputs only one output. In the literature, we find several works on determinacy checking, namely [49, 35, 12]. These works need a mode checker that statically ensures that, in a query with ground inputs, each recursive call is performed with ground inputs.

In our mechanization, and in particular in the *Elpi* language, we adopt a special unification routine for input arguments, called **matching** [2, 62]. Similarly to **unify**, **matching** takes the query term t_1 and the pattern t_2 and returns an optional substitution σ' that does not assign any variable in the query.

The function **matching** has the following property:

Axiom 6.5.1 (**matching_disj**). $\forall \sigma \sigma' t_1 t_2, \text{vars } t_1 \# \text{domf } \sigma \rightarrow \text{vars } t_1 \# \text{vars } t_2 \rightarrow \text{matching } t_1 t_2 \sigma = \sigma' \rightarrow \exists e, \text{domf } \sigma' = \text{domf } \sigma \cup e \wedge e \subseteq \text{vars } t_2.$

If the variables (**vars**) of two terms are disjoint ($\#$) and the variables in the support of σ are disjoint from the variables in t_1 , then, if matching succeeds, only the variables in t_2 are assigned; that is, $t_1 = \sigma' t_2$, and the variables in t_1 are not assigned in σ' .

The **backchain** function uses **matching** or **unify** on the arguments q depending if the argument is in *input* or *output* mode. We assume two important properties of our unification routines.

Axiom 6.5.2 (**unif_trans**). $\forall t_1 t_2 t_3 \sigma, \text{unify } t_1 t_2 \sigma \rightarrow \text{unify } t_2 t_3 \sigma \rightarrow \text{unify } t_1 t_3 \sigma.$

Axiom 6.5.3 (**match_unif**). $\forall t_1 t_2 \sigma, \text{matching } t_1 t_2 \sigma \rightarrow \text{unify } t_1 t_2 \sigma.$

We also assume that refreshing variables does not affect unification. In particular, a refreshing renaming f for a term t is a function from variables to variables that: (i) f is defined on all the variables in t , i.e. all variables get renamed; (ii) f is injective, i.e. does not confuse variables; and (iii) f has non-overlapping domain and codomain, i.e. it maps the variables of the term to a set of completely fresh variables.

Below, we define **refresh_for** to test the validity of a refreshing renaming. Axiom 6.5.4 states the property of term unifiability with respect to term renaming. That is, if the renamings of terms t_1 and t_2 are unifiable under two respective valid refreshing renamings, and they have disjoint variables, then for any pair of renamings z and x that are valid renamings for t_1 and t_2 , respectively, and are disjoint from each other, unification also succeeds. In other words, this theorem accounts for α -equivalence of terms and the unification of terms with disjoint variables.

2909 **Definition 6.5.1** (`refresh_for x t`). $(\text{vars } t \subseteq \text{domf } x) \wedge \text{injective } x \wedge (\text{domf } x \# \text{codomf } x)$.

2910 **Axiom 6.5.4** (`unif_ren`). $\forall x y z w t_1 t_2,$

$$\begin{aligned} & \text{refresh_for } w t_1 \rightarrow \text{refresh_for } y t_2 \rightarrow \text{refresh_for } z t_1 \rightarrow \text{refresh_for } x t_2 \rightarrow \\ & \text{codomf } w \# \text{vars } (\text{rename } y t_2) \rightarrow \text{codomf } z \# \text{vars } (\text{rename } x t_2) \rightarrow \\ & \text{unify } (\text{rename } w t_1) (\text{rename } y t_2) \varepsilon \rightarrow \text{unify } (\text{rename } z t_1) (\text{rename } x t_2) \varepsilon. \end{aligned}$$

2911 In figure 2.4, the rules of the program are equipped with three signatures, one per
 2912 predicate. Besides the arity of each predicate, a signature specifies which arguments are
 2913 inputs and which are outputs, their types, and whether the predicate is deterministic.
 2914 They indicate that `f` and `g` are expected to be functions, as specified by the `func` keyword,
 2915 whereas `r` is a relation, as indicated by the `pred` keyword. The `->` symbol separates inputs
 2916 from outputs; therefore, all three predicates are expected to take an `int` as input and
 2917 return an `int` as output.

2918 The signatures of the program are stored in the `sig` field of the program and are
 2919 encoded as described in [17]: we distinguish data from predicates, and predicates are
 2920 classified into deterministic a non-deterministic.

2921 A program execution behaves deterministically if, given a program, a substitution,
 2922 and an input tree for the `runT` inductive, the execution either fails or, if it succeeds, the
 2923 resulting tree of alternatives is $\square$.

2924 **Definition 6.5.2** (`is_det p σ v t`). $\forall r, \text{runT}' p v \sigma t r \rightarrow r = \square \vee \exists \sigma, r = \cdot(\sigma, \square)$.

2925 The theorem we want to prove is the following one:

2926 **Theorem 6.5.5** (`det_check_call`). $\forall p \sigma t v,$

$$\text{check_P } p \rightarrow \text{tm_is_det } p.(\text{sig}) t \rightarrow \text{is_det } p \sigma v (\text{Todo } (\text{call } t)).$$

2927 Here, `check_P` denotes the function that checks the program with respect to mutual
 2928 exclusion and rule checking, and `tm_is_det` denotes the function that tests whether the
 2929 head of the term refers to a rigid constant with a deterministic type.

2930 The proof of this lemma proceeds by induction on the program execution. However,
 2931 without further work, the induction hypothesis is too weak since it talks about `Todo` trees
 2932 (i.e. atoms), whereas we need it on any tree.

2933 To address this issue, we introduce the notion of deterministic trees (`det_tree`), show
 2934 that a tree satisfying this condition enjoys the desired properties, and then prove the
 2935 following more general theorem:

2936 **Lemma 6.5.6** (`det_check_tree`). $\forall \sigma v p t, \text{check_P } p \rightarrow \text{det_tree } p.(\text{sig}) t \rightarrow \text{is_det } p \sigma v t$.

2937 Our target theorem `det_check_call` is a corollary of `det_check_tree`.

2938 6.6 Related work

2939 We studied a *Prolog* semantics in which input arguments are *matched*, rather than unified,
 2940 against rule heads. This choice agrees with the *Elpi* interpreter, which is in turn inspired
 2941 by B-Prolog’s “matching-clauses” [2, 62]. The two semantics we presented (and proved

equivalent) also apply to standard *Prolog* if one forces all predicate signatures to have only output arguments.

For determinacy analysis, matching input arguments affects the proofs only as follows: axiom 6.5.1 does require the query q to be ground. Adapting our results to standard *Prolog* therefore requires the insertion of two additional hypotheses in theorem 6.5.5. First, one has to require that t has ground inputs. Second, that the program p is well-moded. Well-moded predicates produce ground outputs when given ground inputs; hence, starting from an initial query with ground inputs, well-moded programs behave exactly as if input arguments were matched. Therefore, the proofs in section 6.5 still apply.

The only mechanization of Prolog’s semantics we could find is the one by Pusch in *Isabelle* [46] that is based on the model of Debray and Mishra [11, Sec. 4.3]. Their work starts from that semantics and refines it by introducing some optimizations usually found in the Warren abstract machine [60, 2]. They do not use the semantics to prove properties on the execution of programs as we do in our use case, hence they do not face the difficulty described in section 6.3 that pushed us to develop a more explicit semantics (with respect to cut). In a sense, our work is complementary. By combining both works one can show the tree-based semantics is equivalent to an optimized stack-based one.

Another relevant work is by King [33], but we could not find their *Rocq* source code. Their semantics is denotational and cannot easily cover all the features that [17] covers, but it would have been sufficient for this first step.

The proof technique we use to relate the two semantics is inspired by [4], and we thank Y. Bertot for insightful discussions on the topic. In [4], Bertot mechanizes the correctness of a compiler. He relates the semantics of an imperative language to that of its compiled assembly code and proves their equivalence. His approach uses the compiler as the translation function from one language to the other, but does not introduce a decompiler, just as we do not define a function from stacks to trees. To show that the assembly code behaves like the imperative language, he proceeds by induction on the execution of the assembly program. We adopt the same strategy in the proof of theorem 6.4.3.

6.7 Conclusion

This chapter describes two operational semantics for Prolog with cut. The first semantics is based on And–Or trees: it is well suited to graphically illustrating the scope of cut, and to proving lemmas about the impact of its evaluation when reasoning about disjunctions. The second semantics is based on a stack of alternatives: it is computationally more efficient, but it loses the structural expressiveness of the tree-based semantics. Thanks to a dedicated function, we translate trees into stacks and prove the two semantics equivalent. This stack-based semantics corresponds to the first-order fragment of *Elpi*. Using the tree-based semantics, we mechanize the first-order part of the determinacy checker that has been available in *Elpi* since version 3.0, and we prove that a call to a deterministic predicate returns at most one solution.

The formal development consists of approximately 2,500 lines of specifications and about 5,000 lines of *ssreflect* proofs. Roughly 30% of the code is dedicated to the tree semantics (in particular, the proofs in section 6.2.4), 15% to the stack semantics, and 35%

2984 to the equivalence proof. The remaining 20% is devoted to the determinacy checker. The
2985 mechanization took us approximately 8 months.

2986 To fully model the *Elpi* language and mechanize [17] entirely, we would need to account
2987 for higher-order variables, i.e. variables representing predicates. This extension is ongoing
2988 and largely orthogonal to the present chapter, since cut and higher-order features do not
2989 interact in subtle ways.

CHAPTER 7

Conclusion and Perspectives

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Appendix

APPENDIX A

Simple compiler full implementation

We provide here the complete code for the simple elpi compiler.

```
From elpi Require Import elpi.
Require Import Reals.

Elpi Db tc.db lp:{{
  pred tc term → term.
}}.

Elpi Command Compiler.
Elpi Accumulate Db tc.db.
Elpi Accumulate lp:{{
  shorten std.{rev}.
  shorten coq.{mk-app}.

% Inst Ty Args Prems Res
pred comp term, term, list term, list prop → prop.
comp I {{lp:T → lp:Bo}} Ag P (pi y\ R y) :- !, % r$\to$
  pi x\ comp I Bo [x|Ag] [tc T x | P] (R x).
comp I {{∀ x, lp:(Bo x)}} Ag P (pi y\ R y) :- !, % r$\forall$
  pi x\ comp I (Bo x) [x|Ag] P (R x).
comp I T Ag P (tc T Proof :- [true | Body]) :- % rB
  mk-app I {rev Ag} Proof,
  rev P Body.

pred compile gref →.
compile G :- coq.env.typeof G Ty,
  comp (global G) Ty [] [] R,
  coq.elpi.accumulate _ "tc.db" (clause _ _ R).

main [str S] :- coq.locate S GR, compile GR.
}}.

Elpi Tactic Solver.
Elpi Accumulate Db tc.db.
Elpi Accumulate lp:{{
solve (goal _ _ Ty _ _ as G) S :- tc Ty P, refine P G S.
}}.
```

```
Class Add T := {plus: T → T → T}.

Instance addNat : Add nat := {plus := Nat.add}.
Instance addR : Add R := {plus := Rplus}.
Instance addProd T1 T2 : Add T1 → Add T2 → Add (T1 * T2) :=
  {plus '(x1,y1) '(x2, y2) := (plus x1 x2, plus y1 y2)}.

Elpi Compiler addNat.
Elpi Compiler addR.
Elpi Compiler addProd.

Goal Add (nat * (nat * nat)).
Proof. elpi Solver. Qed.
```


Tritre francais

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Davide FISSORE

Résumé

Les macarons c'est bon par nature, c'est le propre même du macaron que d'être bon. Au vue de l'affluence constante aux divers magasins Ladurée on peut supposer que les macarons qu'on y trouve sont meilleurs qu'ailleurs. Cette thèse vise à donner des idées et pistes sur le pourquoi du comment que les macarons de Ladurée sont si bons.

Mots-clés : Pâtisserie, Macaron, Ladurée.

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Abstract

Macarons are so good.

Keywords: Pastry, Macaron, Ladurée.